

Spin-3/2 TMDs and structure functions

Analytical formalism and independent validation

From spin density matrices to polarized lithium-7 observables

Dustin Keller

Department of Physics, University of Virginia
Charlottesville, Virginia, USA

Revised research note

6 October 2026

Abstract

This note develops the leading-twist quark and gluon structure of a spin-3/2 target from explicit density-matrix and transverse-tensor constructions. Irreducible spin operators, their trace normalization, and the complete population/coherence inversion determine the physical target components. Gauge-invariant correlators and their projections then give minimal bases of 32 quark and 32 gluon TMD coefficients, including fourteen octupole coefficients for each species. The full quark SIDIS response is retained, the fourteen octupole Drell–Yan channels are derived, and modified universality is stated with its Wilson-line, color, scale, and factorization conditions. Physical electron helicity and ordered lepton currents fix the corrected heavy-quark-pair Born contraction and its circular-polarization sign. Positive octupole preparations and joint angular projections provide explicit gluon-response separation. Collinear selection, local moments, Fourier transforms, conditional positivity, nuclear convolution, and ideal NMR calibration retain their analytical assumptions. The computational companion is summarized through its independent construction routes, exact certificates, amplitude comparisons, convolution tests, and run-specific evidence. Established results, target-side extensions, and illustrative models remain distinguished throughout. The analytical examples are preserved as support for the formalism, not as substitutes for a nuclear TMD determination.

Contents

1	Purpose, prerequisites, and relation to previous work	5
1.1	What the reader should know, and what is developed here	5
1.2	Historical foundations and the contribution of this work	5
1.3	Domain, conventions, and the companion manuscript	7
1.4	A practical route through the derivation	8
2	From a scattering measurement to a partonic response	8
2.1	The two resolutions of a TMD measurement	8
2.2	Spin dependence as an amplitude bilinear	9
2.3	Operators, coefficient functions, and structure functions	9
2.4	Three different spin or angular-momentum labels	10
2.5	A spin-one-half warm-up	10
2.6	How to organize a calculation before doing algebra	10
3	Spin-3/2 density matrix and polarization moments	11
3.1	Ensembles, observables, and a useful reference state	11
3.2	Ladder operators and the organization of operator space	11
3.3	Constructing the irreducible spin operators	12
3.4	Deriving the trace subtractions and inverse normalization	12
3.5	Exact definition of the longitudinal/transverse components	14
3.6	Diagonal populations and an octupole-only example	14
3.7	Coherences, spherical tensors, and a projection recipe	15
3.8	A complete matrix reconstruction from populations and coherences	15
4	Kinematics and transverse tensor conventions	16
4.1	Why light-front coordinates are useful	16
4.2	Light-front coordinates and the nuclear momentum fraction	17
4.3	Parton fraction, Bjorken variable, and target-mass counting	17
4.4	Lorentz and Euclidean transverse conventions	18
4.5	Momentum tensors of every needed rank	18
4.6	Scalar invariants and their normalization	19
4.7	Constructing STF tensors instead of memorizing them	19
4.8	Covariant embedding and changes of quantization axis	20
5	Operator correlators and the meaning of a TMD coefficient	20
5.1	From a bilocal field product to a transverse-momentum distribution	20
5.2	Mixed-state expectation values and gauge links	21
5.3	Why the Wilson line is part of the definition	21
5.4	Unsubtracted quark and gluon correlators	22
5.5	Dimensions, good fields, and the leading-twist polarization space	22
5.6	Rapidity subtraction and scales	23
5.7	Ultraviolet and rapidity divergences are different limits	23
5.8	Projection operators and reconstruction	24
5.9	Checking a reconstruction by its projection Gram matrix	24
5.10	How a coefficient is determined from a correlator	24
6	Completeness from transverse rotations, parity, and link reversal	25
6.1	A constructive enumeration procedure	25
6.2	Transverse-helicity matching in two dimensions	25
6.3	Worked coupling problems and the role of reflections	26
6.4	Deriving the parity pattern, including the quark tensor duals	26
6.5	Counting and an independent matrix-space check	27
6.6	Null transverse tensors and redundant coefficient labels	27
6.7	What a completeness check actually establishes	28
6.8	The complete lower-rank gluon conversion	29
6.9	Time reversal and the reversal of Wilson paths	30

6.10	Reading the link-reversal table without discarding allowed terms	30
7	Operator universality and the SIDIS–Drell–Yan connection	31
7.1	Three distinct levels of a cross-process statement	31
7.2	Antiunitary transformation of an arbitrary target state	31
7.3	All fourteen octupole quark relations	32
7.4	Gluon link classes and the meaning of conditional universality	33
7.5	Scale evolution and coefficient-level null tests	33
8	The complete leading-twist quark correlator	33
8.1	Beginning with the unpolarized and vector-polarized channels	34
8.2	Naming convention and the two scalar projections	34
8.3	The complete transverse-spin projection	34
8.4	Constructing the four octupole sectors	35
8.5	Worked construction of every channel for the LTT octupole	35
8.6	Quark distribution labels and normalization conventions	36
9	The complete leading-twist gluon correlator	36
9.1	Linear polarization as interference, not a spin vector	36
9.2	Unpolarized, vector, and quadrupole sectors	37
9.3	The seven octupole linear-polarization terms	37
9.4	Worked octupole examples and normalization of both branches	37
9.5	Gluon distribution labels and normalization conventions	38
10	Projecting and recovering the independent TMD coefficients	38
10.1	Projecting the target and scalar harmonics	38
10.2	Quark vector branches	39
10.3	Gluon linear-polarization branches	39
10.4	Why two target orientations separate the orbital branches	40
11	Physical interpretation of the multipole distributions	40
11.1	A multipole coefficient is a response to the target state	40
11.2	Number and helicity contrasts between target populations	40
11.3	Quark transversity as a contrast of coherences	41
11.4	Momentum-dependent correlations and experimentally distinct questions	42
12	The renormalized collinear limit and inclusive structure functions	42
12.1	Separating angular projection from ultraviolet integration	42
12.2	Angular selection is not a convergent all-orders integral	42
12.3	Six quark and five gluon distributions	43
12.4	Born-level inclusive quark SFs with explicit normalization	43
12.5	First local moments as consistency checks	44
12.6	Deriving the hierarchy of local quark moments	44
12.7	Inclusive limits and conserved quantities as model tests	45
13	Positivity of target states and parton spin matrices	46
13.1	Why a Gram matrix is positive, and what subtraction changes	46
13.2	Target-state positivity and conditional correlator positivity	46
13.3	Constructing the full eight-by-eight matrix	47
13.4	Spectral indices and auxiliary coefficient coordinates	47
13.5	Collinear bounds derived from the spin matrices	48
13.6	A reproducible route from spin matrices to the collinear bounds	49
14	Scale evolution, transverse-distance transforms, and matching	49
14.1	Deriving the tensor Fourier transform and Bessel order	50
14.2	Solving the scale equations and identifying their assumptions	51
14.3	Matching, weighted observables, and the information that remains	51
15	Derivation of the complete quark SIDIS response	51
15.1	What is held fixed in a SIDIS measurement	51

15.2	Process, axes, and fragmentation correlator	52
15.3	Fragmentation momentum and the origin of the convolution	53
15.4	Hard Dirac trace and absolute normalization	53
15.5	Performing the hard trace one polarization channel at a time	54
15.6	Deriving the convolution weights	55
15.7	Lepton factors, phase space, and angular-moment normalization	55
15.8	Complete catalogue of 32 tree-level SIDIS structure functions	56
15.9	Two worked octupole examples	57
15.10	Independence of structure functions and experimental sensitivity	57
15.11	A complete analytic convolution example	58
16	Drell–Yan octupole observables from the same quark correlator	59
16.1	Specify both incoming hadrons and the negative-lepton angles	59
16.2	Derive the distribution–distribution trace before assigning PDFs	60
16.3	Convolution, color factor, and differential normalization	60
16.4	Three number responses and their normalized moments	60
16.5	Four helicity responses and the physical antiquark sign	61
16.6	Seven transverse-spin responses with a Boer–Mulders partner	61
16.7	A predictive cross-process comparison	62
17	Gluon-sensitive structure functions and the Born hard response	62
17.1	Separating the target density from the hard polarization analyzer	62
17.2	Hard-process dependence of a gluon measurement	63
17.3	Deriving the Stokes contraction and its sign conventions	63
17.4	Complete angular kernels for the linear-polarization terms	64
17.5	Physical electron helicity and ordered current tensors	65
17.6	The two-diagram heavy-quark Born analyzer	65
17.7	An explicit derivation of both Ward identities	66
17.8	Building a physical phase-space point from scalar inputs	67
17.9	A reproducible benchmark and the scope of its interpretation	67
17.10	From the Born analyzer to a measured small-imbalance SF	68
17.11	Deriving the active fraction and preserving the imbalance	69
18	Rank-selective target preparation and response tomography	69
18.1	Constructing positive states with only an octupole moment	69
18.2	Deriving the rotated Cartesian components	70
18.3	Continuous inversion of the rank-three response	71
18.4	Discrete preparations and what tomography can identify	72
18.5	A spin–orbital mechanism for a composite target	72
19	Separating all fourteen octupole gluon responses	73
19.1	Fix the hard event while varying the target and recoil directions	73
19.2	Four spin settings and two reduced responses	73
19.3	Derive every angular coefficient	74
19.4	An exact angular certificate and its physical qualifications	74
19.5	Normalized moments and the distinction between a ratio and a TMD	75
19.6	A transverse preparation and the separate rank-one and rank-five terms	75
19.7	Acceptance, conditioning, and the boundary of the Born statement	76
20	Nuclear matrix elements, target calibration, and experimental inference	76
20.1	Nuclear spin structure and an impulse-approximation organization	76
20.2	Setting up a nuclear calculation before projecting TMDs	77
20.3	Vector polarization and independent higher-rank calibration	78
20.4	An ideal NMR population inversion, with its assumptions	78
20.5	Spin temperature is an additional relation among populations	79
20.6	Deriving transition strengths and propagating calibration errors	79
20.7	An explicit octupole-only response combination	80
20.8	A staged determination of TMDs from data	80
20.9	Designing an identifiable structure-function extraction	81

21 Computational implementation and independent validation	81
21.1 Versioned scope and the status of the evidence	82
21.2 Independent construction versus repetition	83
21.3 Completeness, exact certificates, and coefficient recovery	83
21.4 Quark response kernels and independent convolution checks	84
21.5 Reversal relations and matched operator definitions	84
21.6 Gluon responses, physical currents, and numerical diagnostics	85
21.7 Limits, moments, and conditional positivity	86
21.8 Implementation map and evidence integrity	86
21.9 What the computational agreement establishes	88
22 Synthesis and outlook	88
A Rank-four and rank-five transverse momentum tensors	89
B Notation and implementation checklist	90

1 Purpose, prerequisites, and relation to previous work

The central question of this note is operational: how does the partonic response change when a spin-3/2 target is prepared with an octupole polarization independently of its vector and quadrupole moments? Answering it requires more than writing a list of functions. We must specify the target state, define a gauge-invariant operator, determine its independent spin and momentum structures, project out their coefficients, and calculate how a chosen scattering process measures those coefficients. Each of these steps has a different input and a different output.

The note develops this sequence for a general massive spin-3/2 target, with polarized ⁷Li as the motivating nucleus. The ground state of ⁷Li has $J^P = 3/2^-$ [1]. Its intrinsic parity cancels between the bra and ket of a forward matrix element. Thus it does not change the parity classification of the target-side distribution correlators. The nuclear state is an asymptotic QCD state: introducing an elementary Rarita–Schwinger field is neither required nor assumed.

1.1 What the reader should know, and what is developed here

The prerequisites are graduate quantum mechanics, elementary special relativity, and an introductory acquaintance with Dirac fields and Feynman amplitudes. In particular, the reader should know how to multiply matrices, use angular-momentum ladder operators, evaluate a Fourier transform, and read a bilinear such as $\bar{\psi}\gamma^\mu\psi$. The density-matrix, irreducible-tensor, light-front, and TMD machinery is developed explicitly. A first reading can follow the finite-dimensional spin and tensor algebra before returning to the operator-renormalization discussion.

There are three different senses of “determining a TMD.” A symmetry classification determines which scalar coefficient functions are allowed. An operator projection determines the definition of each coefficient in a computed correlator. An experimental extraction determines numerical functions only after the measured response has been related to those operators through a factorization prescription and sufficient data. The first two can be completed without specifying a nuclear wave function; the third cannot be replaced by counting tensors.

$$\rho \longrightarrow (S, T, R) \longrightarrow (\Phi, \Gamma) \longrightarrow (f, g, h) \longrightarrow W^{\mu\nu} \longrightarrow F_\nu \longrightarrow d\sigma. \quad (1.1)$$

The arrows mean, respectively, spin projection, an operator expectation value, a tensor decomposition, a hard-scattering contraction, an observable response decomposition, and a leptonic/phase-space contraction. None is a replacement for the next. The task of this note is to show how to set up and check every arrow within the stated leading-power framework.

1.2 Historical foundations and the contribution of this work

Density matrices and irreducible polarization operators are standard quantum-mechanical tools [2]. The multipole organization of DIS for arbitrary-spin targets predates TMD phenomenology [3]. The lower-spin quark distributions, the spin-one tensor sector, and the usual single-hadron SIDIS hard trace are also established results [4–6]. The spin-3/2 spin algebra used here is compatible with the explicit fragmentation construction of Zhao *et al.* [7]. Their fourteen octupole quark fragmentation functions are an important classification benchmark. They are not the distribution functions of a spin-3/2 incoming target: the operator matrix elements and the role of out-states differ.

Their later SIDIS study explicitly develops the production of a polarized spin-3/2 final-state hadron [8]. Here the rank-three state is instead the incoming target. The quark classification is therefore presented as a distribution-side construction with related tensor algebra, not a first appearance of octupole SIDIS harmonics. The finite- k_T gluon octupole sector and its incoming-target response are the central extensions. Helicity-based organization of parton correlators has

an important methodological precedent in Ref. [9], and the transverse Stokes decomposition is established machinery [10].

Nuclear multipole distributions also have a dynamical precedent: Cloët, Bentz, and Thomas calculated collinear nuclear distributions, including a rank-three quark response in ^{11}B [11]. That is not a finite- k_T gluon octupole construction. The spin-orbital example below illustrates a known type of mechanism rather than claiming that nuclear octupole partonic structure was previously absent.

The quark and gluon GPD analyses and the independent forward-PDF positivity analysis provide additional checks [12–14]. The gluon correlators and helicity matrices for spin at most one supply the lower-rank gluon comparison [15–17]. Spin-one SIDIS has been developed for current fragmentation and in a more general covariant treatment [18, 19]; their kinematic scopes should not be identified merely because both use the word SIDIS.

In particular, the higher-spin extension of the kinematic decomposition in Appendix D of Ref. [19] is distinct from the leading-power partonic correlators and Born convolutions worked out here.

Table 1: Provenance and role of the principal ingredients.
“This work” means the target-side construction developed in this work, not a blanket historical-priority claim.

Ingredient	Established input	Development in this note/work
Spin density matrix	Irreducible angular-momentum operators; explicit spin-3/2 conventions in Ref. [7].	Step-by-step normalization, Cartesian/spherical inversions, and the complete population/coherence map.
Quark tensors	Lower-spin PDFs and spin-3/2 FF tensor classification.	Distribution-side operator construction, link-reversal assignments, and its complete response dictionary in one fixed convention.
Gluon tensors	Lower-spin correlators and helicity matrices in Refs. [16, 17].	Minimal physical two-dimensional basis, explicit trace redundancy, and the target octupole extension.
Completeness	Standard $SO(2)$ representations and matrix linear algebra.	Their application to the complete spin-3/2 target correlators and explicit inverse coefficient maps. The representation theory itself is not new.
Collinear structure	Spin-3/2 forward PDFs/GPDs and positivity relations [12–14].	Re-derivation in the present normalization and a transparent local-moment hierarchy.
Quark SIDIS	Standard electromagnetic leading-power hard trace and fragmentation mechanism [5, 20].	Complete target-rank-three response organization, weights, and worked angular projections.

Ingredient	Established input	Development in this note/work
Drell–Yan	Standard annihilation hard trace and dilepton angular analysis [21].	Incoming-target octupole channels, physical antiquark/beam conventions, and cross-process operator tests.
Gluon hard response	Photon–gluon fusion and gluon-polarization analysis [22, 23].	Coupling the target octupole basis to a fully specified hard polarization matrix. The Born diagrams are standard.
State tomography	Standard spin rotations, linear response, and matrix inversion.	Positive octupole-only probes and a rank-selective measurement construction. Physical realizability of the matrix does not establish apparatus capability.
Nuclear/NMR examples	Impulse-approximation reasoning, magnetic-dipole selection rules, and basic inference.	Explicit instructional examples; no fitted lithium-7 TMD or demonstrated target preparation is inferred.
Computational methods	Established symbolic and array tools [24, 25]; public calculation code in Ref. [26].	Executable teaching constructions, independent representation checks, rank certificates, diagnostic failures, and a reproducibility workflow.

The work-level result is a minimal physical transverse basis of

$$N_q = N_g = 32, \quad N_{K=0,1,2,3} = 2, 6, 10, 14. \quad (1.2)$$

These counts are derived, not taken as premises. In particular, the physical two-dimensional gluon basis is compared carefully with nonminimal published labels. A pure-trace term in an unprojected formula must be absorbed into a trace coefficient; it is not an independent linearly polarized distribution. The note retains that distinction rather than describing the published correlator as a different physical theory.

In the chapters below, a short *Provenance and task* paragraph identifies whether the material is established background, a definition/convention, a construction from this work, or an illustrative assumption. New pedagogical derivations do not automatically imply new physics results. The cited literature establishes the comparisons actually used here; the note does not claim to exhaust every publication on higher-spin observables.

1.3 Domain, conventions, and the companion manuscript

We assume parity-conserving QCD, electromagnetic one-photon exchange, leading power in the hard scale, and target-correlator gauge-link/rapidity prescriptions with no extra preferred transverse vector. There are two physical transverse dimensions. Counts are per quark flavor and per specified gluon link/color class; antiquarks have separate distributions. Extra color or link labels do not add new Lorentz tensors. A complete target-side operator description is the common starting point for SIDIS and low-transverse-momentum Drell–Yan: the partner is a

fragmentation correlator in the first process and a second distribution correlator in the second. Universality relates specified target operators; it does not equate the measured cross sections. The following sections follow that distinction through all quark octupole channels and through the separately specified gluon-sensitive heavy-pair reaction.

For quark SIDIS the observed hadron is unpolarized or spinless and belongs to the current-fragmentation region with $P_{hT}/z \ll Q$. The displayed hard response is at electromagnetic Born order. For gluons we specify $\gamma^* g \rightarrow Q\bar{Q}$ and distinguish its finite hard amplitude from a factorization statement for a measured small-imbalance final state. Factorization assumptions are process dependent [27, 28].

This research note retains the conventions and complete correlator/response catalogues of the research construction. The population contrasts, rotated rank-three probes, local-moment hierarchy, and spin-orbital example are also developed in the companion research manuscript; they are reproduced here with the intermediate steps needed for learning. Additional Gaussian convolution and linear-inversion examples below are explicitly labeled instructional assumptions. They are not results extracted from the research calculations. Computational material is cited through the Li7-TMDs repository [26], separately from manuscript sources.

1.4 A practical route through the derivation

Begin with Sections 2–4 for spin and kinematic foundations. Sections 5–10 construct and invert the TMD basis. Sections 12–14 provide operator limits and consistency tests; Sections 15 and 17 derive scattering responses. Sections 18 and 20 connect the formalism to target preparation and nuclear interpretation. Sections 7 and 16 develop cross-process relations; Section 19 constructs the joint gluon projections. Section 21 records the independent validation logic and versioned scope of the computational companion.

A useful first pass is to work through the density matrix, the transverse STF tensors, and the completeness argument, then reproduce one quark and one gluon octupole term. A second pass starts with the operator projections and derives the SIDIS trace and convolution weights. A third connects collinear limits, scale evolution, positivity, and target preparation. Section 21 states how independent implementations and their evidence test those mathematical interfaces.

Throughout, an equation has four attributes worth recording in a notebook: its index space, its normalization, its assumptions, and its output. For example, ρ is a normalized positive 4×4 matrix in target spin space; Φ is an unnormalized operator correlator with Dirac indices; f is one scalar coefficient; and a structure function is a process-dependent convolution. Their shared appearance in one cross section does not make these objects interchangeable.

2 From a scattering measurement to a partonic response

Provenance and task. This chapter supplies standard scattering and quantum-mechanical background, with the TMD application organized for the present note. The established quark SIDIS framework is Ref. [5]; its spin-3/2 target dependence is derived in later chapters. The purpose here is to identify what is measured before introducing a large tensor basis.

2.1 The two resolutions of a TMD measurement

In inclusive DIS, $\ell + A \rightarrow \ell' + X$, the measured lepton fixes a hard virtuality $Q^2 = -q^2$ and a longitudinal scaling variable $x_{Bj} = Q^2/(2P \cdot q)$. The sum over all unobserved hadrons removes most information about the transverse recoil of the struck parton. In single-hadron SIDIS, a hadron momentum P_h is also recorded. The transverse momentum of that hadron supplies a

second, much smaller resolution scale. At fixed x_{Bj}, z , the TMD regime keeps P_{hT}/z of hadronic size as Q becomes large.

Two expansions must therefore be kept separate. The expansion in $P_{hT}/(zQ)$ is a power expansion of the scattering process. The dependence on P_{hT} itself is retained; it is the information the measurement is intended to resolve. An azimuthal harmonic such as $\cos 3\phi$ is not small merely because it is a third harmonic. Its power counting is set by the operator and hard coupling that generate it, not by the integer in the cosine.

At a schematic level a low-transverse-momentum SIDIS response has the form

$$F_\nu(x_{\text{Bj}}, z, P_{hT}) = \sum_a H_a(Q) \otimes f_a(x, \mathbf{k}_T) \otimes D_a(z, \mathbf{p}_T) \otimes S_a + \text{power corrections.} \quad (2.1)$$

This is a guide to ingredients, not an additional factorization theorem. Depending on the subtraction convention, a vacuum soft factor is divided between the TMD definitions and no separate S_a appears. The hard coefficient is perturbative; the target and fragmentation matrix elements are generally nonperturbative. Beyond Born order the partonic fraction x is integrated against a hard/matching coefficient and is not simply identical to x_{Bj} in every channel.

2.2 Spin dependence as an amplitude bilinear

Let $\mathcal{A}_{X\Lambda}$ be a scattering amplitude with target spin projection Λ and unobserved final configuration X , at specified measured kinematics. Define a spin-space response operator by

$$\widehat{W}_{\Lambda'\Lambda} = \sum_X \mathcal{A}_{X\Lambda'}^* \mathcal{A}_{X\Lambda} d\Pi_X. \quad (2.2)$$

The phase-space measure and measurement constraints are understood in the sum. For a target ensemble with density matrix ρ the response is

$$\mathcal{R}[\rho] = \text{Tr}(\rho \widehat{W}) = \sum_{\Lambda\Lambda'} \rho_{\Lambda\Lambda'} \widehat{W}_{\Lambda'\Lambda}. \quad (2.3)$$

This equation is the reason every target dependence in the correlators must be linear in the independent moments of ρ . A mixture $\rho = w\rho_1 + (1-w)\rho_2$ gives $\mathcal{R} = w\mathcal{R}_1 + (1-w)\mathcal{R}_2$. Introducing an independent term proportional to a product of moments, such as $S_L S_{LL}$, would usually violate this property. Apparent nonlinearities can occur when one restricts preparation to a nonlinear one-parameter family; the underlying response remains linear on density-matrix space.

A diagonal ρ averages probabilities for different incoming spin states. An off-diagonal element multiplies an interference of target helicity amplitudes. Consequently a target-spin coherence can be measured even when it does not change the population of any helicity state. The same logic applied to parton helicities explains quark transversity and gluon linear polarization.

2.3 Operators, coefficient functions, and structure functions

A correlator contains all leading-power target information accessible to its chosen parton fields. Its decomposition has the general form

$$\Phi = \sum_\alpha \mathcal{B}_\alpha(P, k; \rho) F_\alpha(x, k_T^2), \quad (2.4)$$

where the matrices/tensors $\mathcal{B}_\alpha$ are known and the scalar functions F_α are unknown matrix elements. The scalar function is not inferred by looking at the coefficient of one Cartesian

component unless all other contributions to that component have first been removed. The inverse Gram matrix or the harmonic projectors developed below perform that removal.

A structure function, by contrast, is a coefficient of an independent angular/spin response in a measured hadronic tensor or differential cross section. In TMD factorization it is typically a convolution of a target TMD with another nonperturbative factor. Lowercase f, g, h will denote TMDs, calligraphic $\mathcal{F}, \mathcal{G}, \mathcal{T}, \mathcal{H}$ projected correlators, and uppercase F_ν observable structure functions. The superscript g in a label such as F_{30}^L denotes a helicity-type *quark SIDIS response* in the table below; it is not a gluon-species label. A species label is explicit on the lowercase function, for example g_{30}^q or g_{30}^g .

2.4 Three different spin or angular-momentum labels

The target spin is fixed at $J = 3/2$. An irreducible operator acting in its spin space has rank $K = 0, 1, 2, 3$. A component of that operator can carry transverse helicity magnitude $m = 0, \dots, K$. Independently, the partonic projection carries transverse rank r : zero for a number or helicity scalar, one for transverse quark spin, and two for linear gluon polarization. Finally, n is the orbital harmonic rank built from $\mathbf{k}_T$. The ranks obey a transverse-rotation matching rule, not the identity $K = m = n$.

For example, the component S_{LTT} has target rank $K = 3$ but transverse rank $m = 2$. It can couple to quark transverse spin ($r = 1$) through orbital ranks $n = 1, 3$, and to gluon linear polarization ($r = 2$) through $n = 0, 4$. The absence of k_T in the latter low-rank term does not decide whether it survives the straight-link collinear limit; discrete symmetries supply a separate test. This example will be worked out explicitly rather than used as a mnemonic.

2.5 A spin-one-half warm-up

For spin 1/2 write

$$\rho_{1/2} = \frac{1}{2}(I + \mathbf{P} \cdot \boldsymbol{\sigma}), \quad \langle \mathbf{J} \rangle = \frac{1}{2}\mathbf{P}, \quad |\mathbf{P}| \leq 1. \quad (2.5)$$

Here the polarization fraction $\mathbf{P}$ is twice the expectation value of spin. Confusing these conventions generates factors of two even before introducing spin 3/2. The eigenvalues $(1 \pm |\mathbf{P}|)/2$ show why positivity restricts the allowed moments. Because every 2×2 Hermitian matrix is a linear combination of I and σ_i , there is no independent quadrupole or octupole operator for this target.

At leading twist a quark still has four real projected components: number, helicity, and two transverse-spin components. Rotational and parity constraints, applied to the scalar target and its three vector moments, give two unpolarized-target and six vector-target TMDs. These are the familiar eight functions [5]. In a spin-one target, five quadrupole moments add ten coefficients; in spin 3/2, seven additional octupole moments add fourteen. The target moments and the TMD coefficients are different counts: each target-spin component can couple to more than one parton polarization and momentum harmonic.

2.6 How to organize a calculation before doing algebra

Begin by writing the process and the measured variables. Fix the target axis, lepton plane, momentum fractions, and epsilon convention. Choose whether the problem is an operator classification, a model calculation, or a cross-section extraction. Next construct a positive target density matrix and resolve its irreducible moments. Only then select the parton operator, enumerate allowed tensors, and contract with the hard process. This order prevents a desired angular modulation from being mistaken for an allowed independent operator structure.

For each result, perform a dimensional check and two limiting checks. A TMD coefficient here has mass dimension -2 ; a normalized target moment is dimensionless; and a rank- n covariant carries k_T^n/M_A^n . Setting all moments to zero must recover an unpolarized target. Removing the rank three moments must recover the lower-rank tensor structures after any necessary normalization conversion. Angular integration and reversal of the staple direction provide further, logically independent checks.

3 Spin-3/2 density matrix and polarization moments

Provenance and task. Density matrices and irreducible spin operators are standard quantum mechanics [2]; the spin-3/2 normalization used here agrees with the established fragmentation and forward-distribution conventions [7, 14]. We rederive them because every later TMD normalization depends on this choice. The output of this section is an invertible map between a physical 4×4 density matrix and its fifteen real polarization components. The map is not a model for the partonic dynamics.

3.1 Ensembles, observables, and a useful reference state

Choose a fixed quantization axis and an orthonormal spin basis $|J, m_J\rangle$. A pure state is a normalized vector $|\psi\rangle = \sum_{m_J} c_{m_J} |J, m_J\rangle$, whereas an incoherent preparation is described by

$$\rho = \sum_a w_a |\psi_a\rangle \langle \psi_a|, \quad w_a \geq 0, \quad \sum_a w_a = 1. \quad (3.1)$$

The observable average is $\langle X \rangle = \text{Tr}(\rho X)$. Consequently $\rho = \rho^\dagger$, $\text{Tr} \rho = 1$, and $\rho \succeq 0$. Conversely, the spectral decomposition of any matrix satisfying these three conditions constructs a possible ensemble. Different ensembles can give the same ρ and cannot be distinguished by measurements acting only in this spin space.

For $J = 3/2$, the diagonal entries describe four populations. The six complex nondiagonal entries describe relative coherences. Four real diagonal entries and twelve real coherence components give sixteen parameters, of which one is removed by normalization. The unpolarized reference is $\rho_U = I_4/4$. It is invariant under every spin rotation, not merely symmetric under $m_J \mapsto -m_J$ in one chosen axis. For example, equal populations in the two outer levels and zero populations in the inner levels have zero mean spin but are quadrupole polarized. Zero vector polarization is therefore not synonymous with an unpolarized target.

A practical construction starts from ρ_U , adds Hermitian traceless operators, and then checks the resulting eigenvalues. This separates two questions that are often conflated: whether a proposed polarization has the correct transformation properties, and whether its magnitude is compatible with a physical state. Tensor symmetry answers the first; positivity answers the second.

3.2 Ladder operators and the organization of operator space

The ladder identity

$$J_+ |J, m_J\rangle = \sqrt{(J - m_J)(J + m_J + 1)} |J, m_J + 1\rangle \quad (3.2)$$

fixes all nonzero entries of J_+ . For $J = 3/2$ their magnitudes are $\sqrt{3}, 2, \sqrt{3}$. The diagonal generator is fixed by $J_z |J, m_J\rangle = m_J |J, m_J\rangle$. Thus there is no independent normalization to choose once $\hbar = 1$ and the basis phases are fixed. The square roots encode the transition strengths that later reappear in coherence contrasts and in the ideal NMR response.

Under a rotation $D^J(\mathcal{R})$ an operator transforms as $X \mapsto D^J X D^{J\dagger}$. The operator space is therefore the representation $J \otimes J^*$, rather than the state representation J itself. Angular-momentum addition gives one copy of each rank $K = 0, 1, \dots, 2J$. For $J = 3/2$, the scalar, vector, quadrupole, and octupole operator spaces have dimensions 1, 3, 5, 7. This is why a rank-three moment is possible even though no state has a spin projection larger than $3/2$. A rank labels how an *operator* transforms; it is not an additional allowed value of m_J .

3.3 Constructing the irreducible spin operators

Order the basis as $(3/2, 1/2, -1/2, -3/2)$ and set $\hbar = 1$.

$$J_z = \text{diag}(3/2, 1/2, -1/2, -3/2), \quad J_+ = \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad J_- = J_+^\dagger. \quad (3.3)$$

Then $J_x = (J_+ + J_-)/2$, $J_y = (J_+ - J_-)/(2i)$, $[J_a, J_b] = i\epsilon_{abc}J_c$, and $J^2 = 15/4$. A Hermitian 4×4 matrix has 16 real components; normalization $\text{Tr } \rho = 1$ leaves *15 independent polarization parameters*. Rotations organize the operator space as $0 \oplus 1 \oplus 2 \oplus 3$, with dimensions $1+3+5+7 = 16$.

The rank-two and rank-three operators are

$$Q_{ab} = \frac{1}{2}\{J_a, J_b\} - \frac{5}{4}\delta_{ab}I, \quad (3.4)$$

$$O_{abc} = \text{Sym}(J_a J_b J_c) - \frac{41}{60}(\delta_{ab}J_c + \delta_{ac}J_b + \delta_{bc}J_a). \quad (3.5)$$

Sym is the average over all six ordered products, including repeated indices. To derive the octupole subtraction rather than guess it, use $\sum_a J_a J_c J_a = [J(J+1) - 1]J_c$. Contracting the symmetric cubic product gives $[J(J+1) - 1/3]J_c = 41J_c/12$. Contracting the three delta terms gives $5J_c$, hence the coefficient $41/60$. These operators agree with the spin-density convention used in Refs. [7, 14].

Define moments by actual expectation values,

$$S_a = \text{Tr}(\rho J_a), \quad T_{ab} = \text{Tr}(\rho Q_{ab}), \quad R_{abc} = \text{Tr}(\rho O_{abc}). \quad (3.6)$$

Their inverse map is

$$\rho = \frac{1}{4} \left[I + \frac{4}{5}S_a J_a + \frac{2}{3}T_{ab} Q_{ab} + \frac{8}{9}R_{abc} O_{abc} \right]. \quad (3.7)$$

Repeated Cartesian indices in this equation run over all three axes; off-diagonal tensor entries are consequently counted with their multiplicities. One verifies the inverse by tracing with I, J, Q, O . Different irreducible ranks are orthogonal under the trace, and their within-rank trace metrics fix the three coefficients. The inverse can be verified on all 16 Hermitian basis matrices, as in the computational checks [26]. Physical target states require $\rho \succeq 0$, not merely real moments.

3.4 Deriving the trace subtractions and inverse normalization

It is useful to reproduce the numerical coefficients in Eq. (3.5) before using them. A symmetric product of two spin operators contains a rank-zero trace and a rank-two remainder. Since $\sum_a J_a^2 = J(J+1)I$, the trace-free combination is

$$Q_{ab} = \frac{1}{2}\{J_a, J_b\} - \frac{J(J+1)}{3}\delta_{ab}I. \quad (3.8)$$

For $J = 3/2$, $J(J+1)/3 = 5/4$. Notice that these traces run over three spatial directions. The subtraction $1/3$ used here is different from the $1/2$ appearing in a two-dimensional transverse STF projector.

For the cubic operator, define $C_{abc} = \text{Sym}(J_a J_b J_c)$. The identity needed for its trace follows directly from the angular-momentum commutators:

$$\sum_a [J_a, [J_a, J_c]] = 2J_c, \quad (3.9)$$

$$\sum_a [J_a, [J_a, J_c]] = 2J(J+1)J_c - 2 \sum_a J_a J_c J_a. \quad (3.10)$$

Thus $\sum_a J_a J_c J_a = [J(J+1) - 1]J_c$, and symmetrization gives

$$\sum_a C_{aac} = \frac{1}{3} \sum_a (J_a^2 J_c + J_a J_c J_a + J_c J_a^2) = \left[J(J+1) - \frac{1}{3} \right] J_c. \quad (3.11)$$

An isotropic trace subtraction must be proportional to $\delta_{ab}J_c + \delta_{ac}J_b + \delta_{bc}J_a$. Its $a = b$ contraction is $5J_c$. Therefore the required coefficient is $[J(J+1) - 1/3]/5 = 41/60$. Replacing the symmetrized cubic operator simply by $J_a J_b J_c$ would spoil Hermiticity for unequal indices; dropping its trace would mix rank one with rank three.

The inverse density matrix can also be derived without a component-by-component guess. Introduce the identity projectors on three-dimensional symmetric-traceless tensors,

$$(P_2)_{ab,cd} = \frac{1}{2}(\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}) - \frac{1}{3}\delta_{ab}\delta_{cd}. \quad (3.12)$$

For a symmetric rank-three tensor X , the other projector acts as

$$(P_3 X)_{abc} = X_{abc} - \frac{1}{5}(\delta_{ab}X_{ddc} + \delta_{ac}X_{ddb} + \delta_{bc}X_{dda}). \quad (3.13)$$

Rotational invariance implies that within each irreducible rank the trace inner product is a constant times its identity projector. Different ranks are orthogonal. Evaluating one diagonal component fixes the constants:

$$\text{Tr}(J_a J_b) = 5\delta_{ab}, \quad (3.14)$$

$$\text{Tr}(Q_{ab} Q_{cd}) = 6(P_2)_{ab,cd}, \quad (3.15)$$

$$\text{Tr}(O_{abc} O_{def}) = \frac{9}{2}(P_3)_{abc,def}. \quad (3.16)$$

For example, $Q_{zz} = \text{diag}(1, -1, -1, 1)$ has squared Hilbert–Schmidt norm 4, while $(P_2)_{zz,zz} = 2/3$. Also $O_{zzz} = \text{diag}(3, -9, 9, -3)/10$ has squared Hilbert–Schmidt norm $9/5$, while $(P_3)_{zzz,zzz} = 2/5$. These determine 6 and $9/2$.

Now write $\rho = I/4 + aS_a J_a + bT_{ab} Q_{ab} + cR_{abc} O_{abc}$. Taking its trace against J_a , Q_{ab} , and O_{abc} , and using the tracelessness of the moments, gives $a = 1/5$, $b = 1/6$, $c = 2/9$. This is precisely Eq. (3.7). Off-diagonal Cartesian components occur more than once in the repeated-index sums. For example, $T_{xz}Q_{xz}$ and $T_{zx}Q_{zx}$ are both present. Omitting such multiplicities changes the inversion coefficients.

The same calculation yields a useful scalar check,

$$\text{Tr} \rho^2 = \frac{1}{4} + \frac{S_a S_a}{5} + \frac{T_{ab} T_{ab}}{6} + \frac{2R_{abc} R_{abc}}{9}. \quad (3.17)$$

It lies between $1/4$ and 1 for a physical state. Equality to 1 characterizes a pure state only after positivity and unit trace have been established. The converse shortcut is invalid: the matrix $\text{diag}(0.6, 0.3, 0.2, -0.1)$ has unit trace and purity 0.50 but a negative eigenvalue. In calculations, reconstruct ρ and inspect its eigenvalues rather than using purity as the sole constraint.

3.5 Exact definition of the longitudinal/transverse components

We use $a, b, c = x, y, z$ for three-dimensional spin indices and $i, j, k = x, y$ for transverse components relative to the chosen z axis. Here STF_2 removes traces with the two-dimensional metric δ_{ij} ; its explicit tensor construction is given in Section 4. Define

$$\begin{aligned} S_L &= S_z, & S_T^i &= S_i, \\ S_{LL} &= T_{zz}, & S_{LT}^i &= 2T_{zi}, & S_{TT}^{ij} &= 2\text{STF}_2 T_{ij}, \\ S_{LLL} &= R_{zzz}, & S_{LLT}^i &= R_{zzi}, & S_{LTT}^{ij} &= 4\text{STF}_2 R_{zij}, & S_{TTT}^{ijk} &= 4\text{STF}_2 R_{ijk}. \end{aligned} \quad (3.18)$$

The factors 2 and 4 are conventions, not additional polarizations. For complete reproducibility the inverse relations are

$$T_{ij} = \frac{1}{2}S_{TT}^{ij} - \frac{1}{2}\delta_{ij}S_{LL}, \quad T_{zi} = \frac{1}{2}S_{LT}^i, \quad (3.19)$$

$$R_{zij} = \frac{1}{4}S_{LTT}^{ij} - \frac{1}{2}\delta_{ij}S_{LLL}, \quad R_{zzi} = S_{LLT}^i, \quad (3.20)$$

$$R_{ijk} = \frac{1}{4}S_{TTT}^{ijk} - \frac{1}{4}(\delta_{ij}S_{LLT}^k + \delta_{ik}S_{LLT}^j + \delta_{jk}S_{LLT}^i). \quad (3.21)$$

For instance, $R_{zxx} + R_{zyy} = -R_{zzz}$, as three-dimensional tracelessness requires. Treating a raw R_{zij} as a two-dimensional STF tensor without this subtraction would mix LLL and LTT .

We abbreviate the components by A_{Km} :

K	m	A_{Km}
0	0	$A_{00} = 1$
1	0,1	$A_{10} = S_L, \quad A_{11} = S_T$
2	0,1,2	$A_{20} = S_{LL}, \quad A_{21} = S_{LT}, \quad A_{22} = S_{TT}$
3	0,1,2,3	$A_{30} = S_{LLL}, \quad A_{31} = S_{LLT}, \quad A_{32} = S_{LTT}, \quad A_{33} = S_{TTT}$

K is the *three-dimensional spin multipole rank*; m is its *transverse helicity magnitude*. They must not be interchanged. For $m > 0$, $A_{Km,+} = s_{Km} \exp(im\phi_{Km})$; for $m = 0$ the component is a signed real number.

3.6 Diagonal populations and an octupole-only example

Let $p_m = \rho_{mm}$. Taking the diagonal entries of J_z, Q_{zz}, O_{zzz} gives

$$S_L = \frac{3}{2}(p_{3/2} - p_{-3/2}) + \frac{1}{2}(p_{1/2} - p_{-1/2}), \quad (3.22)$$

$$S_{LL} = p_{3/2} + p_{-3/2} - p_{1/2} - p_{-1/2}, \quad (3.23)$$

$$S_{LLL} = \frac{3}{10}[p_{3/2} - p_{-3/2} - 3(p_{1/2} - p_{-1/2})]. \quad (3.24)$$

Conversely,

$$\begin{aligned} p_{3/2} &= \frac{1}{4}(1 + \frac{6}{5}S_L + S_{LL} + \frac{2}{3}S_{LLL}), & p_{1/2} &= \frac{1}{4}(1 + \frac{2}{5}S_L - S_{LL} - 2S_{LLL}), \\ p_{-1/2} &= \frac{1}{4}(1 - \frac{2}{5}S_L - S_{LL} + 2S_{LLL}), & p_{-3/2} &= \frac{1}{4}(1 - \frac{6}{5}S_L + S_{LL} - \frac{2}{3}S_{LLL}). \end{aligned} \quad (3.25)$$

Every right side must be nonnegative. A diagonal state with $S_L = S_{LL} = 0$ and $S_{LLL} = 1/2$ has $(p_{3/2}, p_{1/2}, p_{-1/2}, p_{-3/2}) = (1/3, 0, 1/2, 1/6)$. Reversing the populations changes the octupole sign while leaving the other two moments zero. This octupole-only family allows $|S_{LLL}| \leq 1/2$; the unrestricted diagonal range is $|S_{LLL}| \leq 9/10$. A quoted vector polarization percentage is usually $P_z = \langle J_z \rangle / J = 2S_L/3$, not S_L itself, and does not specify S_{LL} or S_{LLL} .

3.7 Coherences, spherical tensors, and a projection recipe

For an orthonormal spin-operator basis define

$$\tau_{KQ} = \sqrt{\frac{2K+1}{2J+1}} \sum_{m,m'} \langle Jm; KQ | Jm' \rangle |Jm'\rangle \langle Jm|, \quad J = 3/2. \quad (3.26)$$

With standard real Clebsch–Gordan phases, $\text{Tr}(\tau_{KQ}^\dagger \tau_{K'Q'}) = \delta_{KK'} \delta_{QQ'}$, $\tau_{KQ}^\dagger = (-1)^Q \tau_{K,-Q}$, and $\tau_{00} = I/2$. Thus

$$\rho = \sum_{KQ} t_{KQ} \tau_{KQ}, \quad t_{KQ} = \text{Tr}(\tau_{KQ}^\dagger \rho), \quad t_{00} = 1/2. \quad (3.27)$$

For example, $\tau_{10} = J_z/\sqrt{5}$, $\tau_{20} = Q_{zz}/2$, and $\tau_{30} = \sqrt{5} O_{zzz}/3$. The selection $m' = m + Q$ identifies *LLT*, *LTT*, and *TTT* with helicity differences 1, 2, and 3, respectively. Several ranks share a given helicity difference: an individual off-diagonal density matrix entry is not a pure octupole. Projection onto τ_{3Q} , or onto the Cartesian *O* operators, separates it from lower ranks.

Equation (3.26) is also an operational recipe: construct the 4×4 matrices, take traces with a measured or modeled ρ , then use Eq. (3.18). No guessed relation between a target polarization label and a population/coherence is needed.

3.8 A complete matrix reconstruction from populations and coherences

The spherical description is compact, but an explicit Cartesian map is especially useful when preparing a numerical spin density matrix. Label the ordered basis states by 1, 2, 3, 4 and define $c_{ij} = \rho_{ji}$ for $i < j$. Thus c_{ij} is the *lower*-triangular entry; its conjugate occupies the corresponding upper-triangular position. This convention follows from $\text{Tr}(\rho J_+)$: an upper-triangular raising operator contracts with a lower-triangular entry of ρ . Confusing this ordering reverses every coherence phase.

Using the component definitions, the six complex polarization components are

$$A_{11,+} = \sqrt{3}(c_{12} + c_{34}) + 2c_{23}, \quad (3.28)$$

$$A_{21,+} = 2\sqrt{3}(c_{12} - c_{34}), \quad (3.29)$$

$$A_{31,+} = \frac{2\sqrt{3}}{5}(c_{12} + c_{34}) - \frac{6}{5}c_{23}, \quad (3.30)$$

$$A_{22,+} = 2\sqrt{3}(c_{13} + c_{24}), \quad (3.31)$$

$$A_{32,+} = 2\sqrt{3}(c_{13} - c_{24}), \quad (3.32)$$

$$A_{33,+} = 6c_{14}. \quad (3.33)$$

For example, the operators whose averages give $A_{21,+}$, $A_{22,+}$, $A_{32,+}$, and $A_{33,+}$ are, respectively, $\{J_z, J_+\}$, J_+^2 , $\{J_z, J_+^2\}$, and J_+^3 . The operator for $A_{31,+}$ is $O_{zzx} + iO_{zzy}$. These statements can be checked using only the ladder matrix in Eq. (3.3). They also show how ranks with the same helicity change remain independent: the three $\Delta m_J = 1$ combinations weight the outer and middle transitions differently.

Solving these linear equations gives

$$c_{12} = \frac{(3/5)A_{11,+} + A_{31,+}}{2\sqrt{3}} + \frac{A_{21,+}}{4\sqrt{3}}, \quad (3.34)$$

$$c_{34} = \frac{(3/5)A_{11,+} + A_{31,+}}{2\sqrt{3}} - \frac{A_{21,+}}{4\sqrt{3}}, \quad (3.35)$$

$$c_{23} = \frac{1}{5}A_{11,+} - \frac{1}{2}A_{31,+}, \quad (3.36)$$

$$c_{13} = \frac{A_{22,+} + A_{32,+}}{4\sqrt{3}}, \quad (3.37)$$

$$c_{24} = \frac{A_{22,+} - A_{32,+}}{4\sqrt{3}}, \quad (3.38)$$

$$c_{14} = \frac{1}{6}A_{33,+}. \quad (3.39)$$

Together with Eq. (3.25), these equations reconstruct every entry of ρ . They supply an independent way to implement Eq. (3.7). A practical test is to start with an arbitrary positive matrix, extract its moments, reconstruct it by both methods, and compare all sixteen complex entries, including the vanishing imaginary diagonal parts.

Worked example: a coherence that populations cannot detect. Compare the pure state

$$|\psi_\chi\rangle = \frac{|3/2\rangle + e^{i\chi}| -3/2\rangle}{\sqrt{2}} \quad (3.40)$$

with the incoherent mixture of the same two helicities. Both have $p_{3/2} = p_{-3/2} = 1/2$ and $p_{1/2} = p_{-1/2} = 0$. Both therefore give $S_L = 0$, $S_{LL} = 1$, and $S_{LLL} = 0$. But the pure state has $c_{14} = e^{i\chi}/2$ and hence $A_{33,+} = 3e^{i\chi}$, whereas the mixture has $A_{33,+} = 0$. A measurement of the four populations alone cannot distinguish them. Their purities are 1 and 1/2, respectively. Also, the pure state is not an “octupole-only” state, because its quadrupole moment is nonzero. Purity in the quantum-state sense and isolation of an irreducible polarization rank are different concepts.

What has been obtained. The target has now been specified independently of any parton model. The longitudinal components measure population contrasts; the transverse components resolve coherence contrasts. The normalization of each is fixed by an explicit operator. These maps are consequences of standard spin-3/2 quantum mechanics, developed here as the input to the project’s target-side TMD construction. They do not imply that a given laboratory preparation can tune all components independently.

4 Kinematics and transverse tensor conventions

Provenance and task. Light-front coordinates, Fourier transforms, and transverse tensor methods are standard ingredients of TMD physics [5, 15, 28]. Here they are fixed in sufficient detail to translate every later tensor into Cartesian components. The output is a convention sheet: a longitudinal axis, positive transverse metric, orientation tensor, momentum normalization, and a basis of angular harmonics. These are conventions and mathematical tools, not new nonperturbative distributions.

4.1 Why light-front coordinates are useful

Define $v^\pm = (v^0 \pm v^3)/\sqrt{2}$ and $\mathbf{v}_T = (v^1, v^2)$. Then

$$v \cdot w = v^+ w^- + v^- w^+ - \mathbf{v}_T \cdot \mathbf{w}_T, \quad v^2 = 2v^+ v^- - \mathbf{v}_T^2. \quad (4.1)$$

A boost of rapidity η along $+z$ changes v^+ to $e^\eta v^+$ and v^- to $e^{-\eta} v^-$, leaving transverse components unchanged. The ratio k^+/P^+ is therefore unchanged by such a boost. In a frame with a fast target, P^+ is the large component and $P^- = M_A^2/(2P^+)$ follows from its mass shell. The active parton inside an interacting correlator is not assumed on shell; its k^- is integrated over in a TMD definition.

The direction called longitudinal must be stated before polarizations are assigned. We use the target's large-momentum direction. In a collinear DIS frame the virtual photon travels approximately in the opposite direction. A target moment described as longitudinal in a laboratory beam frame must be rotated and, where appropriate, boosted to this spin basis. These transformations act on the entire density matrix, not just the vector polarization.

The mathematical plus sign in v^+ is a light-front component. Later, the notation $A_+ = A^{x\cdots x} + iA^{y\cdots x}$ instead encodes a *complex transverse harmonic*. These two uses of a plus sign are distinct. To make this especially transparent, the dimensionless transverse momentum is denoted $\kappa = (k_x + ik_y)/M_A$ rather than k^+ .

4.2 Light-front coordinates and the nuclear momentum fraction

Use $g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$, $\epsilon^{0123} = +1$, and

$$n^\mu = \frac{1}{\sqrt{2}}(1, 0, 0, -1), \quad \bar{n}^\mu = \frac{1}{\sqrt{2}}(1, 0, 0, 1), \quad n \cdot \bar{n} = 1. \quad (4.2)$$

Thus $v^+ = v \cdot n$, $v^- = v \cdot \bar{n}$, and

$$P^\mu = P^+ \bar{n}^\mu + \frac{M_A^2}{2P^+} n^\mu, \quad k^\mu = x P^+ \bar{n}^\mu + k^- n^\mu + k_T^\mu. \quad (4.3)$$

The fraction is *nuclear* $x = k^+/P^+$. In the massless-quark Born DIS channel it equals $x_{\text{Bj}} = Q^2/(2P \cdot q)$ at leading power; the more general distinction is made explicitly below. A commonly used per-nucleon variable is approximately $x_N = Ax$; using a reference nucleon mass in the target rest frame instead gives $x_N = (M_A/m_N)x$ exactly by that definition. A table or fit must identify which variable it uses.

4.3 Parton fraction, Bjorken variable, and target-mass counting

It is useful to reserve $x_{\text{Bj}} = Q^2/(2P \cdot q)$ for the measured DIS scaling variable and $x = k^+/P^+$ for the argument of the distribution. In the massless-quark Born SIDIS channel, the on-shell outgoing struck quark sets $x = x_{\text{Bj}}$ at leading power. In a higher-order convolution the parton fraction is integrated over; in heavy-quark-pair production it is $x_g = x_{\text{Bj}}(1 + M_{Q\bar{Q}}^2/Q^2)$ at the corresponding Born order. Thus an equality between x and x_{Bj} is a statement about a specified hard process and approximation, not a definition of every TMD.

For a nucleus, P denotes the momentum of the *entire* nucleus. A variable defined by a reference nucleon mass in the target rest frame is $x_N^{\text{exp}} = (M_A/m_N)x_{\text{Bj}}$, approximately Ax_{Bj} . This variable can exceed one without violating nuclear support $x \leq 1$. A nucleon fraction inside an impulse convolution, $x_N = x/y$ with $y = p_N^+/P_A^+$, is a third quantity and is not identical to the experimentally rescaled Bjorken variable.

The target-mass kinematic expansion parameter is

$$\gamma = \frac{2x_{\text{Bj}}M_A}{Q} \simeq \frac{2x_N^{\text{exp}}m_N}{Q}, \quad (4.4)$$

not M_A/Q by itself. This is important when a nuclear mass is used in definitions. A conventionally large M_A in a transverse tensor also does not imply an observable nuclear suppression: the accompanying coefficient is rescaled inversely to keep the matrix element fixed. The mass-convention transformation is discussed in Section 14.

4.4 Lorentz and Euclidean transverse conventions

The Lorentz projector is $g_T^{\mu\nu} = g^{\mu\nu} - \bar{n}^\mu n^\nu - n^\mu \bar{n}^\nu$. It has $g_T^{ij} = -\delta^{ij}$ for $i, j = x, y$. *Every Latin transverse index in the remainder of this note is Euclidean*: it is contracted with δ_{ij} , not with g_{Tij} . We set

$$\epsilon_{xy} = +1, \quad E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \tilde{v}^i = \epsilon_{ia} v^a, \quad \mathbf{k}_T^2 = k_x^2 + k_y^2 = -k_T \cdot k_T. \quad (4.5)$$

For example, $\tilde{v} = (v_y, -v_x)$ is a clockwise quarter-turn. This explicit convention prevents the common sign error of subtracting $+\mathbf{k}_T^2 g_T^{ij}/2$ while treating $\mathbf{k}_T^2$ as positive.

For a rank-two matrix define

$$\text{STF}_2 X^{ij} = \frac{X^{ij} + X^{ji}}{2} - \frac{\delta^{ij}}{2} X^{aa}. \quad (4.6)$$

The factor 1/2 in the symmetrization is part of the definition. In particular, our STF notation is not the unnormalized brace symmetrization used in some references.

4.5 Momentum tensors of every needed rank

Define dimensionless momentum tensors

$$\mathbf{k}_n^{i_1 \dots i_n}(\mathbf{k}_T) = \frac{1}{M_A^n} \text{STF}_2(k^{i_1} \dots k^{i_n}), \quad \mathbf{k}_0 = 1. \quad (4.7)$$

Here STF acts on all indices when the rank exceeds two. The first examples are

$$\mathbf{k}_1^i = k^i/M_A, \quad (4.8)$$

$$\mathbf{k}_2^{ij} = \frac{k^i k^j - \delta^{ij} \mathbf{k}_T^2/2}{M_A^2}, \quad (4.9)$$

$$\mathbf{k}_3^{ijk} = \frac{k^i k^j k^k - \mathbf{k}_T^2 (\delta^{ij} k^k + \delta^{ik} k^j + \delta^{jk} k^i)/4}{M_A^3}. \quad (4.10)$$

A formula that is simpler and less error-prone than writing every rank-four and rank-five component is the following. For any real STF tensor of rank $n > 0$, write

$$A_+ = A^{x \dots x} + i A^{y x \dots x} = a + ib. \quad (4.11)$$

A component containing j indices equal to y is $(-1)^{j/2}a$ for even j , and $(-1)^{(j-1)/2}b$ for odd j . All components are therefore fixed by two real numbers. In particular,

$$(\mathbf{k}_n)_+ = \frac{1}{2^{n-1}} \left(\frac{k_x + i k_y}{M_A} \right)^n, \quad n \geq 1. \quad (4.12)$$

Equations (4.11)–(4.12) explicitly define $\mathbf{k}_4$ and $\mathbf{k}_5$ as well as the lower ranks. Traces vanish because replacing two x indices by two y indices changes a component's sign. Equivalently, $(k_x + i k_y)^n$ is a harmonic polynomial.

For STF tensors with $m > 0$ define a dual on one index,

$$\tilde{A}^{i_1 \dots i_m} = \text{STF}_2(\epsilon_{i_1 a} A^{a i_2 \dots i_m}), \quad (\tilde{A})_+ = -i A_+. \quad (4.13)$$

For an already-STF tensor in two dimensions the right side is already symmetric and traceless. A scalar has no dual.

4.6 Scalar invariants and their normalization

For a target STF tensor A_m define

$$\mathcal{C}_m[A_m] = \mathbf{k}_m^{i_1 \dots i_m} A_m^{i_1 \dots i_m}, \quad \mathcal{S}_m[A_m] = -\mathbf{k}_m^{i_1 \dots i_m} \tilde{A}_m^{i_1 \dots i_m}. \quad (4.14)$$

For $m = 0$, set $\mathcal{C}_0[A_0] = A_0$ and $\mathcal{S}_0 = 0$. Writing $A_{m,+} = s_m e^{im\phi_m}$ and $\kappa = (k_x + ik_y)/M_A$, direct contraction gives

$$\mathcal{C}_m[A_m] = \text{Re}(A_{m,+}^* \kappa^m) = s_m (|\mathbf{k}_T|/M_A)^m \cos m(\phi_k - \phi_m), \quad (4.15)$$

$$\mathcal{S}_m[A_m] = \text{Im}(A_{m,+}^* \kappa^m) = s_m (|\mathbf{k}_T|/M_A)^m \sin m(\phi_k - \phi_m). \quad (4.16)$$

For example, $\mathcal{S}_1[S_T] = -\epsilon_{ij} k^i S_T^j / M_A$. The norm identity $A_m : A_m = 2^{m-1} |A_{m,+}|^2$ explains the factors of two in higher-rank contractions. The amplitude s_m is not silently identified with the tensor's Euclidean norm.

4.7 Constructing STF tensors instead of memorizing them

A completely symmetric rank- n tensor in two dimensions has $n + 1$ components, labeled by the number of y indices. For $n \geq 2$, its trace is a symmetric rank- $(n - 2)$ tensor with $n - 1$ components. Removing the traces therefore leaves two real components. Rank one also has two; rank zero has only one. This elementary counting is the reason the transverse problem is controlled by two helicities, not by all the magnetic substates of a three-dimensional angular momentum.

For rank two, begin with $k^i k^j - c \delta^{ij} \mathbf{k}_T^2$. Contracting the indices gives $(1 - 2c) \mathbf{k}_T^2$, so $c = 1/2$. For rank three, use the symmetric ansatz

$$k^i k^j k^k - c \mathbf{k}_T^2 (\delta^{ij} k^k + \delta^{ik} k^j + \delta^{jk} k^i). \quad (4.17)$$

Its trace over i, j is $(1 - 4c) \mathbf{k}_T^2 k^k$, hence $c = 1/4$. The factors are fixed by the *two-dimensional* trace, not by the three-dimensional spin-tensor subtractions in Eq. (3.5). Division by M_A^n makes $\mathbf{k}_n$ dimensionless.

For higher ranks, a useful generating object is the harmonic polynomial $(k_x + ik_y)^n$. Its real and imaginary parts satisfy the two-dimensional Laplace equation, which is precisely the trace-free condition on the associated symmetric tensor. One can either subtract all traces from $k^{i_1} \dots k^{i_n}$ or reconstruct components from Eq. (4.12). For example,

$$\mathbf{k}_2 = \frac{1}{2M_A^2} \begin{pmatrix} k_x^2 - k_y^2 & 2k_x k_y \\ 2k_x k_y & k_y^2 - k_x^2 \end{pmatrix}. \quad (4.18)$$

At $\mathbf{k}_T = (k, 0)$ the all- x entries of $\mathbf{k}_1, \dots, \mathbf{k}_5$ are $t, t^2/2, t^3/4, t^4/8, t^5/16$, with $t = k/M_A$. This provides a simple normalization test, but testing only this axis does not by itself check the tensor's transformation under rotations.

For a target STF tensor, the same construction proves

$$A_m : B_m = 2^{m-1} \text{Re}(A_{m,+}^* B_{m,+}), \quad m > 0. \quad (4.19)$$

There are $\binom{m}{j}$ components with j indices equal to y . The sum of the even and odd binomial coefficients is 2^{m-1} in each case, which gives the stated factor. Substituting $(\mathbf{k}_m)_+ = \kappa^m / 2^{m-1}$ cancels this factor and yields Eq. (4.16). The cosine and sine invariants consequently have no hidden multiplicity.

A dual is not an additional polarization. For a vector, $E(v_x, v_y) = (v_y, -v_x)$, so $(Ev)_+ = -iv_+$. Applying E to one index of an STF tensor has the same effect on its complex component. It changes its reflection character but preserves its rotation rank and number of components. One index is dualized, not every index. Dualizing every index would introduce a different rank-dependent phase. A scalar has no available index and hence no nonzero dual operation of this kind.

An orientation $A_{m,+} = s_m e^{im\phi_m}$ is defined modulo $2\pi/m$ when $s_m \geq 0$. For a quadrupole this is an axis rather than an oriented arrow; for a rank-three transverse tensor it is an orientation modulo $2\pi/3$. Replacing the tensor amplitude by its Euclidean norm multiplies it by $2^{(m-1)/2}$. Such a change is allowed, but the coefficient functions and observable amplitudes must then be converted consistently.

4.8 Covariant embedding and changes of quantization axis

Introduce a spacelike tetrad orthogonal to P ,

$$e_L^\mu = \frac{P^+}{M_A} \bar{n}^\mu - \frac{M_A}{2P^+} n^\mu, \quad e_x^\mu = (0, 1, 0, 0), \quad e_y^\mu = (0, 0, 1, 0). \quad (4.20)$$

It satisfies $P \cdot e_A = 0$ and $e_A \cdot e_B = -\delta_{AB}$. Set $S^\mu = S_A e_A^\mu$, $T^{\mu\nu} = T_{AB} e_A^\mu e_B^\nu$, and $R^{\mu\nu\alpha} = R_{ABC} e_A^\mu e_B^\nu e_C^\alpha$. This constructs covariant, P -orthogonal polarization tensors from the rest-frame density matrix without inventing extra components. A boost along the chosen z axis introduces no additional spin rotation. For a different preparation axis first rotate $\rho \mapsto D^{3/2}(\mathcal{R})\rho D^{3/2}(\mathcal{R})^\dagger$; for a general change of frame the associated spin-basis/Wigner rotation must also be included. Beam-axis polarization is not automatically photon-axis polarization.

5 Operator correlators and the meaning of a TMD coefficient

Provenance and task. Gauge-invariant bilocal correlators, leading-twist Dirac projections, and TMD subtraction are established QCD constructions [5, 15, 16, 28]. The target's spin changes the matrix element and its allowed polarization dependence, not the fundamental quark or gluon operator. The task is to specify those operators, explain why they have four real leading-twist parton polarization components, and distinguish their coefficients from an experimental fit.

5.1 From a bilocal field product to a transverse-momentum distribution

A number-density operator in elementary quantum mechanics is quadratic in annihilation and creation operators. In a relativistic interacting field theory the corresponding information is contained in a forward matrix element of two fields separated in spacetime. Fourier transformation resolves the momentum flowing between them. Schematically, a fully unintegrated quark correlator has the form

$$\Phi^{(4)}(k) = \int \frac{d^4\xi}{(2\pi)^4} e^{ik \cdot \xi} \langle \bar{\psi}(0) U[0, \xi] \psi(\xi) \rangle_\rho. \quad (5.1)$$

The TMD retains $k^+ = xP^+$ and $\mathbf{k}_T$ while integrating the unobserved k^- . Since

$$\int dk^- e^{ik^- \xi^+} = 2\pi \delta(\xi^+), \quad (5.2)$$

this produces a correlator on the light-front hypersurface $\xi^+ = 0$, with the measure in Eq. (5.6). It does not set $k^- = 0$ or put the active parton on shell. Those are different operations. The analogous reasoning gives the gluon field-strength correlator.

The fields retain their Dirac or transverse polarization indices; the target spin is represented by a different set of indices. Keeping these spaces distinct is essential. A trace over quark Dirac indices selects a *parton* polarization channel. A trace with τ_{KQ} selects a *target* multipole. Neither trace can replace the other. A radial scalar function is defined only after both projections and the transverse harmonic decomposition have been performed.

5.2 Mixed-state expectation values and gauge links

For any operator $\mathcal{O}$ use

$$\langle \mathcal{O} \rangle_\rho = \sum_{\lambda\lambda'} \rho_{\lambda\lambda'} \langle P, \lambda' | \mathcal{O} | P, \lambda \rangle, \quad \langle P', \lambda' | P, \lambda \rangle = 2P^0 (2\pi)^3 \delta^{(3)}(\mathbf{P}' - \mathbf{P}) \delta_{\lambda'\lambda}. \quad (5.3)$$

The correlator is linear in ρ , hence linear in each of S, T, R . Products such as $S_L S_{TT}^{ij}$ are not new independent target structures: allowing them as independent coefficients would violate this linearity.

A Wilson line along a path $\mathcal{P}$ is

$$U_{\mathcal{P}}[a, b] = \mathcal{P} \exp \left[-ig_s \int_a^b dz^\mu A_\mu^c(z) t^c \right], \quad (5.4)$$

with the sign tied to the chosen covariant-derivative convention. A SIDIS distribution uses a future-pointing staple, a Drell–Yan distribution a past-pointing staple. Transverse connectors at large light-cone separation are included. Omitting a Wilson line in a gauge choice is not a proof of its process independence.

5.3 Why the Wilson line is part of the definition

Under a local color transformation $V(x)$, the fields transform as $\psi(x) \mapsto V(x)\psi(x)$ and $\bar{\psi}(x) \mapsto \bar{\psi}(x)V^\dagger(x)$. A product at separated points is not invariant. A transporter with

$$U[0, \xi] \mapsto V(0)U[0, \xi]V^\dagger(\xi) \quad (5.5)$$

makes the bilinear invariant by canceling the two endpoint rotations. Path ordering is necessary because color matrices at distinct positions generally do not commute. The sign in the exponential must be paired with the chosen covariant derivative; changing both consistently leaves the physical operator unchanged.

The path is determined by soft and collinear interactions in the specified scattering process. In the eikonal approximation, successive gluon attachments to a fast outgoing or incoming parton generate ordered products of gauge fields along its direction. Their resummation produces a future- or past-pointing Wilson staple. Thus the line is not an optional embellishment used only to make a formula look gauge invariant: it records interactions that can produce observable correlations. The reversal of its pole prescription is the origin of the familiar SIDIS–Drell–Yan sign relation for link-odd quark distributions [29]. The generalized signs for each target rank are worked out in Section 6.

A fragmentation correlator instead contains a sum over specified hadron out-states. Time reversal changes the status of those states, so it does not impose the same constraint as on a forward distribution matrix element. This is why the spin-3/2 fragmentation classification can check spin and tensor counting without identifying its functions with spin-3/2 PDFs. Universality and factorization of fragmentation and soft factors require their own operator arguments [30].

For a gluon correlator the two paths and any Wilson-loop/color class must be specified together. Distinct such classes can have the same Lorentz decomposition but different functions. Conversely,

a redundant Lorentz tensor does not become independent because a process has a complicated color flow. In a basis of link combinations, time reversal can be diagonalized into even and odd combinations. A link class not closed under the stated Hermitian-conjugation operation must be paired with its conjugate before imposing a real-coefficient parametrization. The real Hermitian correlators used below refer to these physical Hermitian classes or combinations.

5.4 Unsubtracted quark and gluon correlators

For a quark flavor q , with Dirac indices a, b , define

$$\Phi_{ab, \text{unsub}}^{q[U]}(x, \mathbf{k}_T; \rho) = \int \frac{d\xi^- d^2\xi_T}{(2\pi)^3} e^{ixP^+ \xi^- - i\mathbf{k}_T \cdot \xi_T} \langle \bar{\psi}_b^q(0) U[0, \xi] \psi_a^q(\xi) \rangle_\rho \Big|_{\xi^+=0}. \quad (5.6)$$

Color indices are contracted through U . For gluons use

$$\Gamma_{\text{unsub}}^{g[U, U']ij}(x, \mathbf{k}_T; \rho) = \frac{2}{P^+} \int \frac{d\xi^- d^2\xi_T}{(2\pi)^3} e^{ixP^+ \xi^- - i\mathbf{k}_T \cdot \xi_T} \left\langle \text{tr}_c [F^{+i}(0) U[0, \xi] F^{+j}(\xi) U'[\xi, 0]] \right\rangle_\rho \Big|_{\xi^+=0}. \quad (5.7)$$

We use $\text{tr}_c(t^a t^b) = \delta^{ab}/2$. The factor 2 therefore converts the trace to the usual color-contracted field-strength normalization when the links collapse to a single adjoint transporter. The factor $1/P^+$ is essential: with the state normalization above, both correlators have mass dimension -2 . The link pair and any accompanying Wilson-loop/color class must be specified [16].

The fields F^{+i} are physical transverse leading-power gluon operators. The quark analogue is the good field $\psi_+ = \frac{1}{2} \gamma^- \gamma^+ \psi$. Keeping these components, rather than a growing number of transverse powers by themselves, is what defines the leading-twist operator sector. High transverse harmonic rank does not automatically imply a suppression by $1/Q$.

5.5 Dimensions, good fields, and the leading-twist polarization space

The normalization of momentum eigenstates is part of dimensional bookkeeping. With Eq. (5.3), a one-particle ket has mass dimension -1 . A quark bilinear of dimension 3 therefore has a forward matrix element of dimension 1; the three-coordinate TMD integration has dimension -3 . The result has dimension -2 . A product of two field strengths has dimension 4, and its matrix element dimension 2. The coordinate measure and the factor $1/P^+$ in Eq. (5.7) again give dimension -2 . The factor of two in that equation compensates the normalization $\text{tr}_c(t^a t^b) = \delta^{ab}/2$. These factors should be checked before comparing a gluon convention that factors out x , xP^+ , or another normalization.

The large-momentum quark field is selected by

$$\Lambda_+ = \frac{1}{2} \gamma^- \gamma^+, \quad \Lambda_- = \frac{1}{2} \gamma^+ \gamma^-, \quad \Lambda_+ + \Lambda_- = I, \quad \Lambda_\pm^2 = \Lambda_\pm, \quad \Lambda_+ \Lambda_- = 0. \quad (5.8)$$

The good field $\psi_+ = \Lambda_+ \psi$ has two independent spin components. Their Hermitian bilinear is a 2×2 matrix: one number component and three polarization components. In Dirac notation these become the number projection, the helicity projection, and two transverse-spin projections. This explains why the leading-twist reconstruction contains four real quantities rather than all sixteen Dirac bilinears. The bad field and further power-suppressed operator structures enter a higher-twist analysis.

Likewise, F^{+x} and F^{+y} give two physical leading-power gluon polarization components. Their Hermitian matrix has a real trace, a purely imaginary antisymmetric part, and two real symmetric-traceless parts. In the circular basis, the last two parts connect gluon helicities $+1$ and -1 . A quark helicity interference changes helicity by one unit, whereas this gluon interference changes

it by two. This difference will change the orbital ranks but not the dimension of the parton polarization matrix.

A high power of k_T/M_A in an angular tensor does not change the good- field content. Leading power refers to an expansion in the hard scale with k_T held at the TMD scale. Perturbative large- k_T behavior and matching onto collinear twist-three or higher operators are separate questions. One cannot infer a hard-scale suppression by reading an orbital-rank superscript alone.

5.6 Rapidity subtraction and scales

Equations (5.6)–(5.7) are not yet finite renormalized TMD definitions with exactly lightlike staples. A rapidity regulator, a soft subtraction, and UV renormalization are needed. In impact-parameter space write the chosen definition as

$$\tilde{\Phi}_{\text{ren}}^q(x, \mathbf{b}_T; \mu, \zeta) = \lim_{\text{reg} \rightarrow 0} Z_q(\mu, \zeta; \text{reg}) R_q(\mathbf{b}_T; \mu, \zeta; \text{reg}) \tilde{\Phi}_{\text{unsub}}^q(x, \mathbf{b}_T; \text{reg}), \quad (5.9)$$

and similarly for gluons. Here $\tilde{\Phi} = \int d^2\mathbf{k}_T e^{i\mathbf{k}_T \cdot \mathbf{b}_T} \Phi$. R_a is the rapidity counterterm, including the appropriate vacuum Wilson-line soft-factor combination; Z_a is the UV counterterm. In symmetric two-line soft-sharing schemes R_a is often represented schematically by $S_a^{-1/2}$, but that shorthand is not a definition valid for every process/color class. One may, for example, choose a specified soft-subtracted TMD scheme with $\overline{\text{MS}}$ UV renormalization and an explicit rapidity scale ζ . All coefficients below mean that same chosen scheme. The general requirements and scheme dependence are reviewed in Ref. [28].

No tensor coefficient will depend only on x and $\mathbf{k}_T^2$ in a complete phenomenological calculation: its arguments are $(x, \mathbf{k}_T^2; \mu, \zeta; [U])$ or $(x, \mathbf{k}_T^2; \mu, \zeta; [U, U'])$. We suppress these arguments only to keep the tensor algebra readable. An isotropic scalar subtraction does not change the physical tensor count. Evanescent tensors in $2 - 2\epsilon$ transverse dimensions may require a renormalization prescription; they are not additional observable two-dimensional TMDs.

5.7 Ultraviolet and rapidity divergences are different limits

A soft or collinear gluon with fixed transverse momentum can approach arbitrarily large rapidity. For an on-shell massless gluon, $2\ell^+\ell^- = \ell_T^2$. Increasing ℓ^+ while decreasing ℓ^- preserves this invariant but produces a logarithmic integration of the type $\int d\ell^+/\ell^+$. Regulating the invariant mass by ordinary dimensional regularization does not by itself regulate this rapidity range. This is conceptually different from an ultraviolet limit in which all momentum components become large.

A finite TMD definition therefore specifies both a UV prescription and a rapidity prescription, together with a subtraction of the overlap between collinear and soft radiation. The scale μ tracks ultraviolet renormalization; ζ tracks the separation in rapidity. A soft subtraction can be represented by different, perturbatively related schemes. Equation (5.9) deliberately identifies the required factors without pretending to define every regulator by one universal square root. For ordinary two-direction TMD factorization, these issues are standard and do not depend on whether the target has spin 1/2, 1, or 3/2 [28].

For the tensor classification, the relevant assumption is that the chosen regulator and subtraction introduce no additional transverse vector into the target matrix element. Then they do not change the rotation or parity labels of its physical components. The classification can consequently be performed once, while a numerical prediction must choose a definite scheme and matching prescription. Finite subtracted TMDs need not obey every pointwise probability inequality of an unsubtracted cutoff correlator; Section 13 keeps these claims separate.

5.8 Projection operators and reconstruction

Use $\sigma^{\mu\nu} = i[\gamma^\mu, \gamma^\nu]/2$ and $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. Define

$$\mathcal{F}^q = \frac{1}{2} \text{Tr}_D(\Phi^q \gamma^+), \quad \mathcal{G}^q = \frac{1}{2} \text{Tr}_D(\Phi^q \gamma^+ \gamma_5), \quad \mathcal{T}^{q,i} = \frac{1}{2} \text{Tr}_D(\Phi^q i\sigma^{i+} \gamma_5). \quad (5.10)$$

$\mathcal{F}$ is the quark-number projection, $\mathcal{G}$ the helicity projection, and $\mathcal{T}$ the transverse-spin projection. In our Euclidean transverse component convention the leading-twist reconstruction is

$$\Phi_{(2)}^q = \frac{1}{2} [\mathcal{F}^q \gamma^- + \mathcal{G}^q \gamma_5 \gamma^- + \mathcal{T}^{q,i} i\sigma^{i-} \gamma_5]. \quad (5.11)$$

Tracing this expression with the three projectors returns their defining coefficients; the transverse projector has two real components. This explicit inverse is useful when deriving a hard trace rather than relying on remembered azimuthal signs.

For gluons set

$$\Gamma^{g,ij} = \frac{x}{2} [\delta^{ij} \mathcal{F}^g + i\epsilon_{ij} \mathcal{G}^g + \mathcal{H}^{g,ij}], \quad \mathcal{H}^{g,ij} = \mathcal{H}^{g,ji}, \quad \mathcal{H}^{g,ii} = 0. \quad (5.12)$$

Consequently,

$$\mathcal{F}^g = \frac{\delta_{ij} \Gamma^{g,ij}}{x}, \quad \mathcal{G}^g = -\frac{i\epsilon_{ij} \Gamma^{g,ij}}{x}, \quad \mathcal{H}^{g,ij} = \frac{2}{x} \text{STF}_2 \Gamma^{g,ij}. \quad (5.13)$$

They describe unpolarized, circularly polarized, and linearly polarized gluons. Hermiticity makes these four real components real-valued; the explicit i resides in the antisymmetric helicity tensor. In a helicity basis the linear-polarization part connects helicities $+1$ and -1 , hence carries two units of helicity flip. It is not a spin-1/2-like transverse spin vector.

5.9 Checking a reconstruction by its projection Gram matrix

The normalization of a decomposition should be verified by inversion. Let P_A be the four projection matrices γ^+ , $\gamma^+ \gamma_5$, $i\sigma^{x+} \gamma_5$, and $i\sigma^{y+} \gamma_5$. Let R_B be the corresponding reconstruction matrices γ^- , $\gamma_5 \gamma^-$, $i\sigma^{x-} \gamma_5$, and $i\sigma^{y-} \gamma_5$. Their trace Gram matrix is

$$G_{AB} = \frac{1}{4} \text{Tr}_D(P_A R_B) = \delta_{AB}. \quad (5.14)$$

Since $\Phi_{(2)} = \frac{1}{2} \sum_B c_B R_B$, the projection $\frac{1}{2} \text{Tr}_D(P_A \Phi_{(2)})$ returns c_A exactly. For example, $\text{Tr}(\gamma^+ \gamma^-) = 4$ follows from $g^{+-} = 1$; a cross projection containing a single γ_5 and fewer than four independent gamma matrices vanishes. The transverse entries can be checked using the anticommutators and the stated Euclidean component convention. A sign change in a tensor projector must be accompanied by the inverse sign change in its reconstruction matrix.

More generally, for a trial reconstruction basis R_B with invertible G_{AB} , the actual dual projectors contain G^{-1} . This elementary linear-algebra observation is useful when translating published conventions: the matrix of projections, not a remembered function name, fixes normalization. The gluon inverse works in the same way, using $\delta_{ij} \delta_{ij} = 2$, $\epsilon_{ij} \epsilon_{ij} = 2$, and orthogonality of trace, antisymmetric, and symmetric-traceless parts.

5.10 How a coefficient is determined from a correlator

First project the Dirac or gluon indices with Eqs. (5.10) and (5.13). Next resolve the target spin dependence with the matrices τ_{KQ} , or equivalently by substituting Eq. (3.7) into Eq. (5.3). Finally resolve transverse harmonics at fixed $|\mathbf{k}_T|$. For example, a term $A_{30} g_{30}^q$ is obtained from the rank-three, $Q = 0$ component of $\mathcal{G}^q$, not from the entire helicity projection in one longitudinally polarized state. A coefficient multiplying an orbital rank n can be read off after dividing its Fourier harmonic by $(|\mathbf{k}_T|/M_A)^n$ and the stated spin normalization. Section 10 gives an explicit two-branch inverse for transverse-spin and linear-polarization terms. These are operator projection definitions, distinct from fitting convoluted experimental cross sections.

6 Completeness from transverse rotations, parity, and link reversal

Provenance and task. The representation theory of rotations and parity is standard. The project applies it to the target-side spin-3/2 correlator and uses explicit two-dimensional identities to obtain an irredundant gluon basis. The numerical quark count can be compared with the known spin-3/2 fragmentation classification [7]; the lower-spin gluon comparison uses Refs. [16, 17]. The task here is to establish both *spanning* and *independence*, rather than to accept a count merely because a list has the desired length.

6.1 A constructive enumeration procedure

Keep the target rank K , its transverse helicity magnitude m , the parton-channel rank r , and the orbital rank n separate. The target has $K = 0, 1, 2, 3$ and $m = 0, \dots, K$. Choose one target component at a time and one of the projected parton channels. The scalar number and helicity projections have $r = 0$, the quark transverse-spin projection has $r = 1$, and the gluon linear-polarization projection has $r = 2$. Next solve the transverse rotation condition for the possible orbital harmonics. Apply reflection parity to choose an ordinary or dual contraction. Finally remove identities specific to two dimensions and assign one independent radial coefficient to every remaining structure.

This order matters. If a scalar coefficient is allowed to depend on $\mathbf{k}_T \cdot S_T$ or another polarization invariant, it already contains unresolved spin dependence and the counting ceases to be linear in the density matrix. The coefficients in a complete linear decomposition depend on x , $\mathbf{k}_T^2$, scales, and the specified link class. All polarization dependence is in the displayed tensor factors. Additional powers of $\mathbf{k}_T^2$ can be absorbed into those radial functions; they do not generate new angular covariants.

6.2 Transverse-helicity matching in two dimensions

A nonzero-rank STF tensor in two dimensions has only the two helicities $+m$ and $-m$. Under a rotation by θ , its complex component transforms as $A_+ \mapsto e^{im\theta} A_+$. This fact is stronger than the triangle inequality familiar from three-dimensional angular momentum.

Proposition 6.1 (Orbital ranks in the physical transverse plane). *Let a target component have transverse rank m , and let the projected parton operator have nonzero transverse rank r . With scalar coefficients depending on $\mathbf{k}_T^2$, a linear parity-fixed basis needs orbital ranks*

$$\begin{cases} n = r, & m = 0, \\ n = |m - r| \text{ and } n = m + r, & m > 0. \end{cases} \quad (6.1)$$

For a scalar parton projection ($r = 0$), $n = m$; parity selects the allowed scalar or pseudoscalar contraction.

Proof. A product of a target helicity $\pm m$ and an orbital helicity $\pm n$ can have only helicities $\pm m \pm n$. To obtain output $+r$, solve $\pm m \pm n = r$. The nonnegative solutions are the two endpoints in Eq. (6.1). They yield different Fourier orders and are independent at generic nonzero transverse momentum. No intermediate helicity is supplied by a two-dimensional STF tensor. For equal input ranks, the zero-helicity product has a scalar and a pseudoscalar; reflection parity decides which is allowed in a specified channel. Powers of $\mathbf{k}_T^2$ multiplying a covariant can be absorbed into its scalar coefficient and do not add independent structures.

The partonic ranks are $r = 0$ for $\mathcal{F}, \mathcal{G}$, $r = 1$ for $\mathcal{T}^q$, and $r = 2$ for $\mathcal{H}^g$. The complete orbital-rank table is

Target m	$\mathcal{F}, \mathcal{G}$	$\mathcal{T}^q$	$\mathcal{H}^g$
0	0	1	2
1	1	0,2	1,3
2	2	1,3	0,4
3	3	2,4	1,5

There is no $n = 2$ gluon linear-polarization covariant for $m = 2$, and no $n = 3$ one for $m = 3$. In particular, $n = m + r, m + r - 2, \dots, |m - r|$ is *not* a two-dimensional STF rule.

6.3 Worked coupling problems and the role of reflections

For a rank-two target component and a rank-two gluon output, start with $A_+ \mapsto e^{2i\alpha} A_+$ and $\kappa \mapsto e^{i\alpha} \kappa$. A rank-two output must acquire $e^{2i\alpha}$. There are two possibilities:

$$A_+, \quad A_+^* \kappa^4. \quad (6.2)$$

They have orbital ranks zero and four. An apparent intermediate $A_+ \kappa^2$ has helicity four, whereas $A_+^* \kappa^2$ has helicity zero; neither is a rank-two output. Contracting Cartesian indices cannot change this representation-theoretic fact. For $m = 3$ and $r = 2$, the two possibilities are $A_+ \kappa^*$ and $A_+^* \kappa^5$, with orbital ranks one and five. For the same $m = 3$ but a quark rank-one output, they are $A_+ (\kappa^*)^2$ and $A_+^* \kappa^4$. This is the distinction between quark and gluon helicity flip in its most direct form.

Two-dimensional rotations alone do not distinguish a scalar from a pseudoscalar. For $r = 0$, both $\text{Re}(A_+^* \kappa^m)$ and $\text{Im}(A_+^* \kappa^m)$ are invariant under rotations. A reflection fixes which belongs in the number projection and which in the helicity projection. Likewise, multiplying a complex nonzero-rank covariant by i changes its reflection property without changing its rotation rank. This is the origin of the required duals. It is not an additional radial function once the parton and target reflection characters have been specified.

For example, S_{LL} is an even-rank target scalar. The quark transverse-spin projection is an axial vector, so the ordinary vector $S_{LL} k^i / M_A$ has the wrong reflection character. Its dual $S_{LL} \epsilon_{ia} k^a / M_A$ has the correct one. By contrast, S_{LLL} is an odd-rank scalar under the momentum-preserving parity operation, and $S_{LLL} k^i / M_A$ already has the required character. For gluon linear polarization the output is an ordinary rank-two tensor, so these dual assignments are reversed. This reasoning derives the parity pattern independently of a notation dictionary.

6.4 Deriving the parity pattern, including the quark tensor duals

Combine spatial parity with a π rotation about y so that the large target momentum still points along $+z$. On ordinary transverse vectors the operation is the reflection $(x, y) \mapsto (x, -y)$. Spin is axial; therefore its rank- K multipole transforms as

$$A_{Km,+} \mapsto (-1)^K A_{Km,+}^*, \quad \kappa \mapsto \kappa^*. \quad (6.3)$$

It follows immediately that $\mathcal{C}_m$ has reflection sign $(-1)^K$, whereas $\mathcal{S}_m$ has sign $(-1)^{K+1}$. The number projection is a scalar, the helicity projection a pseudoscalar, the quark transverse-spin projection an axial vector, and the gluon linear-polarization projection an ordinary STF rank-two tensor. Hence

Projection	even K	odd K
$\mathcal{F}^{q,g}$	$\mathcal{C}_m$	$\mathcal{S}_m$
$\mathcal{G}^{q,g}$	$\mathcal{S}_m$	$\mathcal{C}_m$
$\mathcal{T}^q$	dual of an ordinary vector covariant	ordinary vector covariant
$\mathcal{H}^g$	ordinary rank-two covariant	dual of a rank-two covariant

Since a scalar has no dual, $\mathcal{F}$ contains no odd- K , $m = 0$ term, and $\mathcal{G}$ contains no even- K , $m = 0$ term. This proves the absence of f_{1LLL} without appealing to analogy. It also proves that all rank-two-target quark transverse-spin terms require a dual. Omitting those duals changes parity; it is not just a sign convention.

6.5 Counting and an independent matrix-space check

At rank K , the combined scalar projections have $2K + 1$ coefficients: one for $m = 0$ and two for each $m = 1, \dots, K$. The helicity-flip projection also has $2K + 1$: one at $m = 0$, two at every $m > 0$. Therefore

$$N_K = 4K + 2, \quad N_{J=3/2} = \sum_{K=0}^3 (4K + 2) = 32 \quad (6.4)$$

for *both* quarks and gluons. The orbital ranks differ, not the number of coefficients.

Target sector	$\mathcal{F}$	$\mathcal{G}$	helicity-flip projection	total
$K = 0$ unpolarized	1	0	1	2
$K = 1$ vector	1	2	3	6
$K = 2$ quadrupole	3	2	5	10
$K = 3$ octupole	3	4	7	14
All ranks	8	8	16	32

An independent upper bound follows from the combined parton–target spin matrix. It is 8×8 Hermitian and has 64 real entries before parity. At a fixed nonzero $\mathbf{k}_T$ along x , reflection exchanges opposite helicities and splits this space into two four-dimensional parity eigenspaces (irrelevant phases can be removed). A parity-invariant Hermitian matrix is block diagonal in these eigenspaces and therefore has $4^2 + 4^2 = 32$ real parameters. Our 32 explicit covariants attain that bound. An independent exact evaluation gives rank 32 for each 64×32 coefficient-to-correlator map [26]. Restricting to target ranks at most two gives 18 independent coefficients, not 19.

6.6 Null transverse tensors and redundant coefficient labels

For two STF matrices write

$$A = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}, \quad B = \begin{pmatrix} c & d \\ d & -c \end{pmatrix}. \quad (6.5)$$

Their product is

$$AB = (ac + bd)I + (ad - bc)E. \quad (6.6)$$

Both $\text{STF}_2(AB)$ and $\text{STF}_2(EAB)$ vanish. This excludes $S_{TT}\mathbf{k}_2$ and $\epsilon\mathbf{k}_2 S_{LTT}$ as independent free-index-STF linear-polarization tensors. For two rank-three STF tensors the same calculation gives

$$A_3^{iab} B_3^{jab} = 2[(ac + bd)\delta^{ij} + (ad - bc)\epsilon_{ij}], \quad (6.7)$$

where $a+ib = (A_3)_+$ and $c+id = (B_3)_+$. Thus $\text{STF}_2[\epsilon_{i\alpha} A_3^{\alpha ab} B_3^{jab}] = 0$, excluding an intermediate-rank TTT gluon covariant. These are exact identities, not model-dependent assumptions.

Comparison with published spin-1 notation. Ref. [16] includes the term

$$S_{TT\alpha}^{\{i} k_T^{j\}\alpha} h_{1TT}^{\perp g}, \quad (6.8)$$

where braces mean a sum, *not* free-index STF projection. Here the two-index momentum object is itself the transverse rank-two STF tensor of Ref. [16],

$$k_T^{j\alpha} = k_T^j k_T^\alpha + \frac{k_T^2}{2} g_T^{j\alpha}, \quad g_T^{j\alpha} = -\delta^{j\alpha}. \quad (6.9)$$

The repeated α contracts this tensor with the target tensor; it is not shorthand for an ordinary vector or an unprojected dyadic. In fact, keeping the same lowered target index and unnormalized braces but using the dyadic instead gives

$$\text{STF}_2(S_{TT\alpha}^{\{i} k_T^{j\}\alpha} k_T^\alpha) = -k_T^2 S_{TT}^{ij}, \quad (6.10)$$

which is generally nonzero. This contrast is explicitly reconstructed in Sec. 21.2.

In physical two dimensions its symmetric part is purely trace, so it is absorbed by the unpolarized-gluon TT coefficient. Keeping that paper's polarization normalization fixed and converting its Minkowski index contractions consistently gives

$$f_{1TT}^{g,\min} = f_{1TT}^{g,B} - h_{1TT}^{\perp g,B}. \quad (6.11)$$

The superscript B denotes the coefficients of Ref. [16]; min denotes the independent trace coefficient before any additional target-polarization rescaling. These are comparison labels, not powers or Wilson-line classes. The complete 18-coordinate conversion is given in Sec. 6.8.

Indeed the published helicity matrix and positivity bound involve precisely this difference [17]. Similarly, the unprojected $S_{LT}^{\{i} k_T^{j\}}$ term contains a trace and an STF part; splitting it shifts the scalar LT coefficient and rescales its linear-polarization partner. For Euclidean S, k ,

$$S^i k^j + S^j k^i = 2 \text{STF}_2(S^i k^j) + \delta^{ij} S \cdot k. \quad (6.12)$$

These are changes of coefficient basis, not missing observables. One must not count the redundant labels of an unprojected decomposition as independent physical TMDs, or claim that its normalization matches ours without doing these redefinitions. This distinction also applies when comparing recent model outputs [31].

6.7 What a completeness check actually establishes

The transverse-helicity construction shows that no further angular representation is available. This is the spanning part of the proof. Independence can be checked in two equivalent ways. Different orbital helicities give different Fourier factors before any particular target state is fixed. At a chosen nonzero momentum, one can instead retain all target-spin components and assemble the linear map from the radial coefficients to the correlator components. Both methods distinguish independent target responses even when two terms share an angular frequency for a special target preparation.

A concrete matrix test uses sixteen independent Hermitian target operators and four real parton projections. Evaluate the response to each target basis operator, flatten the resulting sixty-four real numbers, and use them as one column for each proposed TMD covariant. The result is a

64×32 matrix $V(\mathbf{k}_T)$. If one exact 32×32 minor is nonzero at one admissible nonzero momentum, that minor is not the zero polynomial, proving generic algebraic independence. This does not establish spanning by itself; it must be combined with the helicity argument. Nor does it require full rank at $\mathbf{k}_T = 0$, where many allowed covariants vanish kinematically.

The count can be viewed as a consistency check across target spins. For the same two-state parton polarization space and the stated isotropic transverse setup, the rank- K contribution is $4K + 2$. Summing through $K = 2J$ gives

$$N_J = \sum_{K=0}^{2J} (4K + 2) = 2(2J + 1)^2. \quad (6.13)$$

The cases $J = 0, 1/2, 1, 3/2$ give 2, 8, 18, 32. This formula is a consequence of the same elementary representation counting, not a claim that different processes or link classes contain only one common set of scalar functions. It counts the minimal physical tensor space for one specified class. Extra transverse directions, nonforward momentum transfer, or power-suppressed operators change the problem.

An explicit redundancy test. At $\mathbf{k}_T = (k, 0)$, let $S_{TT} = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ and $\mathbf{k}_2 = \frac{1}{2}t^2 \text{diag}(1, -1)$. Their symmetrized product is $at^2 I$, which has only a trace. Subtracting that trace gives zero, although its unprojected trace is not zero when $a \neq 0$. This simple calculation explains the distinction used in the published- notation comparison. A trace coefficient and the coefficient of this unprojected term enter only through a linear combination. One may shift their separate labels while leaving the correlator unchanged. Only that invariant combination is determined by the matrix element. The reduction removes a redundant coefficient label, not a measured physical response and not a gluon color structure.

6.8 The complete lower-rank gluon conversion

The reduction can be stated as an explicit coefficient dictionary rather than only a count. Let an upright B identify the displayed parametrization of Refs. [16, 17]. Initially match their named components $S_L, S_T, S_{LL}, S_{LT}, S_{TT}$ to the identically named components here, use M_A in both momentum tensors, and match the overall operator normalization. Lower contracted Lorentz transverse indices with $g_{Tij} = -\delta_{ij}$ before expressing free indices as Cartesian components. Every multi-index momentum in the reference is an STF tensor, including the two-index tensor in Eq. (6.9).

The trace and circular projections give

$$f_{00}^g = f_1^{g,B}, \quad f_{11}^g = f_{1T}^{\perp g,B}, \quad (6.14)$$

$$f_{20}^g = f_{1LL}^{g,B}, \quad f_{21}^g = f_{1LT}^{g,B} + h_{1LT}^{g,B}, \quad (6.15)$$

$$f_{22}^g = f_{1TT}^{g,B} - h_{1TT}^{\perp g,B}, \quad g_{10}^g = g_1^{g,B}, \quad (6.16)$$

$$g_{11}^g = g_{1T}^{g,B}, \quad g_{21}^g = g_{1LT}^{g,B}, \quad (6.17)$$

$$g_{22}^g = -g_{1TT}^{g,B}. \quad (6.18)$$

The nine linear-polarization coefficients obey

$$h_{00}^{g[2]} = h_1^{\perp g,B}, \quad h_{10}^{g[2]} = -h_{1L}^{\perp g,B}, \quad (6.19)$$

$$h_{11}^{g[1]} = -h_1^{g,B}, \quad h_{11}^{g[3]} = -h_{1T}^{\perp g,B}, \quad (6.20)$$

$$h_{20}^{g[2]} = h_{1LL}^{\perp g,B}, \quad h_{21}^{g[1]} = 2h_{1LT}^{g,B}, \quad (6.21)$$

$$h_{21}^{g[3]} = -h_{1LT}^{\perp g,B}, \quad h_{22}^{g[0]} = h_{1TT}^{g,B}, \quad (6.22)$$

$$h_{22}^{g[4]} = h_{1TT}^{\perp\perp g,B}. \quad (6.23)$$

These equations give eighteen independent coordinates. In particular, the LT tensor contributes both an STF term and a trace, explaining the factor two and the trace shift, respectively. The common transformation $f_{1TT}^{g,B} \mapsto f_{1TT}^{g,B} + \delta$ and $h_{1TT}^{\perp g,B} \mapsto h_{1TT}^{\perp g,B} + \delta$ leaves the complete correlator unchanged. This is the kernel of the reference-to-minimal coefficient map; it is not an additional constraint between two independently measurable functions.

There are two separate normalization choices. An overall change in the operator definition multiplies every coefficient by the same operator factor. Matching the Cartesian polarization tensors, instead of their named components, is different: the reference has $T_{zz}^B = 2S_{LL}^B/3$ and $T_{zi}^B = S_{LT}^{i,B}/2$. Our definitions have $T_{zz} = S_{LL}$ and $T_{zi} = S_{LT}^i/2$. Thus equal Cartesian tensors require $S_{LL}^B = 3S_{LL}/2$; the two LL coefficients acquire that additional factor 3/2. No analogous rescaling is required for LT or TT . A mass-convention conversion is applied separately as in Eq. (14.5). None of these coordinate comparisons identifies distributions in different target species. The independent validation constructs the original reference tensors and the minimal correlator separately, then checks all projected entries, the map's rank, and the displayed null direction.

6.9 Time reversal and the reversal of Wilson paths

Under the antiunitary PT operation, momenta return to their original directions, a rank- K spin moment acquires $(-1)^K$, and the Wilson staple direction reverses. Number and linear-polarization operators are PT even; helicity and quark transverse spin are PT odd. Applying this to the correlator yields the link-reversal signs

$$\begin{aligned} f_{Km}^{a[U^T]} &= (-1)^K f_{Km}^{a[U]}, & g_{Km}^{a[U^T]} &= (-1)^{K+1} g_{Km}^{a[U]}, \\ h_{Km}^{q[n][U^T]} &= (-1)^{K+1} h_{Km}^{q[n][U]}, & h_{Km}^{g[n][U^T]} &= (-1)^K h_{Km}^{g[n][U]}. \end{aligned} \quad (6.24)$$

For gluons U^T denotes reversal of the full link pair/class, not only one of its paths. These signs reproduce the familiar lower-spin classification [5, 16]. There are 20 T-even and 12 T-odd quark coefficients, and 16 T-even and 16 T-odd gluon coefficients in the minimal spin-3/2 basis.

A minus sign in Eq. (6.24) does not imply a function vanishes at a fixed future-pointing link. It implies a sign change when the appropriate reversed-link function is compared. Nor does the mere appearance of ϵ_{ij} , without examining the whole operator and target transformation, determine naive time-reversal parity.

6.10 Reading the link-reversal table without discarding allowed terms

The antiunitary time-reversal relation compares matrix elements with reversed Wilson paths. It is useful to write a link-even or link-odd combination explicitly:

$$F_{\text{even}} = \frac{1}{2}(F^{[U]} + F^{[U^T]}), \quad F_{\text{odd}} = \frac{1}{2}(F^{[U]} - F^{[U^T]}). \quad (6.25)$$

The sign in Eq. (6.24) identifies which combination carries the displayed tensor. For a fixed future-pointing staple a link-odd function need not vanish. It is zero in a limit only when the relevant operator and its reversed-path counterpart become the same operator under the assumptions used in taking that limit.

For the unpolarized target, the quark $h_{00}^{q[1]}$ is link odd, whereas the gluon $h_{00}^{g[2]}$ is link even. Both describe helicity interference, but the quark transverse-spin operator and the gluon linear-polarization operator have different time-reversal characters. For the octupole target, all three

number-type functions are link odd, all four helicity-type functions are link even, the seven quark helicity-flip functions are link even, and the seven gluon helicity-flip functions are link odd. The appearances of ϵ in the displayed tensors alone would not reproduce this classification.

The two possible zero-orbital-rank exceptions are instructive: $h_{21}^{q[0]}$ and $h_{32}^{g[0]}$ pass the angular test for a collinear contribution but are link odd. A straight-link forward collinear operator removes them by time reversal. These are examples of why rotation, parity, and time reversal must be applied as separate steps rather than bundled into an informal counting rule.

7 Operator universality and the SIDIS–Drell–Yan connection

Provenance and task. Wilson-line reversal and the modified universality of quark TMDs are established operator results [29, 30]. The construction here applies those results to every spin-3/2 target multipole in one explicit convention. For gluons, reversal relates a specified pair of Wilson paths to its reversed pair; it does not equate all possible color-flow classes [32]. This section explains what is predicted at the coefficient level and what additional steps are needed before comparing measured SIDIS and Drell–Yan asymmetries.

7.1 Three distinct levels of a cross-process statement

First distinguish the *tensor basis* from the *operator being expanded*. The same Cartesian spin tensors can be used in two processes whose gauge links are different. The ability to write the same angular form in both processes is a kinematic statement. Equality of the scalar coefficients requires equality of the complete operator definitions, or a symmetry relating them. Finally, equality or a sign relation between operator coefficients is not an equality between cross sections: each process contracts those coefficients with its own hard part and its own partner distribution or fragmentation function.

A useful complete coefficient label is

$$\nu = (\text{species, flavor, particle/antiparticle, } A, \\ K, m, n, \text{projection, } [U], \text{color, } \mu, \zeta). \quad (7.1)$$

The symmetry does not exchange a target with a beam, identify a quark with its antiquark, or set two flavors equal. It preserves those labels while reversing the specified Wilson paths. Omitting the labels from a displayed formula is a typographical abbreviation, not an additional physical assumption. The computational companion records these operator labels separately from the tensor coordinates.

7.2 Antiunitary transformation of an arbitrary target state

For the spin-3/2 representation introduce

$$U_T = \exp(-i\pi J_y), \quad \rho^\Theta = U_T \rho^* U_T^\dagger, \quad \Theta = PT. \quad (7.2)$$

The complex conjugation is essential. It acts on the imaginary coherences of the target state, not just on its four populations. Using $U_T J_a^* U_T^\dagger = -J_a$, symmetrized products give

$$U_T Q_{ab}^* U_T^\dagger = Q_{ab}, \quad U_T O_{abc}^* U_T^\dagger = -O_{abc}. \quad (7.3)$$

Consequently the Cartesian target moments of rank K acquire $(-1)^K$. For a half-integer state $U_T U_T^* = -I$, but applying Eq. (7.2) twice returns ρ : the overall state-vector sign cancels between bra and ket. Neither a minus sign on a state vector nor a changed phase convention for a spherical tensor is a failure of the density-matrix involution.

Under the combined PT operation the compared momentum labels are unchanged: parity and time reversal each reverse ordinary spatial momenta. There is therefore no extra factor $(-1)^n$ from the orbital rank in this comparison. The antiunitary operation does conjugate explicit factors of i , notably the gluon circular-polarization tensor. Together with the transformation of the field projection this gives

$$f_{Km}^{a[U^T]} = (-1)^K f_{Km}^{a[U]}, \quad g_{Km}^{a[U^T]} = (-1)^{K+1} g_{Km}^{a[U]}, \quad (7.4)$$

$$h_{Km}^{q[n][U^T]} = (-1)^{K+1} h_{Km}^{q[n][U]}, \quad h_{Km}^{g[n][U^T]} = (-1)^K h_{Km}^{g[n][U]}. \quad (7.5)$$

These reproduce Eq. (6.24); here their physical application and full operator labels are made explicit. The finite spin and projection algebra can be checked with matrices. The QCD field/link transformation, a symmetry-preserving regulator, and compatible soft subtraction are the analytical inputs to that algebra, rather than outputs of a finite matrix calculation.

7.3 All fourteen octupole quark relations

In standard current-fragmentation SIDIS the distribution staple is future pointing, denoted $[+]$. In small- q_T continuum electromagnetic Drell–Yan it is past pointing, denoted $[-]$. At matched $(x, k_T^2; \mu, \zeta)$ and matched subtraction convention,

$$f_{3m}^{q[-]} = -f_{3m}^{q[+]}, \quad m = 1, 2, 3, \quad (7.6)$$

$$g_{3m}^{q[-]} = g_{3m}^{q[+]}, \quad m = 0, 1, 2, 3, \quad (7.7)$$

$$h_{3m}^{q[n][-]} = h_{3m}^{q[n][+]}, \quad (m, n) \in \mathcal{J}_q^{(3)}, \quad (7.8)$$

where

$$\mathcal{J}_q^{(3)} = \{(0, 1), (1, 0), (1, 2), (2, 1), (2, 3), (3, 2), (3, 4)\}. \quad (7.9)$$

The same statement applies separately to antiquarks. Thus three octupole quark coefficients are link odd and eleven are link even. The three number coefficients correlate an unpolarized quark with a rank-three spin–momentum asymmetry. The helicity and transverse-spin octupole coefficients are link even in this comparison. A sine modulation does not by itself imply that its TMD is link odd: a sine can also arise from the hard analyzer and a parity-required dual.

Table 2: Octupole link-reversal assignments. Each entry refers to a specified operator definition, not to an entire measured asymmetry.

Species/channel	Independent coefficients	Reversal sign
Quark number	$f_{31}^q, f_{32}^q, f_{33}^q$	– (3)
Quark helicity	$g_{30}^q, g_{31}^q, g_{32}^q, g_{33}^q$	+ (4)
Quark transverse spin	$h_{30}^{q[1]}, h_{31}^{q[0,2]}, h_{32}^{q[1,3]}, h_{33}^{q[2,4]}$	+ (7)
Gluon number	$f_{31}^g, f_{32}^g, f_{33}^g$	– (3)
Gluon helicity	$g_{30}^g, g_{31}^g, g_{32}^g, g_{33}^g$	+ (4)
Gluon linear polarization	$h_{30}^{g[2]}, h_{31}^{g[1,3]}, h_{32}^{g[0,4]}, h_{33}^{g[1,5]}$	– (7)

In this table a comma-separated bracket is only a space-saving list of separate orbital ranks. It is not an additional multi-index distribution. The ordinary unpolarized-target linearly polarized

gluon coefficient is link even, whereas the octupole linear-polarization coefficients are link odd. The complete target-parton tensor, rather than the phrase “linear polarization,” determines the transformation.

7.4 Gluon link classes and the meaning of conditional universality

A gluon correlator has two paths with field ordering and conjugation fixed in its definition. For the simplest direction descriptors, full reversal maps

$$[+, +] \longleftrightarrow [-, -], \quad [+, -] \longleftrightarrow [-, +]. \quad (7.10)$$

It does not map $[+, +]$ into $[+, -]$. Additional traced Wilson loops or color contractions require their own explicitly defined transformation; they cannot be discarded because their Lorentz tensor has the same shape. The fourteen coefficients in the gluon row count belong to *each* chosen link/color class.

The standard quark SIDIS–DY sign relation is often called modified universality. Its operator-level symmetry is not a nuclear-model guess. The qualification is in connecting that operator to a particular measurement: one must establish the process factorization, the appropriate Wilson paths, the subtraction prescription, and the kinematic regime. Some hadronic processes with colored final states do not factorize into a product of independently assigned single-hadron TMDs [27]. Calling this “conditional universality” should identify these conditions, not suggest that an arbitrary sign may be fitted for each process. The composite character of ⁷Li does not remove the symmetry of its complete nuclear matrix element; replacing that matrix element by an impulse model is a separate approximation.

7.5 Scale evolution and coefficient-level null tests

Let Σ be the diagonal matrix of reversal signs in a specified impact-parameter tensor-coefficient basis. The relation between the two link definitions is $\tilde{C}^{[-]} = \Sigma \tilde{C}^{[+]}$. An evolution operator $\mathcal{U}$ preserves this relation when

$$\mathcal{U}^{[-]} \Sigma = \Sigma \mathcal{U}^{[+]}. \quad (7.11)$$

In a common two-line scheme with the same scalar, spin-independent counterterms in a fixed parton representation, this reduces to a commutation relation. Arbitrary identical channel-mixing matrices need not commute with Σ ; “use the same matrix twice” is not a sufficient argument. The repository tests the finite algebra with these assumptions exposed. It does not calculate a new Collins–Soper kernel.

At matched arguments the coefficient-level null is $C_\nu^{[-]} - s_\nu C_\nu^{[+]} = 0$. A practical joint analysis inserts these common target coefficients into the separate SIDIS and DY convolutions. Their partner functions, electric-charge weights, transverse momenta, measured angles, and evolution ranges must be retained. Subtracting the two experimental asymmetries directly is not this null test.

8 The complete leading-twist quark correlator

Provenance and task. The $K = 0, 1$ quark structures are the familiar unpolarized and vector-polarized TMDs [5]; the $K = 2$ pattern is the established spin-one extension [4, 6]. The fourteen $K = 3$ target-side coefficients are developed here with explicit normalization and parity, using the known spin-3/2 fragmentation basis as an algebraic comparison [7]. No PDF is identified with a fragmentation function. The task is now to turn the enumeration rule into the complete operator parametrization.

8.1 Beginning with the unpolarized and vector-polarized channels

For an unpolarized target the only number scalar is f_{00}^q . There is no target pseudoscalar for the quark helicity projection. A transverse quark-spin vector can nevertheless be correlated with $\mathbf{k}_T$: reflection parity allows $E\mathbf{k}_T/M_A$, giving the single $h_{00}^{q[1]}$ term. The word unpolarized refers to the *target*; it does not force every correlation involving parton polarization to vanish.

For S_L the helicity projection can be proportional directly to S_L , and the transverse-spin projection can contain $S_L\mathbf{k}_T/M_A$. For S_T the two scalar contractions are $\mathbf{k}_T \cdot S_T/M_A$ and $-\epsilon_{ij}k^i S_T^j/M_A$. Parity assigns them to helicity and number, respectively. A rank-one quark output has two independent vector structures, S_T and $\mathbf{k}_2 \cdot S_T$. This gives six vector- target coefficients and, with the two unpolarized ones, the familiar eight. The same construction, rather than a new independent assumption, then generates the quadrupole and octupole sectors.

The irreducible form $S_T h^{[0]} + \mathbf{k}_2 \cdot S_T h^{[2]}$ is especially helpful. Writing the second term with $k^i k^j$ instead of k_2^{ij} adds a trace proportional to $\mathbf{k}_T^2 S_T^i/2$. In an unprojected notation this trace shifts the coefficient multiplying S_T^i . Consequently a collinear “transversity” may be a linear combination of named functions in another convention. One must compare full tensor products, not identify two functions merely because both are called h_1 .

8.2 Naming convention and the two scalar projections

We use f_{Km}^s and g_{Km}^s with $a = q, g$. For helicity-flip coefficients write $h_{Km}^{a[n]}$, where the bracket $[n]$ is the *explicit orbital rank*. It is not a transverse moment and not a power. This makes the basis unambiguous even when conventional strings of $\perp$ superscripts differ across papers. All functions have mass dimension -2 . The familiar-name dictionary below is local to the explicit tensor definitions, not an assertion that every sign and normalization agrees with every reference.

The common scalar pattern for either parton species is

$$\begin{aligned} \mathcal{F}^a = & f_{00}^s + \mathcal{S}_1[S_T]f_{11}^a + S_{LL}f_{20}^a + \mathcal{C}_1[S_{LT}]f_{21}^a + \mathcal{C}_2[S_{TT}]f_{22}^a \\ & + \mathcal{S}_1[S_{LLT}]f_{31}^a + \mathcal{S}_2[S_{LTT}]f_{32}^a + \mathcal{S}_3[S_{TTT}]f_{33}^a, \end{aligned} \quad (8.1)$$

$$\begin{aligned} \mathcal{G}^a = & S_L g_{10}^a + \mathcal{C}_1[S_T]g_{11}^a + \mathcal{S}_1[S_{LT}]g_{21}^a + \mathcal{S}_2[S_{TT}]g_{22}^a \\ & + S_{LLL}g_{30}^a + \mathcal{C}_1[S_{LLT}]g_{31}^a + \mathcal{C}_2[S_{LTT}]g_{32}^a + \mathcal{C}_3[S_{TTT}]g_{33}^a. \end{aligned} \quad (8.2)$$

Equations (4.14) and (4.16) define every contraction and sign. For example, $\mathcal{F}_{33}^q = -\mathbf{k}_3^{abc}\epsilon_{ad}S_{TTT}^{abc}f_{33}^q$, whereas $\mathcal{G}_{33}^q = \mathbf{k}_3^{abc}S_{TTT}^{abc}g_{33}^q$. They are distinct pseudoscalar/scalar couplings to the same target coherence.

8.3 The complete transverse-spin projection

Write $\mathcal{T}^q = \sum_{K=0}^3 \mathcal{T}_{(K)}^q$. Its 16 terms are

$$\mathcal{T}_{(0)}^{q,i} = \epsilon_{ia}\mathbf{k}_1^a h_{00}^{q[1]}, \quad (8.3)$$

$$\mathcal{T}_{(1)}^{q,i} = S_L k_1^i h_{10}^{q[1]} + S_T^i h_{11}^{q[0]} + \mathbf{k}_2^{ia} S_T^a h_{11}^{q[2]}, \quad (8.4)$$

$$\begin{aligned} \mathcal{T}_{(2)}^{q,i} = & \epsilon_{ia} [S_{LL}\mathbf{k}_1^a h_{20}^{q[1]} + S_{LT}^a h_{21}^{q[0]} + \mathbf{k}_2^{ab} S_{LT}^b h_{21}^{q[2]} \\ & + S_{TT}^{ab} \mathbf{k}_1^b h_{22}^{q[1]} + \mathbf{k}_3^{abc} S_{TT}^{bc} h_{22}^{q[3]}], \end{aligned} \quad (8.5)$$

$$\begin{aligned} \mathcal{T}_{(3)}^{q,i} = & S_{LLL}\mathbf{k}_1^i h_{30}^{q[1]} + S_{LLT}^i h_{31}^{q[0]} + \mathbf{k}_2^{ia} S_{LLT}^a h_{31}^{q[2]} \\ & + S_{LTT}^{ia} \mathbf{k}_1^a h_{32}^{q[1]} + \mathbf{k}_3^{iab} S_{LTT}^{ab} h_{32}^{q[3]} \\ & + S_{TTT}^{iab} \mathbf{k}_2^b h_{33}^{q[2]} + \mathbf{k}_4^{iabc} S_{TTT}^{abc} h_{33}^{q[4]}. \end{aligned} \quad (8.6)$$

A superscript [0] means no explicit transverse momentum. It does not by itself guarantee a nonzero collinear PDF; link-reversal parity must also be checked. The positive signs in these equations define our coefficients after converting to Euclidean components.

8.4 Constructing the four octupole sectors

For S_{LLL} , $m = 0$: the scalar number projection is parity forbidden, the helicity projection is $S_{LLL}g_{30}^q$, and a transverse-spin vector requires exactly one $\mathbf{k}_1$. For S_{LLT} , $m = 1$: one scalar and one pseudoscalar use $\mathbf{k}_1$, while the vector terms are S_{LLT} and $\mathbf{k}_2 \cdot S_{LLT}$. For S_{LTT} , $m = 2$: the scalar contractions use $\mathbf{k}_2$, and the vector endpoints are $S_{LTT} \cdot \mathbf{k}_1$ and $\mathbf{k}_3 : S_{LTT}$. Finally, S_{TTT} , $m = 3$, couples through $\mathbf{k}_3$ in the scalar projections and through $S_{TTT} : \mathbf{k}_2$ and $\mathbf{k}_4 : S_{TTT}$ in the vector projection.

Thus the four octupole components contribute 2, 4, 4, 4 coefficients, respectively. Their $3 + 4 + 7 = 14$ partition among $\mathcal{F}, \mathcal{G}, \mathcal{T}$ follows from the rotation and parity construction. It agrees in algebraic count with the spin-3/2 quark fragmentation classification, but fragmentation's out-state and link/time-reversal properties must be treated separately [7].

8.5 Worked construction of every channel for the LTT octupole

Take $K = 3, m = 2$, and write

$$S_{LTT} = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}, \quad A_{32,+} = a + ib. \quad (8.7)$$

The number and helicity projections require orbital rank two. Their explicit dimensionless spin factors are

$$\mathcal{C}_2[S_{LTT}] = \frac{a(k_x^2 - k_y^2) + 2bk_xk_y}{M_A^2}, \quad (8.8)$$

$$\mathcal{S}_2[S_{LTT}] = \frac{2ak_xk_y - b(k_x^2 - k_y^2)}{M_A^2}. \quad (8.9)$$

Because K is odd, the number coefficient f_{32}^q multiplies $\mathcal{S}_2$ and the helicity coefficient g_{32}^q multiplies $\mathcal{C}_2$. Neither scalar can be discarded by choosing $b = 0$ in a single preparation: a and b are independent components of the target response space.

The quark transverse-spin output has rank one. Its lower orbital branch is the ordinary contraction

$$(S_{LTT} \cdot \mathbf{k}_1)^i = \frac{1}{M_A} (ak_x + bk_y, bk_x - ak_y)^i. \quad (8.10)$$

Its complex component is $A_{32,+}\kappa^*$, which transforms with helicity $2 - 1 = 1$. The upper branch is $\mathbf{k}_3^{iab} S_{LTT}^{ab}$; using Eq. (4.19) gives complex component $\frac{1}{2} A_{32,+}^* \kappa^3$, with helicity $-2 + 3 = 1$. There is no parity-required dual for odd K . Thus the complete response to this one target component is

$$\mathcal{F}_{32}^q = \mathcal{S}_2[S_{LTT}] f_{32}^q, \quad (8.11)$$

$$\mathcal{G}_{32}^q = \mathcal{C}_2[S_{LTT}] g_{32}^q, \quad (8.12)$$

$$\mathcal{T}_{32,+}^q = A_{32,+}\kappa^* h_{32}^{q[1]} + \frac{1}{2} A_{32,+}^* \kappa^3 h_{32}^{q[3]}. \quad (8.13)$$

This four-coefficient example contains the entire construction in a small problem: fix the target rank, solve the rotation condition, apply parity, and normalize the Cartesian contractions. Repeating it for the other three octupole components gives the fourteen terms already displayed. Nothing in this algebra fixes their radial shapes.

8.6 Quark distribution labels and normalization conventions

(K, m)	f_{Km}^q	g_{Km}^q	$h_{Km}^{q[n]}$: rank n and local name
$(0, 0)$	f_1	—	$[1] : h_1^\perp$
$(1, 0)$	—	g_{1L}	$[1] : h_{1L}^\perp$
$(1, 1)$	$f_{1T}^\perp$	g_{1T}	$[0] : h_1, [2] : h_{1T}^\perp$
$(2, 0)$	f_{1LL}	—	$[1] : h_{1LL}^\perp$
$(2, 1)$	f_{1LT}	g_{1LT}	$[0] : h_{1LT}, [2] : h_{1LT}^\perp$
$(2, 2)$	f_{1TT}	g_{1TT}	$[1] : h_{1TT}^\perp, [3] : h_{1TT}^{\perp\perp}$
$(3, 0)$	—	g_{1LLL}	$[1] : h_{1LLL}^\perp$
$(3, 1)$	$f_{1LLT}^\perp$	$g_{1LLT}^\perp$	$[0] : h_{1LLT}, [2] : h_{1LLT}^\perp$
$(3, 2)$	$f_{1LTT}^\perp$	$g_{1LTT}^\perp$	$[1] : h_{1LTT}^\perp, [3] : h_{1LTT}^{\perp\perp}$
$(3, 3)$	$f_{1TTT}^\perp$	$g_{1TTT}^\perp$	$[2] : h_{1TTT}^\perp, [4] : h_{1TTT}^{\perp\perp}$

This dictionary is a naming aid. Equations (8.1)–(8.6) and the spin normalizations, rather than a name alone, define the functions. A comparison to another convention must compare the full coefficient–tensor product. In particular a change $A_{Km} \rightarrow cA_{Km}$ requires the associated coefficient to change by $1/c$.

9 The complete leading-twist gluon correlator

Provenance and task. The trace, circular, and linear gluon projections and their lower-spin Wilson-line dependence are established [15–17]. The work here supplies a minimal physical transverse reorganization and the target-side octupole extension. The scalar patterns are shared with the quark decomposition because they have the same transverse rank and reflection character. Their scalar functions remain independent quark and gluon matrix elements.

9.1 Linear polarization as interference, not a spin vector

Choose transverse basis vectors $\hat{x}, \hat{y}$ and circular combinations $e_\pm = (\hat{x} \mp i\hat{y})/\sqrt{2}$. Their relative phases are a convention fixed consistently with $E_{xy} = +1$. A real polarization vector at angle α is an equal-amplitude superposition of the two circular states with a relative phase 2α . Consequently linear polarization is described by a symmetric-traceless tensor with angular character $e^{\pm 2i\alpha}$. Rotating the physical linear axis by π gives the same state. This explains the rank-two parton channel without invoking a target model.

For an unpolarized target the available rank-two momentum tensor is k_2 , giving $h_{00}^{g[2]}$. For a vector target S_T , the two allowed branches have orbital ranks one and three. For a quadrupole S_{TT} they have ranks zero and four. The rank-zero quadrupole term is particularly important: a target with a two-unit helicity coherence can compensate the gluon helicity flip even without transverse orbital momentum. This is the origin of a collinear gluon transversity for targets of spin at least one. It is not an analogue of the spin-1/2 quark spin vector.

The trace and helicity parts are exactly the scalar constructions (8.1)–(8.2) with the independent gluon functions f^g, g^g . Only the helicity-flip part changes from rank one to rank two. For a rank-two STF tensor write $(\tilde{B})^{ij} = \text{STF}_2(\epsilon_{ia} B^{aj})$.

9.2 Unpolarized, vector, and quadrupole sectors

The 1, 3, and 5 linearly polarized terms in the first three sectors are

$$\mathcal{H}_{(0)}^{g,ij} = k_2^{ij} h_{00}^{g[2]}, \quad (9.1)$$

$$\mathcal{H}_{(1)}^{g,ij} = [S_L k_2 h_{10}^{g[2]} + \widetilde{\text{STF}_2(S_T k_1)} h_{11}^{g[1]} + (k_3 \cdot S_T) h_{11}^{g[3]}]^{ij}, \quad (9.2)$$

$$\begin{aligned} \mathcal{H}_{(2)}^{g,ij} = & S_{LL} k_2^{ij} h_{20}^{g[2]} + \text{STF}_2(S_{LT}^i k_1^j) h_{21}^{g[1]} + k_3^{ija} S_{LT}^a h_{21}^{g[3]} \\ & + S_{TT}^{ij} h_{22}^{g[0]} + k_4^{ijab} S_{TT}^{ab} h_{22}^{g[4]}. \end{aligned} \quad (9.3)$$

The free indices on the bracket in Eq. (9.2) are i, j before and after dualization; $(k_3 \cdot S_T)^{ij} = k_3^{ija} S_T^a$. Every term is real and STF. Together with the scalar projections, these sectors contain $2 + 6 + 10 = 18$ independent gluon TMDs. They are an irredundant reorganization of the lower-spin operator structures, not a verbatim reuse of the nonminimal coefficient notation discussed in Section 6.6.

9.3 The seven octupole linear-polarization terms

Define the ordinary STF tensor

$$\begin{aligned} B_{(3)}^{ij} = & S_{LLL} k_2^{ij} h_{30}^{g[2]} + \text{STF}_2(S_{LLT}^i k_1^j) h_{31}^{g[1]} + k_3^{ija} S_{LLT}^a h_{31}^{g[3]} \\ & + S_{LTT}^{ij} h_{32}^{g[0]} + k_4^{ijab} S_{LTT}^{ab} h_{32}^{g[4]} \\ & + S_{TTT}^{ija} k_1^a h_{33}^{g[1]} + k_5^{ijabc} S_{TTT}^{abc} h_{33}^{g[5]}. \end{aligned} \quad (9.4)$$

Then

$$\mathcal{H}_{(3)}^{g,ij} = \tilde{B}_{(3)}^{ij}. \quad (9.5)$$

This is the complete octupole linear-polarization sector. The $m = 2$ component has orbital ranks 0, 4, not 0, 2, 4; the $m = 3$ component has ranks 1, 5, not 1, 3, 5. In particular the TTT low-rank term is $S_{TTT}^{ija} k^a / M_A$, with one power of mass in the denominator, not two. Adding its three f and four g functions gives 14 independent octupole gluon TMDs.

The complete correlator is obtained by substituting $\mathcal{H}^g = \sum_{K=0}^3 \mathcal{H}_{(K)}^g$ and Eqs. (8.1)–(8.2) into Eq. (5.12). No term is left implicit beyond the explicitly defined STF and dual operations.

9.4 Worked octupole examples and normalization of both branches

Return to S_{LTT} with $A_{32,+} = a + ib$. The gluon output has rank two, so its ordinary lower and upper branches are S_{LTT}^{ij} and $k_4^{ijab} S_{LTT}^{ab}$. Their complex components are $A_{32,+}$ and $\frac{1}{4} A_{32,+}^* \kappa^4$. Since the target rank is odd while gluon linear polarization is an ordinary tensor, dualize the output:

$$\mathcal{H}_{32,+}^g = -i \left[A_{32,+} h_{32}^{g[0]} + \frac{1}{4} A_{32,+}^* \kappa^4 h_{32}^{g[4]} \right]. \quad (9.6)$$

The $\frac{1}{4}$ follows from k_4 's normalization and the contraction of two STF indices. It is not an extra convention for the hard cross section. The putative rank-two orbital branch constructed from $k_2 S_{LTT}$ has vanishing free-index STF part, as proved above.

For S_{TTT} , let $A_{33,+}$ denote its transverse helicity-three component. Contracting one index with k_1 gives a rank-two tensor with complex component $A_{33,+} \kappa^*$; contracting the three target indices with k_5 gives $\frac{1}{4} A_{33,+}^* \kappa^5$. Therefore

$$\mathcal{H}_{33,+}^g = -i \left[A_{33,+} \kappa^* h_{33}^{g[1]} + \frac{1}{4} A_{33,+}^* \kappa^5 h_{33}^{g[5]} \right]. \quad (9.7)$$

The lower branch contains one power of k_T/M_A , despite coupling to a rank-three target. The gluon operator has supplied two units of helicity flip, leaving only one to be compensated orbitally. This example makes clear why target rank and momentum rank cannot be used interchangeably. The trace and helicity responses to these same target components are already determined by $\mathcal{S}_m$ and $\mathcal{C}_m$. Combining those two scalars with the two linear-polarization branches gives four radial functions for each $m > 0$. For $m = 0$ parity allows only one scalar channel and one linear-polarization channel. Thus the octupole gluon partition is $2 + 4 + 4 + 4 = 14$, with the same total as the quark sector but different orbital ranks and link-reversal properties.

9.5 Gluon distribution labels and normalization conventions

(K, m)	scalar names	$h_{Km}^{g[n]}$: rank n and local name
$(0, 0)$	f_1^g	$[2] : h_1^{\perp g}$
$(1, 0)$	g_{1L}^g	$[2] : h_{1L}^{\perp g}$
$(1, 1)$	$f_{1T}^{\perp g}, g_{1T}^g$	$[1] : h_1^g, [3] : h_{1T}^{\perp g}$
$(2, 0)$	f_{1LL}^g	$[2] : h_{1LL}^{\perp g}$
$(2, 1)$	f_{1LT}^g, g_{1LT}^g	$[1] : h_{1LT}^g, [3] : h_{1LT}^{\perp g}$
$(2, 2)$	f_{1TT}^g, g_{1TT}^g	$[0] : h_{1TT}^g, [4] : h_{1TT}^{\perp\perp g}$
$(3, 0)$	g_{1LLL}^g	$[2] : h_{1LLL}^{\perp g}$
$(3, 1)$	$f_{1LLT}^{\perp g}, g_{1LLT}^{\perp g}$	$[1] : h_{1LLT}^g, [3] : h_{1LLT}^{\perp g}$
$(3, 2)$	$f_{1LTT}^{\perp g}, g_{1LTT}^{\perp g}$	$[0] : h_{1LTT}^g, [4] : h_{1LTT}^{\perp\perp g}$
$(3, 3)$	$f_{1TTT}^{\perp g}, g_{1TTT}^{\perp g}$	$[1] : h_{1TTT}^g, [5] : h_{1TTT}^{\perp\perp g}$

There are no independent $h_{22}^{g[2]}$, $h_{32}^{g[2]}$, or $h_{33}^{g[3]}$ in this physical STF basis. The symbol f_{1TT}^g in this table is the minimal trace coefficient; it is not separately the first of the two redundant published labels in Eq. (6.11).

10 Projecting and recovering the independent TMD coefficients

Provenance and task. The inversions below are elementary linear algebra applied to the explicit coefficient basis of this work. They supply operational operator definitions and are independent of any assumed TMD shape. They do not invert the fragmentation convolution of an experimental cross section. The desired input is a computed spin-dependent correlator, or its linear response to a complete set of target matrices.

10.1 Projecting the target and scalar harmonics

Suppose a parton projection has been represented by a target-space operator $\hat{R}(\mathbf{k}_T)$ such that $R[\rho](\mathbf{k}_T) = \text{Tr}[\rho \hat{R}(\mathbf{k}_T)]$. Resolve its rank- K part with $\text{Tr}[\tau_{KQ}^\dagger \hat{R}]$. This first step prevents a vector response from being mistaken for an octupole response merely because both have the same transverse helicity. Convert the resulting spin components to the A_{Km} normalization and then resolve the azimuth of $\mathbf{k}_T$ at fixed $t = |\mathbf{k}_T|/M_A$.

For a single scalar response $R(\phi_k) = s_{Km} t^m F_{Km} \cos[m(\phi_k - \phi_{Km})]$ with $m > 0$, orthogonality gives

$$F_{Km} = \frac{1}{\pi s_{Km} t^m} \int_0^{2\pi} d\phi_k R(\phi_k) \cos[m(\phi_k - \phi_{Km})]. \quad (10.1)$$

Replace both cosines by sines for the pseudoscalar branch. The $m = 0$ coefficient is the average $\int d\phi_k R/(2\pi A_{K0})$. The different factors π and 2π come from the norms of nonconstant and

constant Fourier functions. Equation (10.1) is to be applied to a target-projected response, not to the total correlator with all ranks simultaneously present.

The formal use of a unit target component in a derivative need not correspond to a standalone physical state. A physical implementation of the same linear response is always available locally: for a Hermitian traceless basis matrix X , choose

$$\rho_{\pm} = I/4 \pm \epsilon X, \quad 0 < \epsilon \leq \frac{1}{4\|X\|_{\text{op}}}. \quad (10.2)$$

Both matrices are positive, and $[R(\rho_+) - R(\rho_-)]/(2\epsilon) = \text{Tr}(X\hat{R})$. Thus basis derivatives can be assembled from differences of allowed states. Their practical preparation and calibration remain experimental questions.

The matrix/tensor construction already defines all coefficients. This section shows how to solve for them from a computed correlator, which also independently checks normalizations used in observables.

10.2 Quark vector branches

For one target component with $m > 0$, remove the parity-required dual from $\mathcal{T}^q$ and call the resulting complex vector V_+ . The low- and high-rank terms in Eqs. (8.4)–(8.6) satisfy

$$V_+ = A_{Km,+}(\kappa^*)^{m-1} h_{Km}^{q[m-1]} + \frac{1}{2} A_{Km,+}^* \kappa^{m+1} h_{Km}^{q[m+1]}. \quad (10.3)$$

For $m = 0$, $V_+ = A_{K0} \kappa h_{K0}^{q[1]}$. The actual transverse-spin component is $\mathcal{T}_+ = V_+$ for odd K , and $\mathcal{T}_+ = -iV_+$ for even K .

To invert Eq. (10.3), take $\mathbf{k}_T$ along x with $t = |\mathbf{k}_T|/M_A > 0$, and evaluate the linear response first to $A_{Km,+} = 1$, then to $A_{Km,+} = i$. Denote the two results v_1, v_i . They obey

$$v_1 = t^{m-1} h_- + \frac{1}{2} t^{m+1} h_+, \quad \frac{v_i}{i} = t^{m-1} h_- - \frac{1}{2} t^{m+1} h_+, \quad (10.4)$$

so that

$$h_- = \frac{v_1 + v_i/i}{2t^{m-1}}, \quad h_+ = \frac{v_1 - v_i/i}{t^{m+1}}. \quad (10.5)$$

Unit target components here denote basis derivatives or linear combinations of physical density matrices; they need not themselves be isolated physical states with every other moment zero.

10.3 Gluon linear-polarization branches

For $B_+ = B^{xx} + iB^{xy}$ before the odd- K dual, the explicit Cartesian contractions give

$$\begin{aligned} B_+(m=0) &= \frac{1}{2} A_{K0} \kappa^2 h_{K0}^{g[2]}, \\ B_+(m=1) &= \frac{1}{2} A_{K1,+} \kappa h_{K1}^{g[1]} + \frac{1}{4} A_{K1,+}^* \kappa^3 h_{K1}^{g[3]}, \\ B_+(m=2) &= A_{K2,+} h_{K2}^{g[0]} + \frac{1}{4} A_{K2,+}^* \kappa^4 h_{K2}^{g[4]}, \\ B_+(m=3) &= A_{K3,+} \kappa^* h_{K3}^{g[1]} + \frac{1}{4} A_{K3,+}^* \kappa^5 h_{K3}^{g[5]}. \end{aligned} \quad (10.6)$$

The actual $\mathcal{H}_+$ is B_+ for even K and $-iB_+$ for odd K . For $m > 0$ define $c_m = (1/2, 1, 1)$ for $m = (1, 2, 3)$ and $n_- = |m - 2|$. The same two target responses give

$$h_- = \frac{b_1 + b_i/i}{2c_m t^{n_-}}, \quad h_+ = \frac{2(b_1 - b_i/i)}{t^{m+2}}. \quad (10.7)$$

Every coefficient is therefore uniquely recoverable at generic $k_T \neq 0$. At $k_T = 0$, some tensor structures vanish and one must use a limiting/radial definition; this is not a new independent TMD.

10.4 Why two target orientations separate the orbital branches

For a real momentum along x , the complex target choices $A_+ = 1$ and $A_+ = i$ produce a sum and a difference of the low- and high- orbital branches. The reason is conjugation: $A_+^* = A_+$ for the first choice but $A_+^* = -A_+$ for the second. This is the whole mechanism of Eqs. (10.5) and (10.7). For example, at $t = 1$ in the quark case,

$$\begin{pmatrix} v_1 \\ v_i/i \end{pmatrix} = \begin{pmatrix} 1 & 1/2 \\ 1 & -1/2 \end{pmatrix} \begin{pmatrix} h_- \\ h_+ \end{pmatrix}. \quad (10.8)$$

Its determinant is -1 . One target orientation would give only one linear combination; two independent orientations recover both radial coefficients. In a numerical implementation, first remove the parity-required dual, then apply this real 2×2 inverse. Doing those operations in the opposite convention without accounting for the $-i$ phase creates spurious imaginary functions.

Near $k_T = 0$, divisions by t^n can become ill-conditioned. The definitions still refer to the coefficient of a covariant, but numerical extraction should use nonzero momenta, analytic small- k_T behavior, or a regular radial parametrization. The full correlator may be finite while particular coefficient conventions are not determined by its value exactly at the origin. At perturbative level, distribution-valued terms also require a regulator and cannot be reconstructed by naive pointwise division.

11 Physical interpretation of the multipole distributions

Provenance and task. The population and coherence contrasts in this section develop the physical interpretation of the present research formalism. They use standard spin matrix elements and agree with the forward spin-3/2 operator content discussed in Refs. [12–14]. Their purpose is to identify what an independent rank-three response measures, rather than to attach an unexplained name to each tensor. We first discuss the collinear coefficients; their operator reduction and its qualifications are established in Section 12.

11.1 A multipole coefficient is a response to the target state

A target TMD can be regarded as a reduced coefficient in the linear spin-space response of a bilocal QCD operator. Varying S_{LLL} at fixed lower moments changes the helicity projection by $\delta S_{LLL} g_{30}^q$ or $\delta S_{LLL} g_{30}^g$. It does not change it by a cube of the vector polarization. Although O_{abc} contains cubic products of spin generators, its expectation is *linear* in ρ . The corresponding TMD is not a product of three helicity distributions.

A spin multipole is also distinct from a spatial charge multipole. A rank-three spin response does not by itself imply a static octupole charge deformation, three active partons, or a unique orbital partial wave. It specifies how a matrix element changes under rotations and variations of the target density matrix. The internal mechanism must be calculated from hadronic or nuclear dynamics. Section 18 gives an explicit spin–orbital example in which ordinary spin-1/2 constituents participate in a rank-three nuclear response.

11.2 Number and helicity contrasts between target populations

For a pure target helicity $\Lambda = 3/2$ or $1/2$, define the number responses U_3, U_1 and helicity responses G_3, G_1 . In the normalization of Eq. (3.18), the collinear projections give, for either

parton species,

$$U_3 = f_1 + f_{1LL}, \quad U_1 = f_1 - f_{1LL}, \quad (11.1)$$

$$G_3 = \frac{3}{2}g_1 + \frac{3}{10}g_{1LLL}, \quad G_1 = \frac{1}{2}g_1 - \frac{9}{10}g_{1LLL}. \quad (11.2)$$

The factors are eigenvalues of J_z, Q_{zz}, O_{zzz} . Negative target helicities give the same number responses and opposite helicity responses. Where a positive parton spectral interpretation applies, the two helicity densities in a fixed target state are $(U_\Lambda + G_\Lambda)/2$ and $(U_\Lambda - G_\Lambda)/2$. The linear operator relations remain meaningful even in a renormalized scheme where such pointwise densities are not probabilities.

Adding and subtracting the first two equations gives f_1 and f_{1LL} . Inverting the helicity 2×2 system gives

$$f_1 = \frac{1}{2}(U_3 + U_1), \quad f_{1LL} = \frac{1}{2}(U_3 - U_1), \quad (11.3)$$

$$g_1 = \frac{1}{5}(3G_3 + G_1), \quad g_{1LLL} = \frac{1}{3}(G_3 - 3G_1). \quad (11.4)$$

The quadrupole distribution compares number responses of the outer and inner helicity sectors. The octupole helicity distribution instead measures a departure from a helicity response proportional to target helicity: exact proportionality would require $G_3 = 3G_1$ at every x . A nonzero g_{1LLL} detects independent structure beyond that one-parameter description. Its lowest local moments may vanish while its x dependence is nonzero.

For a generic diagonal ensemble, the helicity response is $S_L g_1 + S_{LLL} g_{1LLL}$. One measurement of it gives one equation for two functions. Two preparations can separate them only when the matrix of their (S_L, S_{LLL}) values has rank two. Opposite signs of the same preparation do not generally supply a second independent ratio. The octupole-only pair discussed later is a particularly clean algebraic construction of an independent response.

At finite transverse momentum the interpretation of $g_{30}(x, k_T^2)$ is the same after the target-rank projection is performed. Other $m > 0$ components bring momentum-dependent coherences rather than additional diagonal population moments. Keeping these distinctions prevents an ordinary helicity measurement in one polarized sample from being relabeled as an octupole measurement.

11.3 Quark transversity as a contrast of coherences

At orbital rank zero, both S_T and S_{LLT} can compensate the one-unit quark helicity flip. They differ in irreducible target rank. The corresponding spin-space raising operator is

$$h_1 J_+ + h_{1LLT} (O_{zzx} + iO_{zzy}). \quad (11.5)$$

The ladder coefficients show that the two outer transition amplitudes and the middle transition amplitude are

$$a = \sqrt{3} \left(h_1 + \frac{2}{5} h_{1LLT} \right), \quad b = 2h_1 - \frac{6}{5} h_{1LLT}. \quad (11.6)$$

Dividing out their ordinary spin-ladder strengths and taking a difference isolates the octupole:

$$h_{1LLT} = \frac{a}{\sqrt{3}} - \frac{b}{2}, \quad h_1 = \frac{3a}{5\sqrt{3}} + \frac{b}{5}. \quad (11.7)$$

Thus ordinary transversity gives the common ladder-weighted response, whereas octupole transversity measures a difference between outer and middle coherences. One off-diagonal density-matrix entry cannot by itself project onto rank three. The matrix combinations in Eq. (3.33) are required to separate its vector, quadrupole, and octupole content.

A gluon linear-polarization operator connects helicities differing by two units. Its zero-orbital-rank partners are S_{TT} and S_{LTT} , not S_{LLT} . The quadrupole partner survives as the collinear gluon transversity h_{1TT}^g . The octupole partner is allowed in a link-odd TMD but is removed in the straight-link collinear limit. This is a time-reversal restriction; the target certainly has helicity-two coherences, as Eq. (3.33) demonstrates.

11.4 Momentum-dependent correlations and experimentally distinct questions

The three octupole number functions f_{3m} , $m = 1, 2, 3$, multiply sine-type momentum correlations. They describe an unpolarized parton's angular response to a rank-three target preparation and generalize the Sivers-type momentum asymmetry. The four g_{3m} describe helicity responses to the four longitudinal/transverse octupole components. The seven quark h functions correlate those components with transverse quark-spin interference; the seven gluon h functions correlate them with linear gluon polarization.

These interpretations specify different questions to ask of a target. A number projection requires no outgoing spin analyzer. A helicity projection requires a hard measurement sensitive to the parton's helicity, such as the antisymmetric lepton tensor in the examples here. A quark transverse-spin projection requires another chiral-odd factor, here the Collins fragmentation function. A gluon linear-polarization projection requires a nonzero symmetric-traceless hard response. Existence of a target TMD does not guarantee that every reaction has the analyzer needed to measure it.

Finally, the same azimuthal frequency may be produced by different ranks. Ordinary vector-target pretzelosity produces a third-harmonic SIDIS term. A target octupole also generates third harmonics, and its highest quark Collins branch generates a fifth. To assign a rank, one must identify the dependence on the calibrated target moments and the appropriate beam/hard channel, not only a Fourier frequency.

12 The renormalized collinear limit and inclusive structure functions

Provenance and task. The forward distribution content and its multipole organization have independent antecedents in arbitrary-spin DIS and spin-3/2 GPD/PDF analyses [3, 12–14]. The task here is to derive that limit from the present explicit tensor basis, explain which operations are legitimate, and obtain local- operator tests of a model. Agreement with a known collinear limit is an important check but cannot determine all finite- k_T functions.

12.1 Separating angular projection from ultraviolet integration

A rank- n harmonic has a factor $e^{\pm in\phi_k}$, whose angular integral vanishes for $n > 0$. This is an exact statement of rotation invariance. The radial integral is a different issue. For example, the illustrative tail $F(k_T^2) = C/(k_T^2 + \Lambda^2)$ gives

$$\int_{|\mathbf{k}_T| < R} d^2\mathbf{k}_T F(k_T^2) = \pi C \ln \frac{R^2 + \Lambda^2}{\Lambda^2}, \quad (12.1)$$

which diverges as $R \rightarrow \infty$. This simple example is not a claim that all polarized TMDs have the same tail. It shows why a formal angular reduction cannot turn an ultraviolet-divergent integration into a finite renormalized PDF.

The reliable sequence is to identify the zero-orbital-rank operator structures, impose the symmetries of the straight-link collinear operator, and define its matrix elements with UV renormalization. The relation to a soft-subtracted TMD is then a matching statement at small transverse separation. This order preserves both the tensor selection and the renormalization prescription.

12.2 Angular selection is not a convergent all-orders integral

For a rotationally invariant transverse regulator and a radial coefficient, $\int d\phi_k \mathbf{k}_n = 0$ for $n > 0$. This angular statement is exact. The additional radial integral over arbitrarily large $|\mathbf{k}_T|$ need not

converge: perturbative tails and UV counterterms matter. Consequently one must not interpret $f(x; \mu) = \int d^2\mathbf{k}_T f(x, \mathbf{k}_T^2; \mu, \zeta)$ as a universal finite identity for subtracted TMDs.

Define the collinear distribution instead by the renormalized light-ray operator at $\xi_T = 0$, with the appropriate straight Wilson line, for example

$$\Phi_{\text{col}}^q(x; \mu) = \left[\int \frac{d\xi^-}{2\pi} e^{ixP^+\xi^-} \langle \bar{\psi}(0) U_{\text{straight}}[0, \xi^-] \psi(\xi^-) \rangle_\rho \right]_{\text{UV ren}}. \quad (12.2)$$

The relation to a TMD is supplied by a small- b_T expansion and matching, not by discarding its rapidity subtraction. The following survival test combines angular rank with the symmetry of this straight-link collinear operator.

12.3 Six quark and five gluon distributions

The rank-zero quark candidates are $f_{00}^q, f_{20}^q, g_{10}^q, g_{30}^q, h_{11}^{q[0]}, h_{21}^{q[0]}, h_{31}^{q[0]}$. The $h_{21}^{q[0]}$ candidate is T odd and therefore has no straight-link collinear PDF. Thus

$$\{f_1^q, f_{1LL}^q, g_1^q, g_{1LLL}^q, h_1^q, h_{1LLT}^q\} \quad (12.3)$$

are the six surviving collinear quark distributions. Explicitly,

$$\begin{aligned} \mathcal{F}_{\text{col}}^q &= f_1^q + S_{LL} f_{1LL}^q, \\ \mathcal{G}_{\text{col}}^q &= S_L g_1^q + S_{LLL} g_{1LLL}^q, \\ \mathcal{T}_{\text{col}}^{q,i} &= S_T^i h_1^q + S_{LLT}^i h_{1LLT}^q. \end{aligned} \quad (12.4)$$

In particular, h_{1LLT}^q is an additional rank-three target response in a single-quark helicity-flip channel; it is not the usual vector-target transversity with a different label.

The gluon rank-zero candidates are the same four scalar/helicity functions and $h_{22}^{g[0]}, h_{32}^{g[0]}$. The latter LTT coefficient is T odd. The five surviving functions are

$$\{f_1^g, f_{1LL}^g, g_1^g, g_{1LLL}^g, h_{1TT}^g\}, \quad (12.5)$$

with

$$\mathcal{F}_{\text{col}}^g = f_1^g + S_{LL} f_{1LL}^g, \quad \mathcal{G}_{\text{col}}^g = S_L g_1^g + S_{LLL} g_{1LLL}^g, \quad \mathcal{H}_{\text{col}}^{g,ij} = S_{TT}^{ij} h_{1TT}^g. \quad (12.6)$$

The absence of a collinear h_{1LLT}^g is a time-reversal constraint, not an inability of spin 3/2 to support a helicity difference two. These forward counts agree with the independent spin-3/2 PDF/GPD analysis [13, 14]. The omitted T-odd *TMDs* need not vanish pointwise at nonzero transverse momentum.

12.4 Born-level inclusive quark SFs with explicit normalization

For the azimuth-independent, diagonal target multipoles define

$$\begin{aligned} F_{1,A}^{(0)} &= \frac{1}{2} \sum_q e_q^2 (f_1^q + f_1^{\bar{q}}), & F_{1,A}^{(2)} &= \frac{1}{2} \sum_q e_q^2 (f_{1LL}^q + f_{1LL}^{\bar{q}}), \\ g_{1,A}^{(1)} &= \frac{1}{2} \sum_q e_q^2 (g_1^q + g_1^{\bar{q}}), & g_{1,A}^{(3)} &= \frac{1}{2} \sum_q e_q^2 (g_{1LLL}^q + g_{1LLL}^{\bar{q}}). \end{aligned} \quad (12.7)$$

At massless quark Born order the transverse number response is $2x[F_{1,A}^{(0)} + S_{LL} F_{1,A}^{(2)}]$, and the helicity response is $2x[S_L g_{1,A}^{(1)} + S_{LLL} g_{1,A}^{(3)}]$. The corresponding number-sector $F_2^{(K)} = 2x F_1^{(K)}$

for $K = 0, 2$. These equations *define* our target-multipole SF normalization. A b_1 -type tensor SF in another convention requires the corresponding normalization conversion; it should not be identified by name alone. The chiral-odd quark PDFs do not enter inclusive electromagnetic DIS at this order. Gluon Wilson coefficients and additional allowed photon-helicity structures require their own higher-order hard calculation; Eq. (12.7) is not that calculation.

One can recover the stated Born number response from SIDIS by summing hadron species with the energy weight and using the fragmentation momentum sum rule. This is different from setting $D_1 = 1$ in a single identified-hadron cross section.

12.5 First local moments as consistency checks

There are useful constraints beyond counting functions. Integrating a collinear quark light-ray operator over its full momentum-fraction support produces its local current. The vector current's forward charge is independent of the target spin; the spatial axial current is a rank-one rotation operator; an antisymmetric local tensor current also has only rank-one spatial components. The Wigner–Eckart theorem therefore excludes rank-two or rank-three target pieces from the corresponding first moments. With the usual positive- x antiquark conventions this gives, whenever the moments exist or are defined by the corresponding renormalized local operators,

$$\begin{aligned}\int_0^1 dx [f_{1LL}^q - f_{1LL}^{\bar{q}}] &= 0, \\ \int_0^1 dx [g_{1LLL}^q + g_{1LLL}^{\bar{q}}] &= 0, \\ \int_0^1 dx [h_{1LLT}^q - h_{1LLT}^{\bar{q}}] &= 0.\end{aligned}\tag{12.8}$$

These are local-operator selection rules, not assumptions about a nuclear wave function and not statements that the functions vanish at each x . For the vector case they constrain the valence combination, not the sea-inclusive tensor structure function by itself. A direct analogue of the axial first-moment argument must *not* be imposed on the gluon helicity integral: its gauge-invariant operator content is different. Small- x convergence and the chosen local-operator renormalization should be checked before using a numerical integral as a sum-rule test.

12.6 Deriving the hierarchy of local quark moments

The first-moment conditions above are special cases of a local-operator selection rule. This extension is developed in the companion research manuscript; here we give the intermediate reasoning and its relation to the established multipole operator analysis [3, 12, 13]. The statement concerns collinear quark light-ray operators and their local moments, not arbitrary weighted moments of a TMD.

For a quark distribution extended to $-1 < x < 1$, its N th Mellin moment has weight x^{N-1} . A derivative of its Fourier exponential supplies a power of xP^+ . Equivalently, differentiating the gauge-linked bilocal at zero separation turns ordinary derivatives into covariant derivatives. The resulting leading-twist local number and helicity operators are

$$\mathcal{O}_{V,N}^{\mu_1 \dots \mu_N} = \bar{q} \gamma^{\{\mu_1} iD^{\mu_2} \dots iD^{\mu_N\}} q - \text{Lorentz traces},\tag{12.9}$$

$$\mathcal{O}_{A,N}^{\mu_1 \dots \mu_N} = \bar{q} \gamma^{\{\mu_1} \gamma_5 iD^{\mu_2} \dots iD^{\mu_N\}} q - \text{Lorentz traces}.\tag{12.10}$$

The braces symmetrize the displayed indices; their normalization is chosen consistently in defining the corresponding reduced moment. The transversity operator starts with $i\sigma^{\alpha\mu_1}\gamma_5$ and has $N-1$ derivatives with the appropriate mixed index symmetry. Forward matrix elements of total derivatives vanish by translation invariance.

To determine the largest target spin rank, work in the target rest frame and decompose the *local operator* under ordinary rotations. A four-vector has a rotational scalar time component and a rank-one spatial component. An axial vector has the corresponding parity-reversed components but the same maximum rotational rank. An antisymmetric tensor also contains two rotational vectors: its $0i$ components and the dual of its ij components. It is not an independent rotational rank-two object. Each extra covariant derivative has maximum rank one. Therefore the maximum rotational rank of any of these N th-moment quark operators is N .

The Wigner–Eckart theorem states that the target-spin dependence of a matrix element of a rank- j operator lies in the rank- j spin-operator subspace. Since different ranks are trace orthogonal, no target rank $K > N$ can occur. Thus

$$\text{a rank-}K \text{ local quark moment vanishes for } N < K. \quad (12.11)$$

Contracting the Lorentz indices with a fixed lightlike direction selects a linear combination of the local tensor’s components. It does not create a reduced matrix element of a rotational rank absent from that tensor. The existence of the nonlocal light-ray direction therefore does not invalidate this local selection argument.

The antiquark combination must be handled before using this result. With the usual positive- x antiquark definitions,

$$f^q(-x) = -f^{\bar{q}}(x), \quad h^q(-x) = -h^{\bar{q}}(x), \quad g^q(-x) = g^{\bar{q}}(x), \quad x > 0. \quad (12.12)$$

Changing integration variable in the negative- x part gives

$$\int_{-1}^1 dx x^{N-1} f^q(x) = \int_0^1 dx x^{N-1} [f^q + (-1)^N f^{\bar{q}}], \quad (12.13)$$

$$\int_{-1}^1 dx x^{N-1} g^q(x) = \int_0^1 dx x^{N-1} [g^q + (-1)^{N-1} g^{\bar{q}}], \quad (12.14)$$

and transversity has the number-channel sign. These signs are independent of the target-rank normalization. At $N = 1$, they reproduce the three relations in Eq. (12.8). At $N = 2$, rank three is still forbidden, giving

$$\int_0^1 dx x [g_{1LLL}^q - g_{1LLL}^{\bar{q}}] = 0, \quad (12.15)$$

$$\int_0^1 dx x [h_{1LLT}^q + h_{1LLT}^{\bar{q}}] = 0. \quad (12.16)$$

The first locally *allowed* octupole quark moment has $N = 3$, that is, an x^2 weight. Allowed does not mean nonzero in every model. Conversely, vanishing $N = 1, 2$ moments do not imply that the nonlocal distribution vanishes pointwise.

There are three qualifications worth preserving in an implementation. First, the quark–antiquark combination changes with N ; the two zero moments do not automatically constrain the same positive- x function. Second, these are moments defined by renormalized local operators. A numerical integral over incomplete support or an unregulated small- x singularity is not that definition. Third, a first moment of gluon helicity does not follow from a local quark axial current, so the same argument cannot be transplanted to it. For a continuous, integrable, nonzero combination with a vanishing unweighted integral, contributions of both signs are necessary. This is a useful model check, but it fixes neither a node position nor a distribution’s magnitude.

12.7 Inclusive limits and conserved quantities as model tests

The inclusive functions in Eq. (12.7) are charge-weighted flavor sums. Their number and helicity normalization differs from the lowercase PDFs by the conventional factor $1/2$ and by the flavor

sum. A tensor structure function called b_1 in another spin normalization must be translated by comparing the complete target response. Its label alone does not determine that conversion.

For the entire nucleus, integrating the total QCD momentum density with weight x gives one, independently of the target spin state. Applying the rank-two target projection to this identity gives a zero sum of the momentum-weighted tensor quark, antiquark, and gluon distributions. It does not force any individual species contribution to vanish. Electromagnetic partons must be included when the chosen scheme includes them. These constraints, the local moment hierarchy, and positivity in an appropriate input prescription form complementary tests of a nuclear calculation; none replaces the calculation itself.

13 Positivity of target states and parton spin matrices

Provenance and task. Density-matrix positivity is standard quantum mechanics. Positivity relations for forward spin-3/2 distributions and lower-spin gluon TMDs have been studied independently [14, 17]. We derive the relevant matrices and inequalities in the explicit spin normalization used here. The task is to distinguish physical target states, partonic Gram matrices in an appropriate prescription, and positive hard analyzers. These are related linear-algebra problems but not interchangeable physical assumptions.

13.1 Why a Gram matrix is positive, and what subtraction changes

If $M_{ij} = \sum_X a_{Xi}^* a_{Xj}$ with a nonnegative measure, then for any vector v ,

$$v^\dagger M v = \sum_X \left| \sum_i a_{Xi} v_i \right|^2 \geq 0. \quad (13.1)$$

This proves positivity directly, without diagonalizing a matrix. Its application to an unsubtracted parton correlator requires a spectral representation with the specified gauge-link and regulator properties. A generic color-link combination need not individually have a probability interpretation. Moreover, UV counterterms and rapidity subtractions are not in general sums of nonnegative probabilities. One must therefore state the prescription in which a pointwise partonic bound is imposed.

The target matrix is different: $\rho \succeq 0$ is an exact statement about the prepared ensemble. The finite Born hard matrix is also different: it is explicitly built from physical amplitudes at a specified phase-space point. Neither statement alone proves positivity of every separately subtracted TMD coefficient. This distinction is particularly important when fitting functions allowed to change sign.

For a Hermitian 2×2 matrix $M = \begin{pmatrix} a & c \\ c^* & b \end{pmatrix}$, positivity is equivalent to $a \geq 0$, $b \geq 0$, and $ab \geq |c|^2$. Applying this criterion to the parton matrices in Eqs. (13.2) and (17.1) gives the fixed-target bounds. They constrain the *complete* projection in a physical target state. They cannot be split into independent bounds on every scalar coefficient without further optimization over allowed target states.

13.2 Target-state positivity and conditional correlator positivity

Target positivity is exact: $\rho \succeq 0$ with $\text{Tr } \rho = 1$. It restricts simultaneous choices of S, T, R independently of a partonic approximation. Correlator positivity has a different status. Before subtractions, a suitable spectral/Gram representation gives positive matrices. Those inequalities are useful for a cutoff model or a positivity-preserving input prescription. Arbitrary

UV/rapidity subtractions or fixed-order $\overline{\text{MS}}$ distributions need not preserve pointwise probability interpretations. All correlator/PDF bounds in this section are conditional on the positive Gram representation, not asserted as scheme-independent pointwise laws for every renormalized TMD [17, 28].

For a fixed physical target state the quark Gram matrix can be written

$$D_q(\rho) = \frac{1}{2} \begin{pmatrix} \mathcal{F}^q + \mathcal{G}^q & \mathcal{T}^{q,x} + i\mathcal{T}^{q,y} \\ \mathcal{T}^{q,x} - i\mathcal{T}^{q,y} & \mathcal{F}^q - \mathcal{G}^q \end{pmatrix}. \quad (13.2)$$

Its eigenvalues, and those of Eq. (17.1), imply

$$\begin{aligned} \mathcal{F}^q &\geq \sqrt{(\mathcal{G}^q)^2 + (\mathcal{T}^{q,x})^2 + (\mathcal{T}^{q,y})^2}, \\ \mathcal{F}^g &\geq \sqrt{(\mathcal{G}^g)^2 + (\mathcal{H}^{g,xx})^2 + (\mathcal{H}^{g,xy})^2}. \end{aligned} \quad (13.3)$$

These constrain combinations of TMDs evaluated in the specified state. They do not justify assigning $|f_{Km}| \leq f_{00}$ separately to every coefficient with an arbitrary target normalization.

13.3 Constructing the full eight-by-eight matrix

A complete finite-dimensional positivity test is available without listing many redundant inequalities. In the full decompositions replace each target moment by its *defining spin operator*: $S_i \rightarrow J_i$, $S_{LL} \rightarrow Q_{zz}$, $S_{LT}^i \rightarrow 2Q_{zi}$, $S_{TT}^{ij} \rightarrow 2\text{STF}_2 Q_{ij}$, $S_{LLL} \rightarrow O_{zzz}$, and the remaining octupoles according to Eq. (3.18). Denote the resulting four-dimensional operators by $\hat{F}, \hat{G}, \hat{T}_i, \hat{H}_{ij}$. There are no factors from the inverse density-matrix expansion in this replacement: by construction $\text{Tr}(\rho\hat{F}) = \mathcal{F}$.

Then the joint quark and Cartesian-gluon Gram matrices are

$$\begin{aligned} \mathbb{M}_q &= \frac{1}{2} \begin{pmatrix} \hat{F} + \hat{G} & \hat{T}_x + i\hat{T}_y \\ \hat{T}_x - i\hat{T}_y & \hat{F} - \hat{G} \end{pmatrix}, \\ \mathbb{M}_g &= \frac{1}{2} \begin{pmatrix} \hat{F} + \hat{H}_{xx} & \hat{H}_{xy} + i\hat{G} \\ \hat{H}_{xy} - i\hat{G} & \hat{F} - \hat{H}_{xx} \end{pmatrix}. \end{aligned} \quad (13.4)$$

Their indices are parton polarization $\otimes$ target spin. Taking the target expectation reproduces Eqs. (13.2) and (17.1). A spectral representation has the form

$$\mathbb{M}_{i\Lambda,j\Lambda'} = \sum_X \mathcal{A}_{X;i\Lambda}^* \mathcal{A}_{X;j\Lambda'} [\text{nonnegative momentum measure}], \quad (13.5)$$

so it is positive semidefinite. Equivalently, all its eigenvalues or all principal minors are nonnegative. Testing only 2×2 minors is generally necessary but not sufficient for a generic finite- k_T 8×8 matrix. The explicit matrix construction avoids mistaking an incomplete set of pairwise bounds for a full positivity theorem.

13.4 Spectral indices and auxiliary coefficient coordinates

The order of the two spin factors must be fixed before applying a positivity test. Use parton polarization as the outer index and target helicity as the inner index in Eq. (13.4). For removal amplitudes $\mathcal{A}_{X;i\Lambda}$, a positive spectral prescription gives

$$\mathbb{M}_{i\Lambda,j\Lambda'} = \sum_X \int d\mu_X \mathcal{A}_{X;i\Lambda}^* \mathcal{A}_{X;j\Lambda'}, \quad d\mu_X \geq 0. \quad (13.6)$$

For a pure target $|c\rangle = \sum_{\Lambda} c_{\Lambda} |\Lambda\rangle$, linearity of the removal amplitude gives

$$D_{ij}(c) = \sum_{\Lambda\Lambda'} c_{\Lambda}^* \mathbb{M}_{i\Lambda,j\Lambda'} c_{\Lambda'}, \quad (13.7)$$

$$D_{ij}(\rho) = \sum_{\Lambda\Lambda'} \rho_{\Lambda'\Lambda} \mathbb{M}_{i\Lambda,j\Lambda'} = \text{Tr}_A(\rho \mathbb{M}_{ij}). \quad (13.8)$$

The last expression means the target trace of the ij block. It is consistent with replacing moments by their defining operators, and uses no inverse-density weights. For arbitrary parton coordinates e_i , $e^\dagger D(c) e$ is a sum of absolute squares. Hence positive joint matrices give positive fixed-target matrices; the converse does not follow from a finite collection of target-state tests. Complex target coherences are essential for detecting an erroneous transpose.

The repository also uses an auxiliary matrix-coordinate map for basis rank and inversion. In its documented normalization,

$$\mathbb{M}_{\text{source}} = \frac{1}{2} T_p(\mathbb{M}_{\text{aux}}), \quad [T_p(X)]_{i\Lambda,j\Lambda'} = X_{j\Lambda,i\Lambda'}. \quad (13.9)$$

Here T_p transposes only the parton factor. It is an invertible real-linear transformation of the Hermitian matrix coordinates, so the corresponding row transformation preserves the rank of a 64×32 coefficient map. It is not a unitary similarity of each 8×8 matrix and need not preserve that matrix's rank, spectrum, or positive semidefiniteness. Thus auxiliary coordinates can establish coefficient independence, while spectral positivity and physical target contractions must use the source ordering in Eqs. (13.6)–(13.8).

A simple diagnostic is the two-qubit projector on $(|00\rangle + |11\rangle)/\sqrt{2}$. Its partial transpose has eigenvalues $\{1/2, 1/2, 1/2, -1/2\}$, whereas the projector itself is positive and has rank one [33]. This is a matrix counterexample, not a nuclear TMD state. The validation also uses source-consistent finite- k_T Gram examples, so the distinction is checked within the implemented coefficient space as well as in the abstract matrix example.

13.5 Collinear bounds derived from the spin matrices

For either species, abbreviate the collinear functions by $f = f_1$, $f_Q = f_{1LL}$, $g = g_1$, and $g_O = g_{1LLL}$. Define

$$U_3 = f + f_Q, \quad U_1 = f - f_Q, \quad G_3 = \frac{3}{2}g + \frac{3}{10}g_O, \quad G_1 = \frac{1}{2}g - \frac{9}{10}g_O. \quad (13.10)$$

The subscript 3 or 1 denotes the $|m_J| = 3/2$ or $1/2$ target state, not a moment of the PDF. The diagonal entries are $U_3 \pm G_3$ and $U_1 \pm G_1$, so

$$U_3 \geq |G_3|, \quad U_1 \geq |G_1|. \quad (13.11)$$

For quarks write $h = h_1^q$, $h_O = h_{1LLT}^q$. The off-diagonal parton block is $hJ_+ + h_O(O_{zzx} + iO_{zzy})$ and

$$O_{zzx} + iO_{zzy} = \begin{pmatrix} 0 & 2\sqrt{3}/5 & 0 & 0 \\ 0 & 0 & -6/5 & 0 \\ 0 & 0 & 0 & 2\sqrt{3}/5 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (13.12)$$

The outer transitions have amplitude $\sqrt{3}(h + 2h_O/5)$, and the middle transition has amplitude $2h - 6h_O/5$. Their 2×2 blocks therefore require

$$\begin{aligned} 3 \left(h + \frac{2}{5}h_O \right)^2 &\leq (U_3 + G_3)(U_1 - G_1), \\ \left| 2h - \frac{6}{5}h_O \right| &\leq U_1 + G_1. \end{aligned} \quad (13.13)$$

For gluons, transform to the circular Gram basis $(1, -i)/\sqrt{2}, (1, i)/\sqrt{2}$. The helicity-flip block is $h_{1TT}^g J_+^2$, since $S_{TT}^{xx} + iS_{TT}^{xy} \rightarrow Q_{xx} - Q_{yy} + 2iQ_{xy} = J_+^2$. Its two nonzero matrix entries are $2\sqrt{3}h_{1TT}^g$, giving

$$12(h_{1TT}^g)^2 \leq (U_3 + G_3)(U_1 + G_1). \quad (13.14)$$

After permuting rows/columns, the collinear matrices contain only these 2×2 blocks and isolated diagonal entries. Thus Eqs. (13.11)–(13.13), or Eqs. (13.11) and (13.14), are necessary and sufficient for these *real collinear Gram matrices* to be positive. They are consistent with the spin-3/2 positivity analysis of Ref. [14]; the derivation here fixes the numerical factors using the same explicitly defined spin moments.

13.6 A reproducible route from spin matrices to the collinear bounds

To reproduce the collinear inequalities, start with the diagonal spin operators

$$\hat{F} = fI + f_Q Q_{zz}, \quad \hat{G} = gJ_z + g_O O_{zzz}. \quad (13.15)$$

Their eigenvalues give $U_3 \pm G_3$ and $U_1 \pm G_1$ in the diagonal parton blocks. Next compute the nonzero entries of the helicity-flip block. For quarks they are the ladder-weighted outer amplitudes $\sqrt{3}(h + 2h_O/5)$ and the middle amplitude $2h - 6h_O/5$. Permute the basis so that each connected pair appears in its own 2×2 block. Applying $ab \geq |c|^2$ to the outer blocks gives the first inequality in Eq. (13.13); the middle block has equal diagonal entries and gives the second. The isolated diagonal entries give Eq. (13.11).

For gluons, changing from Cartesian to circular polarization diagonalizes the circular part. The helicity-flip block is proportional to J_+^2 , whose two nonzero entries are $2\sqrt{3}$. Its squared magnitude gives $12(h_{1TT}^g)^2$ and hence Eq. (13.14). These numerical factors are consequences of the spin normalization, not adjustable phenomenological constants. They also show why a spin-1/2 gluon collinear transversity is absent: there is no two-unit target helicity transition.

In the real collinear case these blocks exhaust the matrix, so the diagonal and block inequalities are sufficient as well as necessary. At finite transverse momentum the general 8×8 matrix has more couplings. Pairwise minors alone are then insufficient; evaluate its full eigenvalue spectrum or all principal minors. Positive expectations on a restricted family of product spin states are also weaker than positivity of the full joint Gram matrix. The complete spin-space construction keeps that distinction visible.

14 Scale evolution, transverse-distance transforms, and matching

Provenance and task. TMD evolution and short-distance matching are established QCD tools [28]; the Fourier and Bessel methods used below are standard and have been applied to weighted SIDIS observables [34]. The target multipole decomposition does not introduce a new independent rapidity kernel for each spin rank in the ordinary isotropic TMD operator setup. The task is to explain which radial coefficient evolves, how its transverse harmonic is transformed, and why a finite- b_T operator expansion is not a naive integral over all k_T .

The polynomial rank n distinguishes azimuthal transformation properties; it is not an independent choice of factorization scale. At nonzero b_T , a leading-twist TMD in a fixed quark or gluon representation obeys the usual rapidity and UV evolution equations, schematically

$$\frac{\partial \tilde{F}^a}{\partial \ln \sqrt{\zeta}} = \mathcal{K}_a(b_T; \mu) \tilde{F}^a, \quad \frac{\partial \tilde{F}^a}{\partial \ln \mu} = \gamma_F^a(\alpha_s, \zeta/\mu^2) \tilde{F}^a, \quad \frac{d\mathcal{K}_a}{d \ln \mu} = -\gamma_K^a. \quad (14.1)$$

Here $\tilde{F}^a$ denotes the appropriately defined impact-parameter coefficient of the operator, not the logarithm of a possibly sign-changing polarization function. These operator evolution factors do

not acquire a new independent target-spin label for each K . Quark and gluon representations have different kernels, and matching coefficients depend on the channel [28].

The tensor Fourier transform can be computed without guessing extra powers of b_T :

$$\int d^2\mathbf{k}_T e^{i\mathbf{k}_T \cdot \mathbf{b}_T} \mathbf{k}_n^{i_1 \dots i_n} F(x, \mathbf{k}_T^2) = \frac{(-i)^n}{M_A^n} \text{STF}_2(\partial_{b^{i_1}} \dots \partial_{b^{i_n}}) \tilde{F}(x, b_T^2). \quad (14.2)$$

For a harmonic κ^n the equivalent radial transform contains

$$2\pi i^n e^{in\phi_b} \int_0^\infty dk k (k/M_A)^n J_n(kb_T) F(x, k^2). \quad (14.3)$$

Thus the explicit rank- n harmonic is preserved by Fourier transformation, although the scalar radial coefficient is differentiated.

At small b_T the operator has an expansion

$$\tilde{\mathcal{O}}_{Km}^a(x, b_T; \mu, \zeta) = \sum_j C_{Km}^{a \leftarrow j}(b_T; \mu, \zeta) \otimes \langle \mathcal{O}_j \rangle_{Km} + \text{higher powers of } b_T \Lambda_{\text{QCD}}. \quad (14.4)$$

The sum includes the operator types required by that channel. It is not always a sum over only twist-two collinear PDFs; in particular, T-odd TMD matching can involve multiparton/twist-three matrix elements. A vanishing rank-zero collinear PDF therefore does not eliminate the corresponding finite- k_T TMD.

Using M_A in Eq. (4.7) is a normalization choice. If the same rank- n covariant is normalized with a reference mass M_0 , its scalar coefficient changes as

$$F_{[n]}^{(M_0)} = (M_0/M_A)^n F_{[n]}^{(M_A)}. \quad (14.5)$$

No observable changes. Large M_A does not, by itself, predict a physical A^{-n} suppression of a nuclear TMD. Nor does an orbital harmonic rank uniquely identify a particular partial wave, a number of correlated nucleons, or a number of exchanged gluons.

14.1 Deriving the tensor Fourier transform and Bessel order

With the convention $\tilde{F}(\mathbf{b}_T) = \int d^2\mathbf{k}_T e^{i\mathbf{k}_T \cdot \mathbf{b}_T} F(\mathbf{k}_T)$, differentiation gives $\partial_{b^i} e^{i\mathbf{k}_T \cdot \mathbf{b}_T} = i k^i e^{i\mathbf{k}_T \cdot \mathbf{b}_T}$. Moving n derivatives outside the integral and projecting their free indices to STF therefore produces Eq. (14.2). The factor is $(-i)^n$, not i^n at this derivative stage. For a radial function, the angular integral can alternatively be performed first:

$$\int_0^{2\pi} d\phi_k e^{ikb \cos(\phi_k - \phi_b)} e^{in\phi_k} = 2\pi i^n e^{in\phi_b} J_n(kb). \quad (14.6)$$

There is no contradiction between these phases: radial derivatives of the order-zero Fourier transform supply the compensating signs. The two formulas provide independent normalization checks. The transverse harmonic determines the Bessel order n ; it does not change into an arbitrary mixture of ranks under an isotropic Fourier transform.

For a covariant $\mathbf{k}_n F$, the coefficient multiplying the corresponding b -space STF tensor is a derivative or Bessel-weighted transform of F . It is not generally the plain order-zero transform of the scalar function carrying the same name. Evolution should be applied to the proper b -space operator coefficient. Treating all momentum-space functions as identical scalar Fourier transforms would miss the rank- dependent normalization and derivatives.

14.2 Solving the scale equations and identifying their assumptions

At fixed nonzero b_T , the equations in Eq. (14.1) are multiplicative for the appropriate leading-twist coefficient in the standard two-direction TMD setup. For example, evolve first from ζ_0 to ζ at μ_0 , then from μ_0 to μ at fixed ζ :

$$\tilde{F}^a(\mu, \zeta) = \tilde{F}^a(\mu_0, \zeta_0) \exp \left[\mathcal{K}_a(b_T; \mu_0) \ln \frac{\sqrt{\zeta}}{\sqrt{\zeta_0}} + \int_{\mu_0}^{\mu} \frac{d\mu'}{\mu'} \gamma_F^a(\alpha_s(\mu'), \zeta/\mu'^2) \right]. \quad (14.7)$$

Here x, b_T , the polarization channel, and the link prescription are held fixed. The consistency condition $\partial \gamma_F^a / \partial \ln \sqrt{\zeta} = -\gamma_K^a$ ensures agreement with reversing the order of the two paths, together with $d\mathcal{K}_a/d \ln \mu = -\gamma_K^a$. This expression evolves a coefficient, not the logarithm of a possibly negative function. A sign-changing initial polarization function causes no mathematical problem in a linear evolution equation.

The short-distance coefficients and the long-distance rapidity kernel need perturbative and nonperturbative inputs in their appropriate regions. They have not been calculated afresh for every channel in this note. In more complicated colored-final-state factorizations, soft factors can carry extra color and directional structure. The simple evolution equation should be used only for the corresponding well-defined target TMD operator, not promoted to a complete evolution formula for an unspecified multiparton process.

14.3 Matching, weighted observables, and the information that remains

At small b_T , the bilocal operator can be expanded into collinear operators with calculable Wilson coefficients. A given TMD channel need not match only onto a twist-two PDF; link-odd channels, for example, involve appropriate multiparton correlations. Operator selection and power counting determine the allowed terms. The absence of a straight-link rank-zero PDF therefore does not imply that a finite- k_T TMD or its measurable harmonic vanishes.

Finite- b_T or Bessel-weighted observables can turn a transverse convolution into products of transforms and organize distinct harmonics cleanly [34]. They do not remove flavor sums, fragmentation uncertainties, limited acceptance, or the need to match the large-transverse-momentum part of a physical cross section. Taking $b_T \rightarrow 0$ restores precisely the short-distance and UV issues that invalidate unregulated polynomial moments. A calculation should state its b_T range, scale prescription, and treatment of the large- P_{hT} region before interpreting such a weighted measurement as a TMD determination.

15 Derivation of the complete quark SIDIS response

Provenance and task. The leading-power handbag hard scattering, the two fragmentation projections for an unpolarized hadron, and the ordinary SIDIS angular conventions are established [5, 20, 35]. Spin-one target generalizations are also known [18, 19]. The work here applies the specified target-side spin-3/2 basis to derive the full thirty-two-response catalogue, including its octupole sector. The hard trace is standard; the multipole-resolved catalogue and convention-explicit weights are the project's construction. The task is to connect each lowercase TMD to a measurable uppercase structure function without skipping the hard contraction or momentum conservation.

15.1 What is held fixed in a SIDIS measurement

In inclusive DIS one observes only the scattered lepton and sums over hadronic final states. In SIDIS an identified hadron is also observed, with energy fraction z and transverse momentum

$\mathbf{P}_{hT}$. This adds a fragmentation matrix element. The observed $\mathbf{P}_{hT}$ receives contributions from both the incoming parton's transverse momentum and fragmentation; it is not the incoming $\mathbf{k}_T$ itself. Consequently a measured SIDIS structure function is a convolution, even before radiative recoil is considered.

The approximation here is leading power for $P_{hT}/z \ll Q$ in the current-fragmentation region. The detected hadron is spinless or its polarization is unobserved. We use electromagnetic one-photon exchange and the massless-quark Born hard interaction. These choices fix which parton polarizations the process analyzes and the lepton depolarization factors. Other hadrons, spin analyses, or hard processes require their own fragmentation or hard tensors; the target basis itself remains available for those extensions.

To set up a calculation, record $(x_{Bj}, y, z, P_{hT}, \phi_h)$, the lepton helicity, the target density matrix in the adopted axes, and the fragmentation momentum convention. At Born order in this channel set the distribution fraction $x = x_{Bj}$. Evaluate the target tensor factors from ρ , not from an assumed equality between vector and octupole polarization. Only then contract with the hard and fragmentation projections.

15.2 Process, axes, and fragmentation correlator

Consider

$$\ell(l, \lambda) + A(P, \rho) \rightarrow \ell(l') + h(P_h) + X, \quad q = l - l', \quad Q^2 = -q^2, \quad y = \frac{P \cdot q}{P \cdot l}, \quad z = \frac{P \cdot P_h}{P \cdot q}. \quad (15.1)$$

The lepton helicity is $\lambda = \pm 1$. Use a right-handed frame with the large target momentum along $+z$, the virtual photon along $-z$, and $+x$ along the lepton's transverse momentum. Azimuths increase from $+x$ toward $+y$. Write $\hat{h} = \mathbf{P}_{hT}/|\mathbf{P}_{hT}| = (\cos \phi_h, \sin \phi_h)$ and $\hat{h}_\perp = (-\sin \phi_h, \cos \phi_h)$. These are explicit *native* angles. A convention with $+z$ along the photon and the same $+x$ has the opposite y axis and opposite azimuths. Longitudinal and mixed-index spin components must also be rotated when comparing such conventions; changing the sign of a sine alone is not a complete convention conversion. The standard SIDIS normalization used below follows Ref. [5].

Let $\mathbf{p}_T$ be the outgoing quark transverse momentum *relative to the detected-hadron direction*, and let M_h be the hadron mass. At leading power, transverse momentum conservation is

$$\mathbf{k}_T - \mathbf{p}_T = \mathbf{P}_{hT}/z. \quad (15.2)$$

The alternative hadron momentum relative to its quark is $\mathbf{j}_T = -z\mathbf{p}_T$; these two fragmentation variables must not be mixed. For the unpolarized detected hadron the unsubtracted correlator is

$$\begin{aligned} \Delta_{ab, \text{unsub}}(z, \mathbf{p}_T) &= \frac{1}{2z} \sum_X \int \frac{d\xi^+ d^2\xi_T}{(2\pi)^3} e^{ip^- \xi^+ - i\mathbf{p}_T \cdot \xi_T} \\ &\times \langle 0 | U[\infty, \xi] \psi_a(\xi) | hX; \text{out} \rangle \langle hX; \text{out} | \bar{\psi}_b(0) U[0, \infty] | 0 \rangle \Big|_{\xi^- = 0}. \end{aligned} \quad (15.3)$$

with $p^- = P_h^-/z$, the standard color contraction, and the same consistent soft-sharing and UV scheme as the distribution correlator. Its two leading-twist projections define

$$\frac{1}{2} \text{Tr}_D(\Delta \gamma^-) = D_1, \quad \mathcal{D}^i \equiv \frac{1}{2} \text{Tr}_D(\Delta i\sigma^{i-} \gamma_5) = -\frac{\epsilon_{ia} p^a}{M_h} H_1^\perp. \quad (15.4)$$

Thus

$$\Delta_{(2)} = \frac{1}{2} [D_1 \gamma^+ + \mathcal{D}^i i\sigma^{i+} \gamma_5]. \quad (15.5)$$

Equation (15.4) fixes the Collins-function sign used in this note. At regulated tree level the familiar collinear convention is $D_1(z) = z^2 \int d^2\mathbf{p}_T D_1(z, \mathbf{p}_T^2)$; the z^2 is the Jacobian from $\mathbf{p}_T$ to $\mathbf{j}_T$. UV matching is required beyond that formal integral, just as for a PDF.

15.3 Fragmentation momentum and the origin of the convolution

There are two common choices of fragmentation transverse momentum. The variable $\mathbf{p}_T$ used here is the outgoing quark momentum relative to the observed-hadron axis. The hadron momentum relative to its quark is $\mathbf{j}_T = -z\mathbf{p}_T$. At leading power, adding transverse momenta gives

$$\mathbf{P}_{hT} = z\mathbf{k}_T + \mathbf{j}_T = z(\mathbf{k}_T - \mathbf{p}_T). \quad (15.6)$$

This proves the sign and the factor of z in Eq. (15.2). It also gives $d^2j_T = z^2 d^2p_T$. A change of momentum variable must therefore change the fragmentation normalization and its delta function, not only the argument of an azimuthal weight.

For an unpolarized detected hadron, the number projection defines D_1 . The transverse-spin projection can be proportional to the pseudovector $\epsilon_{ia}p^a$, defining the Collins function with the sign in Eq. (15.4). Although the final hadron is unpolarized, its momentum can analyze an interfering transverse spin of the fragmenting quark. This is the physical content of a Collins correlation [20]. In the adopted leading-power massless hard interaction, chirality is conserved. A single chiral-odd quark transversity projection therefore cannot enter a cross section alone; it is paired with the chiral-odd Collins projection. The number and helicity projections instead couple to D_1 .

A useful dimensional check is that the TMD PDF and fragmentation TMD each have dimension -2 . Two transverse integrations contribute $+4$ and the transverse delta function contributes -2 . Thus a dimensionless weight produces a structure function of dimension -2 . The factors $1/M_A^n$ and $1/M_h$ belong to that weight. They are not missing powers of the hard scale Q .

15.4 Hard Dirac trace and absolute normalization

Define the convolution, summing quarks and antiquarks separately, by

$$C[wfD] = x \sum_{a=q,\bar{q}} e_a^2 \int d^2\mathbf{k}_T d^2\mathbf{p}_T \delta^{(2)}\left(\mathbf{k}_T - \mathbf{p}_T - \frac{\mathbf{P}_{hT}}{z}\right) w(\mathbf{k}_T, \mathbf{p}_T) f^a(x, \mathbf{k}_T^2) D^a(z, \mathbf{p}_T^2). \quad (15.7)$$

No unknown spin factor is hidden in this definition. A dimensionless weight gives a structure function of mass dimension -2 .

At tree level the leading-power electromagnetic hadronic tensor is obtained from $\text{Tr}_D(\Phi\gamma^\mu\Delta\gamma^\nu)$. For transverse photon indices, direct multiplication of Eqs. (5.11) and (15.5) gives

$$\text{Tr}_D(\Phi\gamma^i\Delta\gamma^j) = \delta^{ij}\mathcal{F}^q D_1 + i\epsilon_{ij}\mathcal{G}^q D_1 - (\mathcal{T}^{q,i}\mathcal{D}^j + \mathcal{T}^{q,j}\mathcal{D}^i - \delta^{ij}\mathcal{T}^q \cdot \mathcal{D}). \quad (15.8)$$

For example, the $\mathcal{F}D_1$ trace is δ^{ij} , the $\mathcal{G}D_1$ trace is $i\epsilon_{ij}$, and the $\mathcal{T}_x\mathcal{D}_x$ trace is $\text{diag}(-1, 1)$. These three checks fix both the antisymmetric sign and the sign of the Collins contraction. The full set of traces can be checked by exact symbolic evaluation [26].

Let w^{ij} be the convolution of the right side of Eq. (15.8). In the conventional hadronic-tensor normalization $\mathbb{W}^{\mu\nu} = 2M_A W^{\mu\nu}$, its leading transverse part is $\mathbb{W}^{ij} = (2z/x)w^{ij}$. The covariant leptonic tensor for a massless incident lepton is

$$L_{\mu\nu} = 2(l_\mu l'_\nu + l_\nu l'_\mu - g_{\mu\nu} l \cdot l' + i\lambda\epsilon_{\mu\nu\alpha\beta} l^\alpha l'^\beta). \quad (15.9)$$

At leading power in our axes, its normalized transverse part is

$$\mathcal{L}_{ij} = \frac{1}{2}[\delta_{ij} + \varepsilon \text{diag}(1, -1)_{ij} - i\lambda D_\ell \epsilon_{ij}], \quad \varepsilon = \frac{1-y}{1-y+y^2/2}, \quad D_\ell = \sqrt{1-\varepsilon^2}. \quad (15.10)$$

For instance, $L_{xx} = 4l_T^2 + Q^2$, $L_{yy} = Q^2$, $l_T^2 = Q^2(1-y)/y^2$, and $L_{xy} = -i\lambda Q^2(2-y)/y$. Its trace is $4Q^2(1-y+y^2/2)/y^2$. Using the standard phase-space relation $d\sigma/d\mathcal{P} = \alpha_{\text{em}}^2 y \mathbb{W}^{\mu\nu} L_{\mu\nu}/(8zQ^4)$ therefore gives

$$\frac{d\sigma}{d\mathcal{P}} = \mathcal{N} \mathcal{L}_{ij} w^{ij}, \quad \mathcal{N} = \frac{\alpha_{\text{em}}^2}{xyQ^2} (1-y+y^2/2), \quad d\mathcal{P} = dx dy d\psi dz d\phi_h dP_{hT}^2. \quad (15.11)$$

Here ψ is the overall lepton-plane azimuth in the standard phase space. Fixed-axis target moments are evaluated relative to that plane. Target-mass kinematic prefactors may be restored consistently; doing so alone does not calculate subleading-power structure functions.

Substitution of $\mathcal{D} = -E\mathbf{p}_T H_1^\perp/M_h$ into Eq. (15.8) yields the useful master result

$$\frac{1}{\mathcal{N}} \frac{d\sigma}{d\mathcal{P}} = \mathcal{C}[\mathcal{F}^q D_1] + \lambda D_\ell \mathcal{C}[\mathcal{G}^q D_1] + \varepsilon \mathcal{C} \left[\frac{\mathcal{T}^{q,x} p_y + \mathcal{T}^{q,y} p_x}{M_h} H_1^\perp \right]. \quad (15.12)$$

The final numerator is $\text{Im}(\mathcal{T}_+^q p_+)$. This equation plus the full correlator already determines all 32 responses with no guessed azimuthal sign. A chiral-odd h term requires the chiral-odd Collins function; an unpolarized D_1 alone cannot analyze it. At this order $F_{UU,L} = 0$ and the twist-three photon interference terms are outside the stated approximation.

15.5 Performing the hard trace one polarization channel at a time

The handbag contraction can be written with its Dirac indices exposed:

$$\Phi_{ab}(\gamma^\mu)_{bc} \Delta_{cd}(\gamma^\nu)_{da} = \text{Tr}_D(\Phi \gamma^\mu \Delta \gamma^\nu). \quad (15.13)$$

Substitute the reconstruction matrices rather than inserting a list of remembered angular functions. Linearity means it is sufficient to compute a small table of basis traces. The number–number entry is

$$\frac{1}{4} \text{Tr}_D(\gamma^- \gamma^i \gamma^+ \gamma^j) = \delta^{ij}, \quad i, j = x, y. \quad (15.14)$$

The factor $1/4$ is the product of the two reconstruction factors $1/2$. The trace identity $\text{Tr}(\gamma^\alpha \gamma^\beta \gamma^\gamma \gamma^\delta) = 4(g^{\alpha\beta} g^{\gamma\delta} - g^{\alpha\gamma} g^{\beta\delta} + g^{\alpha\delta} g^{\beta\gamma})$ and $g_T^{ij} = -\delta^{ij}$ determine the positive sign. The helicity–number entry is $i\epsilon_{ij}$ in the stated γ_5 convention. For the transverse-spin entries one obtains

$$-(\delta^{ia} \delta^{jb} + \delta^{ib} \delta^{ja} - \delta^{ij} \delta^{ab}) \mathcal{T}^a \mathcal{D}^b. \quad (15.15)$$

It is symmetric and traceless in the photon indices. This reproduces Eq. (15.8) and shows which photon polarization tensor analyzes each quark channel.

The normalized transverse lepton tensor can then be contracted without any further spin assumptions. Its trace part gives $\mathcal{F} D_1$. Its antisymmetric part gives $(-i\lambda D_\ell/2) \epsilon_{ij} (i\epsilon_{ij} \mathcal{G} D_1) = \lambda D_\ell \mathcal{G} D_1$. Its symmetric-traceless part gives $\varepsilon(-\mathcal{T}_x \mathcal{D}_x + \mathcal{T}_y \mathcal{D}_y)$. Since $\mathcal{D}_x = -p_y H_1^\perp/M_h$ and $\mathcal{D}_y = p_x H_1^\perp/M_h$, the last expression is

$$\frac{\varepsilon}{M_h} (\mathcal{T}_x p_y + \mathcal{T}_y p_x) H_1^\perp. \quad (15.16)$$

This derivation fixes the sign of the master Collins term. Reversing the azimuthal axis or the Collins-function convention changes signs coherently across the correlator, hard trace, and observable weights. Changing only a final sine function is not a convention conversion.

15.6 Deriving the convolution weights

Rotate integration variables into the observed-hadron azimuth:

$$u = \frac{k_x + ik_y}{M_A} e^{-i\phi_h} = a + ib, \quad v = \frac{p_x + ip_y}{M_h} e^{-i\phi_h} = c + id. \quad (15.17)$$

Here $a = \hat{h} \cdot \mathbf{k}_T / M_A$, $b = \hat{h}_\perp \cdot \mathbf{k}_T / M_A$, $c = \hat{h} \cdot \mathbf{p}_T / M_h$, and $d = \hat{h}_\perp \cdot \mathbf{p}_T / M_h$. The momentum delta function and radial TMD/FF coefficients are unchanged under $(b, d) \mapsto (-b, -d)$. An imaginary reduced weight is odd under this reflection and integrates to zero. All surviving weights can therefore be written as real parts:

$$\begin{aligned} w_m &= \text{Re}(u^m), \quad m = 0, 1, 2, 3, \\ b_0 &= \text{Re}(uv), \\ \ell_m &= \text{Re}[(u^*)^{m-1}v], \quad m = 1, 2, 3, \\ r_m &= \frac{1}{2} \text{Re}[u^{m+1}v], \quad m = 1, 2, 3. \end{aligned} \quad (15.18)$$

The factor $1/2$ in r_m comes from the STF contraction in Eq. (10.3), not from a measured asymmetry normalization. For direct implementation these are

$$\begin{aligned} w_0 &= 1, \quad w_1 = a, \quad w_2 = a^2 - b^2, \quad w_3 = a^3 - 3ab^2, \\ b_0 &= ac - bd, \quad \ell_1 = c, \quad \ell_2 = ac + bd, \quad \ell_3 = (a^2 - b^2)c + 2abd, \\ r_1 &= \frac{1}{2}[(a^2 - b^2)c - 2abd], \\ r_2 &= \frac{1}{2}[(a^3 - 3ab^2)c - (3a^2b - b^3)d], \\ r_3 &= \frac{1}{2}[(a^4 - 6a^2b^2 + b^4)c - (4a^3b - 4ab^3)d]. \end{aligned} \quad (15.19)$$

For example, $b_0 = [2(\hat{h} \cdot \mathbf{k}_T)(\hat{h} \cdot \mathbf{p}_T) - \mathbf{k}_T \cdot \mathbf{p}_T] / (M_A M_h)$, and $\ell_2 = \mathbf{k}_T \cdot \mathbf{p}_T / (M_A M_h)$.

The angular arguments generated by a target component (K, m) are

$$\delta_m = m(\phi_h - \phi_{Km}), \quad \theta_m^- = (2 - m)\phi_h + m\phi_{Km}, \quad \theta_m^+ = (m + 2)\phi_h - m\phi_{Km}. \quad (15.20)$$

For $m = 0$, the only Collins angle is $2\phi_h$. The low-rank vector branch in Eq. (10.3) gives θ_m^- and ℓ_m ; the high-rank branch gives θ_m^+ and r_m . Odd K gives sines because the master trace takes an imaginary part. Even K adds the dual $-i$, giving *minus cosines* instead.

15.7 Lepton factors, phase space, and angular-moment normalization

Write $A(y) = 1 - y + y^2/2$. For $0 < y < 1$ the leading-power polarization parameters obey

$$\varepsilon = \frac{1 - y}{A(y)}, \quad D_\ell = \frac{y(2 - y)}{2A(y)}, \quad \varepsilon^2 + D_\ell^2 = 1. \quad (15.21)$$

These factors belong to the hard lepton analyzer, not to the definitions of target TMDs. A partial incident beam polarization multiplies the helicity term by its calibrated polarization fraction. No beam polarization is required for the number or Collins terms in this leading-order example.

The phase-space convention also matters for absolute normalization. For a target-rest-frame massless lepton with fixed incident energy, $d^3l'/(2E') = E' dE' d\cos\theta_\ell d\psi/2$. Using $y = 1 - E'/E$ and $Q^2 = 2EE'(1 - \cos\theta_\ell)$ gives

$$\frac{d^3l'}{2E'} = \frac{1}{4} dy dQ^2 d\psi. \quad (15.22)$$

For the detected hadron at fixed transverse momentum in the leading- power current region, $d^3P_h/(2E_h) = dz dP_{hT}^2 d\phi_h/(4z)$. Combining the flux, one-photon propagator, chosen hadronic-tensor normalization, and the Jacobian from Q^2 to x_{Bj} gives the prefactor in Eq. (15.11), as in

the standard SIDIS normalization [5]. Integrating the overall lepton-plane angle ψ supplies a factor 2π only when the target/acceptance configuration makes the integrand independent of that angle. It must not be dropped silently.

A structure-function coefficient and a normalized experimental angular moment are also different. For a uniformly accepted distribution $\sigma(\phi) = \sigma_0[1 + a \sin(n\phi)]$, $n > 0$,

$$\langle \sin(n\phi) \rangle = \frac{\int d\phi \sin(n\phi) \sigma(\phi)}{\int d\phi \sigma(\phi)} = \frac{a}{2}. \quad (15.23)$$

Thus the Fourier amplitude is twice the raw normalized sine moment under these conditions. Target amplitudes, beam factors, multiple harmonics, and nonuniform acceptance belong in the actual extraction. The table below gives structure functions before dividing by an unpolarized cross section or applying this moment normalization.

15.8 Complete catalogue of 32 tree-level SIDIS structure functions

In Table 3, a row contributes

$$s_{Km} d(y, \lambda) [\text{angular factor}] [\text{SF}] \quad \text{to} \quad \mathcal{N}^{-1} d\sigma/d\mathcal{P}, \quad (15.24)$$

where $s_{00} = 1$, $s_{K0} = A_{K0}$, and $s_{Km} = s_{Km}$ for $m > 0$. The table's scalar functions do not contain this target amplitude or the displayed depolarization factor. A label F^U , F^L , or F^T is a *structure function*, whereas the lowercase letter inside its convolution is a TMD. We abbreviate $H = H_1^\perp$ only in the table. The angles $\delta_m, \theta_m^\pm$ use that row's (K, m) .

Table 3: All 32 independent leading-twist quark SIDIS responses in the native axes of this note. Target amplitudes multiply rows according to Eq. (15.24). Every convolution includes flavor charges and the momentum delta function in Eq. (15.7).

(K, m)	label	$d(y, \lambda)$	angular factor	SF coefficient
$K = 0$: unpolarized target (2 responses)				
(0, 0)	F_{00}^U	1	1	$\mathcal{C}[f_{00}^q D_1]$
(0, 0)	$F_{00}^{T[1]}$	ε	$\cos 2\phi_h$	$-\mathcal{C}[b_0 h_{00}^{q[1]} H]$
$K = 1$: vector target (6 responses)				
(1, 0)	F_{10}^L	λD_ℓ	1	$\mathcal{C}[g_{10}^q D_1]$
(1, 0)	$F_{10}^{T[1]}$	ε	$\sin 2\phi_h$	$\mathcal{C}[b_0 h_{10}^{q[1]} H]$
(1, 1)	F_{11}^U	1	$\sin \delta_1$	$\mathcal{C}[w_1 f_{11}^q D_1]$
(1, 1)	F_{11}^L	λD_ℓ	$\cos \delta_1$	$\mathcal{C}[w_1 g_{11}^q D_1]$
(1, 1)	$F_{11}^{T[0]}$	ε	$\sin \theta_1^-$	$\mathcal{C}[\ell_1 h_{11}^{q[0]} H]$
(1, 1)	$F_{11}^{T[2]}$	ε	$\sin \theta_1^+$	$\mathcal{C}[r_1 h_{11}^{q[2]} H]$
$K = 2$: quadrupole target (10 responses)				
(2, 0)	F_{20}^U	1	1	$\mathcal{C}[f_{20}^q D_1]$
(2, 0)	$F_{20}^{T[1]}$	ε	$\cos 2\phi_h$	$-\mathcal{C}[b_0 h_{20}^{q[1]} H]$
(2, 1)	F_{21}^U	1	$\cos \delta_1$	$\mathcal{C}[w_1 f_{21}^q D_1]$
(2, 1)	F_{21}^L	λD_ℓ	$\sin \delta_1$	$\mathcal{C}[w_1 g_{21}^q D_1]$
(2, 1)	$F_{21}^{T[0]}$	ε	$\cos \theta_1^-$	$-\mathcal{C}[\ell_1 h_{21}^{q[0]} H]$
(2, 1)	$F_{21}^{T[2]}$	ε	$\cos \theta_1^+$	$-\mathcal{C}[r_1 h_{21}^{q[2]} H]$
(2, 2)	F_{22}^U	1	$\cos \delta_2$	$\mathcal{C}[w_2 f_{22}^q D_1]$
(2, 2)	F_{22}^L	λD_ℓ	$\sin \delta_2$	$\mathcal{C}[w_2 g_{22}^q D_1]$

(K, m)	label	$d(y, \lambda)$	angular factor	SF coefficient
$(2, 2)$	$F_{22}^{T[1]}$	ε	$\cos \theta_2^-$	$-\mathcal{C}[\ell_2 h_{22}^{q[1]} H]$
$(2, 2)$	$F_{22}^{T[3]}$	ε	$\cos \theta_2^+$	$-\mathcal{C}[r_2 h_{22}^{q[3]} H]$
$K = 3$: octupole target (14 responses)				
$(3, 0)$	F_{30}^L	λD_ℓ	1	$\mathcal{C}[g_{30}^q D_1]$
$(3, 0)$	$F_{30}^{T[1]}$	ε	$\sin 2\phi_h$	$\mathcal{C}[b_0 h_{30}^{q[1]} H]$
$(3, 1)$	F_{31}^U	1	$\sin \delta_1$	$\mathcal{C}[w_1 f_{31}^q D_1]$
$(3, 1)$	F_{31}^L	λD_ℓ	$\cos \delta_1$	$\mathcal{C}[w_1 g_{31}^q D_1]$
$(3, 1)$	$F_{31}^{T[0]}$	ε	$\sin \theta_1^-$	$\mathcal{C}[\ell_1 h_{31}^{q[0]} H]$
$(3, 1)$	$F_{31}^{T[2]}$	ε	$\sin \theta_1^+$	$\mathcal{C}[r_1 h_{31}^{q[2]} H]$
$(3, 2)$	F_{32}^U	1	$\sin \delta_2$	$\mathcal{C}[w_2 f_{32}^q D_1]$
$(3, 2)$	F_{32}^L	λD_ℓ	$\cos \delta_2$	$\mathcal{C}[w_2 g_{32}^q D_1]$
$(3, 2)$	$F_{32}^{T[1]}$	ε	$\sin \theta_2^-$	$\mathcal{C}[\ell_2 h_{32}^{q[1]} H]$
$(3, 2)$	$F_{32}^{T[3]}$	ε	$\sin \theta_2^+$	$\mathcal{C}[r_2 h_{32}^{q[3]} H]$
$(3, 3)$	F_{33}^U	1	$\sin \delta_3$	$\mathcal{C}[w_3 f_{33}^q D_1]$
$(3, 3)$	F_{33}^L	λD_ℓ	$\cos \delta_3$	$\mathcal{C}[w_3 g_{33}^q D_1]$
$(3, 3)$	$F_{33}^{T[2]}$	ε	$\sin \theta_3^-$	$\mathcal{C}[\ell_3 h_{33}^{q[2]} H]$
$(3, 3)$	$F_{33}^{T[4]}$	ε	$\sin \theta_3^+$	$\mathcal{C}[r_3 h_{33}^{q[4]} H]$

15.9 Two worked octupole examples

Highest quark Collins harmonic. For $h_{33}^{q[4]}$, Eq. (10.3) gives $\mathcal{T}_+ = \frac{1}{2} A_{33,+}^* \kappa^4 h_{33}^{q[4]}$. Multiplying by p_+/M_h and extracting the imaginary part gives

$$\frac{s_{33}}{2} \text{Im} \left[e^{i(5\phi_h - 3\phi_{33})} u^4 v \right] h_{33}^{q[4]}. \quad (15.25)$$

Reflection symmetry removes $\text{Im}(u^4 v)$ from the convolution, leaving

$$\varepsilon s_{33} \sin(5\phi_h - 3\phi_{33}) \mathcal{C}[r_3 h_{33}^{q[4]} H_1^\perp]. \quad (15.26)$$

The fifth harmonic comes from rank four in the target-side momentum and rank one in Collins fragmentation. It is still a leading-power TMD contribution.

A response with no hadron azimuth in its angular factor. For $h_{32}^{q[1]}$, $\mathcal{T}_+ = A_{32,+} \kappa^* h_{32}^{q[1]}$. The same operation gives

$$\varepsilon s_{32} \sin(2\phi_{32}) \mathcal{C}[\ell_2 h_{32}^{q[1]} H_1^\perp]. \quad (15.27)$$

The dependence on the target tensor's orientation relative to the lepton plane is physical even though ϕ_h has canceled from the trigonometric factor. The convolution remains differential in P_{hT}^2 . Integrating over all transverse momenta is a different operation and cannot be inferred merely from this absence of ϕ_h .

15.10 Independence of structure functions and experimental sensitivity

Thirty-two responses are independent in the joint space of target multipoles, beam helicity, and azimuths. They are not 32 distinct Fourier frequencies of a single data set with one fixed target state. For example, g_{10}^q and g_{30}^q share an azimuth-independent beam-helicity response and require independent variation of S_L and S_{LL} . The $K = 1$ and $K = 3$ $m = 1$ channels likewise share

several angular forms. A third harmonic alone cannot distinguish ordinary pretzelosity from a TTT octupole response.

The hard kernel contains no independent leading-order gluon TMD for this single-hadron channel: the photon strikes a quark. Gluons can enter perturbative matching and evolution, or a different hard process. Replacing f^q by f^g and e_q^2 by a guessed color factor in Table 3 is not a derivation.

15.11 A complete analytic convolution example

A solvable radial example is useful for learning the delta function, normalizations, and harmonic weights. The following Gaussians are *educational test functions*; they are not fitted distributions, not a lithium-7 prediction, and not assumed to satisfy all QCD evolution, positivity, or moment conditions. Let $a, b > 0$ have units of momentum squared and, for one flavor, take

$$f(\mathbf{k}_T^2) = \frac{F_0}{\pi a} e^{-\mathbf{k}_T^2/a}, \quad D(\mathbf{p}_T^2) = \frac{D_0}{z^2 \pi b} e^{-\mathbf{p}_T^2/b}. \quad (15.28)$$

Then $\int d^2 k_T f = F_0$ and $z^2 \int d^2 p_T D = D_0$ in this finite example. Set $\mathbf{q}_h = \mathbf{P}_{hT}/z$, $q_h = |\mathbf{q}_h|$, and $\Sigma = a + b$. Use the delta function to set $\mathbf{p}_T = \mathbf{k}_T - \mathbf{q}_h$ and complete the square:

$$\frac{\mathbf{k}_T^2}{a} + \frac{(\mathbf{k}_T - \mathbf{q}_h)^2}{b} = \frac{\Sigma}{ab} \left(\mathbf{k}_T - \frac{a}{\Sigma} \mathbf{q}_h \right)^2 + \frac{q_h^2}{\Sigma}. \quad (15.29)$$

The elementary two-dimensional Gaussian integral gives

$$\mathcal{C}[fD] = \frac{x e_q^2 F_0 D_0}{z^2 \pi \Sigma} e^{-q_h^2/\Sigma} \equiv \mathcal{C}_0. \quad (15.30)$$

Flavor sums can be restored after evaluating the individual terms. The observed width in P_{hT} is $z^2 \Sigma$, not Σ , because $q_h = P_{hT}/z$ in this momentum convention.

To compute a weighted convolution, regard the completed square as a normalized conditional Gaussian. In the frame with $\hat{h} = \hat{x}$,

$$\langle k_x \rangle = \mu_k = \frac{a q_h}{\Sigma}, \quad \langle p_x \rangle = \mu_p = -\frac{b q_h}{\Sigma}, \quad \langle k_y \rangle = \langle p_y \rangle = 0. \quad (15.31)$$

The same fluctuation $\delta \mathbf{k}$ appears in $\mathbf{k}_T$ and $\mathbf{p}_T$; its component variances are $ab/(2\Sigma)$. In complex notation $\delta = \delta k_x + i \delta k_y$,

$$\langle \delta^n \rangle = 0 \quad (n > 0), \quad \langle \delta^* \delta \rangle = \frac{ab}{\Sigma}. \quad (15.32)$$

The first identity follows from rotation symmetry of the centered Gaussian. It is the reason pure STF moments depend only on the conditional mean, while mixed conjugate products also contain the variance.

For the scalar harmonic weights this gives

$$\frac{\mathcal{C}[w_m f D]}{\mathcal{C}_0} = \left(\frac{a q_h}{M_A \Sigma} \right)^m, \quad m = 0, 1, 2, 3. \quad (15.33)$$

For the rank-two Collins weight and the low- $m = 2$ weight,

$$\frac{\mathcal{C}[b_0 f D]}{\mathcal{C}_0} = -\frac{ab q_h^2}{M_A M_h \Sigma^2}, \quad (15.34)$$

$$\frac{\mathcal{C}[\ell_2 f D]}{\mathcal{C}_0} = \frac{ab}{M_A M_h \Sigma} \left(1 - \frac{q_h^2}{\Sigma} \right). \quad (15.35)$$

For $b_0 = \text{Re}(k_+ p_+)/ (M_A M_h)$ there is no centered second complex moment, so only $\mu_k \mu_p$ survives. For $\ell_2 = \text{Re}(k_+^* p_+)/ (M_A M_h)$ the nonzero $\langle \delta^* \delta \rangle$ adds the first term in parentheses. This explicitly explains how a response with no ϕ_h in its angular factor can still have nontrivial radial dependence and a zero as a function of P_{hT} in a specified model.

The high branches are equally simple:

$$\frac{\mathcal{C}[r_m f D]}{\mathcal{C}_0} = -\frac{a^{m+1} b q_h^{m+2}}{2M_A^{m+1} M_h \Sigma^{m+2}}, \quad m = 1, 2, 3. \quad (15.36)$$

For the low branches, $\mathcal{C}[\ell_1 f D]/\mathcal{C}_0 = \mu_p/M_h$, and for $m = 2, 3$,

$$\frac{\mathcal{C}[\ell_m f D]}{\mathcal{C}_0} = \frac{\mu_k^{m-1} \mu_p + (m-1) \mu_k^{m-2} a b / \Sigma}{M_A^{m-1} M_h}. \quad (15.37)$$

Inserting a chosen radial h coefficient and Collins function of the form in Eq. (15.28) makes these identities a direct check of the corresponding SIDIS row. The amplitudes and widths then belong to those test functions, not automatically to f_1, D_1 .

Several checks are immediate. The factor z^{-2} comes from the fragmentation convention. The scalar harmonics vanish as q_h^m at the origin. The highest Collins branch vanishes as q_h^{m+2} . Integrating the ℓ_2 result over all $\mathbf{q}_h$ gives zero in this finite Gaussian example, even though the angular factor itself has no ϕ_h dependence. This last result follows from integrating the full radial convolution and is not a general license to ignore the UV qualifications of QCD TMD integrals.

16 Drell–Yan octupole observables from the same quark correlator

Provenance and task. The polarized electromagnetic Drell–Yan hard contraction and angular analysis are established [21]; tensor-polarized targets have lower-spin precedents [36]. Here the complete incoming rank-three distribution is inserted into that contraction. The purpose is to derive, rather than obtain by renaming a SIDIS angle, the three number, four helicity, and seven transverse-spin octupole channels.

16.1 Specify both incoming hadrons and the negative-lepton angles

Consider

$$A(P_A, \rho) + B(P_B) \longrightarrow \ell^-(l_-) + \ell^+(l_+) + X, \quad q = l_- + l_+, \quad Q^2 = q^2 > 0, \quad q_T \ll Q. \quad (16.1)$$

The hard channel is $q\bar{q} \rightarrow \gamma^* \rightarrow \ell^- \ell^+$. Ordinary continuum DY at this Born order does not directly analyze an initial gluon TMD. Take A along $+z$ and B along $-z$; define

$$x_A = \frac{Q}{\sqrt{s}} e^{\mathcal{Y}}, \quad x_B = \frac{Q}{\sqrt{s}} e^{-\mathcal{Y}}, \quad s = (P_A + P_B)^2. \quad (16.2)$$

The subscript B here labels the second hadron and is distinct from x_{Bj} in SIDIS. The hadron masses remain in the TMD tensor normalizations, even though their hard-kinematic power corrections are neglected. Replacing Q by $\sqrt{Q^2 + q_T^2}$ changes this relation only beyond the retained hard-power accuracy.

Write $\mathbf{q}_T = q_T(\cos \varphi, \sin \varphi)$ and $\hat{h} = \mathbf{q}_T/q_T$. In the Collins–Soper dilepton rest frame choose Z along $\hat{P}_A^* - \hat{P}_B^*$, X opposite $\hat{P}_A^* + \hat{P}_B^*$, and $Y = Z \times X$. Stars denote the unit hadron directions after boosting. The angles (θ_ℓ, ϕ_ℓ) belong to the negative lepton. At leading power the transverse axes are the recoil-aligned $(\hat{h}, \hat{h}_\perp)$ axes of the hadron frame. The recoil azimuth is undefined at $q_T = 0$; Cartesian forward expressions may have a limit there, but an angular projector requiring $\hat{h}$ must not assign a fictitious direction.

16.2 Derive the distribution–distribution trace before assigning PDFs

Replace the fragmentation correlator by the annihilation-antiquark correlator $\bar{\Phi}_B$. Its leading projections use γ^- , $\gamma^- \gamma_5$, and $i\sigma^{i-} \gamma_5$, and its reconstruction uses the opposite light-front directions to Φ_A . Both distribution staples are $[-]$. Direct multiplication gives the symmetric transverse trace

$$\mathcal{H}_{\text{DY}}^{(ij)} = \delta^{ij} (\mathcal{F}_A \bar{\mathcal{F}}_B + \mathcal{G}_A \bar{\mathcal{G}}_B) - \mathcal{T}_A^i \bar{\mathcal{T}}_B^j - \mathcal{T}_A^j \bar{\mathcal{T}}_B^i + \delta^{ij} \mathcal{T}_A \cdot \bar{\mathcal{T}}_B. \quad (16.3)$$

The symmetric projection is required because the final lepton spins are summed. A number–helicity product is antisymmetric in photon indices and therefore drops out. There are four distribution projection components on each side, so a direct implementation checks all sixteen pairs and all four transverse photon-index choices, including zeros.

The final-lepton spin sum yields

$$\frac{L_{ij}}{2Q^2} = \delta_{ij} - \sin^2 \theta_\ell e_i e_j, \quad \mathbf{e} = (\cos \phi_\ell, \sin \phi_\ell). \quad (16.4)$$

Contracting its isotropic part with the transverse-spin tensor gives zero, because that tensor is STF. Contracting the remaining part gives $\sin^2 \theta_\ell \Re[e^{-2i\phi_\ell} \mathcal{T}_{A,+} \bar{\mathcal{T}}_{B,+}]$. The result is

$$(1 + \cos^2 \theta_\ell) (\mathcal{F}_A \bar{\mathcal{F}}_B + \mathcal{G}_A \bar{\mathcal{G}}_B) + \sin^2 \theta_\ell \Re[e^{-2i\phi_\ell} \mathcal{T}_{A,+} \bar{\mathcal{T}}_{B,+}]. \quad (16.5)$$

The plus sign multiplying the *annihilation-correlator* $\bar{\mathcal{G}}_B$ is not yet the longitudinal analyzing sign written in terms of physical antiquark helicity PDFs.

16.3 Convolution, color factor, and differential normalization

For physical scalar distributions define

$$\begin{aligned} \mathcal{C}_{\text{DY}}[wa_A b_B] &= \frac{1}{N_c} \sum_q e_q^2 \int d^2 k_{AT} d^2 k_{BT} \delta^{(2)}(\mathbf{q}_T - \mathbf{k}_{AT} - \mathbf{k}_{BT}) w \\ &\quad \times \left[a_A^{q[-]} b_B^{\bar{q}[-]} + a_A^{\bar{q}[-]} b_B^{q[-]} \right]. \end{aligned} \quad (16.6)$$

The arguments of the two factors are (x_A, k_{AT}^2) and (x_B, k_{BT}^2) ; their rapidity scales satisfy $\zeta_A \zeta_B = Q^4$ in the stated two-line scheme. There is no fragmentation function, no SIDIS factor x_A , and no fragmentation Jacobian in Eq. (16.6). The factor $1/N_c$ corresponds to color-summed distribution correlators. The two annihilation orderings must remain separately present when the beam and target or their quark and antiquark inputs are unequal.

With

$$d\mathcal{P}_{\text{DY}} = dQ^2 d\mathcal{Y} d^2 q_T d\cos \theta_\ell d\phi_\ell, \quad \mathcal{N}_{\text{DY}} = \frac{\alpha_{\text{em}}^2}{4sQ^2}, \quad (16.7)$$

the cross section is $\mathcal{N}_{\text{DY}}$ times the flavor convolution of Eq. (16.5). As a normalization check, $\int d\Omega_\ell (1 + \cos^2 \theta_\ell) = 16\pi/3$. With $N_c = 3$ this produces the collinear Born coefficient $4\pi\alpha_{\text{em}}^2/(9sQ^2)$ after the transverse integral in a regulated Born example. This arithmetic checks flux, color, and angular normalization. It is not an identification of an unregulated subtracted TMD integral with a renormalized PDF.

16.4 Three number responses and their normalized moments

For an unpolarized partner B , $\bar{\mathcal{G}}_B = 0$. Its Boer–Mulders projection can remain nonzero; averaging over ϕ_ℓ removes its transverse-spin contribution. Rotate both integration momenta into the recoil direction:

$$u_A = \frac{k_{Ax} + ik_{Ay}}{M_A} e^{-i\varphi}, \quad v_B = \frac{k_{Bx} + ik_{By}}{M_B} e^{-i\varphi}. \quad (16.8)$$

The delta function and radial coefficients are invariant when both perpendicular components change sign. Thus the reduced weights are $w_m = \Re(u_A^m)$, and

$$F_{3m}^{\text{DY}} = \mathcal{C}_{\text{DY}}[w_m f_{3m}^A f_1^B], \quad m = 1, 2, 3, \quad F_{UU}^{\text{DY}} = \mathcal{C}_{\text{DY}}[f_{00}^A f_1^B]. \quad (16.9)$$

For opposite octupole-only preparations with $A_{3m,+} = \eta O \alpha_m e^{im\psi}$, Eq. (16.9) gives the dilepton-azimuth-averaged rate

$$\bar{\sigma}_\eta = \mathcal{N}_{\text{DY}}(1 + \cos^2 \theta_\ell) \left[F_{UU}^{\text{DY}} + \eta O \sum_{m=1}^3 \alpha_m \sin m(\varphi - \psi) F_{3m}^{\text{DY}} \right]. \quad (16.10)$$

The physical states and signed α_m are constructed in Secs. 18 and 19. Using their actual moments, rather than declaring one matrix element to be an octupole, removes the lower-rank responses. A full two-angle moment is

$$\frac{2}{O} \frac{\int d\varphi d\psi \sin m(\varphi - \psi) (\bar{\sigma}_+ - \bar{\sigma}_-)}{\int d\varphi d\psi (\bar{\sigma}_+ + \bar{\sigma}_-)} = \alpha_m \frac{F_{3m}^{\text{DY}}}{F_{UU}^{\text{DY}}}. \quad (16.11)$$

The factor two removes the mean of a squared sine, $1/2$. A zero α_m means that this preparation does not identify that coefficient; a zero denominator makes this normalized moment undefined.

16.5 Four helicity responses and the physical antiquark sign

A longitudinally polarized spin-1/2 partner supplies a helicity analyzer. Let λ_B be its polarization along its own momentum. The annihilation convention gives $\bar{\mathcal{G}}_B = -\lambda_B g_{1L}^{\bar{q}/B}$. This follows from the charge-conjugated field projection and the opposite beam direction, not from changing a Wilson staple. The other annihilation ordering gives the same negative physical product, with the independent target antiquark coefficient. Define

$$G_{3m}^{\text{DY}} = \mathcal{C}_{\text{DY}}[\Re(u_A^m) g_{3m}^A g_{1L}^B], \quad m = 0, 1, 2, 3. \quad (16.12)$$

Its rank-three rate is

$$-\lambda_B \eta O \mathcal{N}_{\text{DY}}(1 + \cos^2 \theta_\ell) \left[\alpha_0 G_{30}^{\text{DY}} + \sum_{m=1}^3 \alpha_m \cos m(\varphi - \psi) G_{3m}^{\text{DY}} \right]. \quad (16.13)$$

For a longitudinal octupole and $\lambda_B = \pm 1$,

$$\frac{1}{4} \sum_{\lambda_B, \eta = \pm 1} \lambda_B \eta \bar{\sigma}_{\lambda_B, \eta} = -O \mathcal{N}_{\text{DY}}(1 + \cos^2 \theta_\ell) G_{30}^{\text{DY}}. \quad (16.14)$$

The negative longitudinal analyzing power and the link-even relation $g_{30}^{q[-]} = g_{30}^{q[+]}$ are compatible: they are statements about different factors. A direct massless $q\bar{q} \rightarrow \ell^- \ell^+$ current calculation tests the former without redefining the target function.

16.6 Seven transverse-spin responses with a Boer–Mulders partner

In common transverse axes the unpolarized, minus-moving partner has $\bar{\mathcal{T}}_{B,+} = i v_B h_1^{\perp \bar{q}/B}$. The epsilon sign relative to a plus-moving hadron follows from interchanging the light-front directions. Multiplication of this analyzer by the target branches gives

$$H_{30}^{[1],\text{DY}} = \mathcal{C}_{\text{DY}}[\Re(u_A v_B) h_{30}^{A[1]} h_1^{\perp B}], \quad (16.15)$$

$$H_{3m}^{[m-1],\text{DY}} = \mathcal{C}_{\text{DY}}[\Re((u_A^*)^{m-1} v_B) h_{3m}^{A[m-1]} h_1^{\perp B}], \quad (16.16)$$

$$H_{3m}^{[m+1],\text{DY}} = \mathcal{C}_{\text{DY}}[\tfrac{1}{2} \Re(u_A^{m+1} v_B) h_{3m}^{A[m+1]} h_1^{\perp B}], \quad m = 1, 2, 3. \quad (16.17)$$

All functions inside each convolution have past-pointing distribution links. The brackets on H and h record the target orbital rank, not those links. Put $\Delta_m = m(\varphi - \phi_{3m})$. The measured contribution is

$$\begin{aligned} \frac{1}{\mathcal{N}_{\text{DY}}} \frac{d\sigma}{d\mathcal{P}_{\text{DY}}} \Big|_{K=3,h} &= \sin^2 \theta_\ell \left\{ S_{LLL} \sin(2\phi_\ell) H_{30}^{[1],\text{DY}} \right. \\ &\quad + \sum_{m=1}^3 s_{3m} \sin(2\phi_\ell + \Delta_m) H_{3m}^{[m-1],\text{DY}} \\ &\quad \left. + \sum_{m=1}^3 s_{3m} \sin(2\phi_\ell - \Delta_m) H_{3m}^{[m+1],\text{DY}} \right\}. \end{aligned} \quad (16.18)$$

For example, the high TTT target branch is $\frac{1}{2}s_{33}e^{i\Delta_3}u_A^4h_{33}^{A[4]}$. Multiplying by $iv_B e^{-2i\phi_\ell}$ and taking the real part produces $\sin(2\phi_\ell - \Delta_3)$ after the odd reduced weight has integrated out. This two-line derivation fixes both the phase and the factor $1/2$. A guessed replacement of the SIDIS Collins angle would not fix them.

Together Eqs. (16.9), (16.12), and (16.17) account for all fourteen octupole channels. Their coefficients are not fourteen directly measured densities. Flavor mixtures, partner inputs, target calibration, and acceptance remain in the extraction. The target h functions are link even, but the Boer–Mulders partner is link odd and is not a Collins fragmentation function. The repository’s separate row and convolution comparisons make these distinctions executable.

16.7 A predictive cross-process comparison

Insert $f_{3m}^{q[-]} = -f_{3m}^{q[+]}$ in Eq. (16.6); retain its DY momentum delta function and partner distributions. The result is a prediction using the target function constrained in SIDIS, not the negative of its SIDIS structure function $\mathcal{C}[w_m f_{3m}^{q[+]} D_1]$. Even with exact sign reversal at the target operator level, different companion functions can give different magnitudes or flavor-weighted signs for the observed asymmetries. Independent synthetic convolution comparisons in Sec. 21.4 illustrate this distinction without claiming an empirical test. Beyond the displayed Born kernels, a joint analysis needs the process-specific hard factors, evolution, recoil convolution, matching, and power corrections.

17 Gluon-sensitive structure functions and the Born hard response

Provenance and task. The Born subprocess $\gamma^* g \rightarrow Q\bar{Q}$ and its use as an analyzer of gluon polarization are established [22]. The present construction inserts the full spin-3/2 target correlator into a four-component hard-polarization response. The new target multipoles belong to the long-distance matrix element; the familiar two Feynman diagrams are not new interactions. The task is to show exactly how to build the hard matrix, verify it, and combine it with the target basis for a specified small-imbalance measurement.

17.1 Separating the target density from the hard polarization analyzer

The same target correlator can be interrogated by different final states. The hard process decides which components of the gluon polarization matrix it can see. A useful way to preserve this separation is to retain the incoming gluon’s two Cartesian polarization amplitudes until the very last contraction, rather than immediately averaging its spin. Then the hard process supplies a 2×2 Hermitian matrix and the target supplies another, with their index ordering fixed by the field operator definition.

This separation is particularly important for a high-spin target. Adding more target multipoles does not automatically create additional parton analyzers. For example, a beam-averaged hard process with no antisymmetric response cannot measure an octupole gluon helicity coefficient, even though the coefficient is allowed in the target. Likewise, integrating over a final-state angle can remove linear- polarization sensitivity. Such zeros are properties of the chosen measurement and should not be interpreted as missing target TMDs.

17.2 Hard-process dependence of a gluon measurement

The gluon correlator is an operator decomposition. It is not by itself a cross section. A measurement specifies the link/color class, kinematic fractions, hard polarization analyzer, final-state measurement, and any soft/jet factors. Heavy-quark or dijet production in lepton-hadron scattering provides a standard gluon-sensitive example [22]. The following construction gives all target-spin responses once that analyzer is supplied, without assuming that one observable accesses every coefficient.

Use the Cartesian Gram correlator

$$D_g = \Gamma^g/x = \frac{1}{2} \begin{pmatrix} \mathcal{F}^g + \mathcal{H}^{g,xx} & \mathcal{H}^{g,xy} + i\mathcal{G}^g \\ \mathcal{H}^{g,xy} - i\mathcal{G}^g & \mathcal{F}^g - \mathcal{H}^{g,xx} \end{pmatrix}. \quad (17.1)$$

The field ordering in Eq. (5.7) is conjugate-amplitude index first. For amplitudes $\mathcal{M}_i$ with incoming Cartesian gluon polarization i , use the matching hard matrix $B_{ij} = \sum_X \mathcal{M}_i \mathcal{M}_j^*$. The contraction is

$$\mathcal{W}_g = \text{Tr}_2(BD_g) = \sum_{ij} B_{ij}(D_g)_{ji}. \quad (17.2)$$

Writing it as a matrix trace fixes the index ordering and the circular polarization sign. For example, a physical polarization vector e_i has Gram matrix $e_i^* e_j$, and Eq. (17.2) returns $|\sum_i \mathcal{M}_i e_i|^2$.

Every Hermitian hard matrix has the decomposition

$$B = b_U I + b_G iE + b_C \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + b_S \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (17.3)$$

where

$$b_U = \frac{1}{2}(B_{xx} + B_{yy}), \quad b_G = \text{Im } B_{xy}, \quad b_C = \frac{1}{2}(B_{xx} - B_{yy}), \quad b_S = \text{Re } B_{xy}. \quad (17.4)$$

Consequently

$$\mathcal{W}_g = b_U \mathcal{F}^g + b_G \mathcal{G}^g + b_C \mathcal{H}^{g,xx} + b_S \mathcal{H}^{g,xy} = b_U \mathcal{F}^g + b_G \mathcal{G}^g + \text{Re}(b_+^* \mathcal{H}_+^g), \quad (17.5)$$

with $b_+ = b_C + ib_S$. Equations (8.1), (8.2), and (10.6) now supply *every one of the 32 terms*. The b 's are process-dependent hard coefficients, not new target TMDs. If $b_G = 0$ in the selected measurement, that measurement has no g_{Km}^g sensitivity regardless of how many such functions exist.

17.3 Deriving the Stokes contraction and its sign conventions

Let D be a conjugate-amplitude-first gluon polarization matrix. For a pure incoming polarization e_i , $D_{ij} = e_i^* e_j$. With $B_{ij} = \mathcal{M}_i \mathcal{M}_j^*$,

$$\text{Tr}(BD) = \sum_{ij} \mathcal{M}_i \mathcal{M}_j^* e_j^* e_i = \left| \sum_i \mathcal{M}_i e_i \right|^2. \quad (17.6)$$

This explicitly verifies the contraction. Using $\sum_{ij} B_{ij} D_{ij}$ instead would transpose one polarization matrix and change the circular term. A basis change is allowed, but it must act on both matrices with this ordering respected.

The four Hermitian basis matrices are I , iE , σ_3 , and σ_1 . Since $iE = -\sigma_2$ in the usual Pauli convention, the sign of the circular coefficient differs from a formula written directly as a coefficient of σ_2 . These basis matrices have pairwise trace norm two, so multiplying Eq. (17.3) by the corresponding basis matrix and tracing extracts b_U, b_G, b_C, b_S . This is the same Gram-inversion principle used for the Dirac projections. It leads to Eq. (17.5) without an additional factor of two because D_g already contains $1/2$.

For a physical Born amplitude, B is positive semidefinite. Its two eigenvalues are

$$\lambda_{\pm}(B) = b_U \pm \sqrt{b_G^2 + b_C^2 + b_S^2}. \quad (17.7)$$

When $b_U > 0$, the dimensionless analyzing vector

$$\mathcal{A}_g = \frac{(b_G, b_C, b_S)}{b_U}, \quad |\mathcal{A}_g| \leq 1 \quad (17.8)$$

characterizes the hard process's selectivity. The bound follows from an amplitude Gram matrix; it does not assume that a soft-subtracted TMD is pointwise positive. If $b_U = 0$, positivity forces all three polarization coefficients to vanish as well.

Writing $b_C + ib_S = b_{\text{lin}} e^{2i\phi_B}$ turns the last two terms in the contraction into the real part of two rank-two complex tensors. For an even target rank the result is a cosine; for an odd target rank the required $-i$ dual changes it to a sine. This is the direct origin of the gluon angular-kernel table, including its half- and quarter-normalizations. The analyzer angle ϕ_B is computed from the hard matrix. It need not coincide with a heavy-quark or jet azimuth.

17.4 Complete angular kernels for the linear-polarization terms

At fixed hard kinematics put $b_+ = b_{\text{lin}} \exp(2i\phi_B)$ with $b_{\text{lin}} = |b_+|$. When $b_+ = 0$, the entire linear-polarization response vanishes and ϕ_B is irrelevant. Let $t = |\mathbf{k}_T|/M_A$. For a row in Table 4, the contribution is

$$b_{\text{lin}} s_{Km} c t^n h_{Km}^{g[n]} \begin{cases} \cos \Theta, & K \text{ even,} \\ \sin \Theta, & K \text{ odd.} \end{cases} \quad (17.9)$$

The odd- K sine follows from $\text{Re}[-i \exp(i\Theta)] = \sin \Theta$. For each row, allowed target ranks are $K = m, \dots, 3$, with $K = 0$ also allowed for $m = 0$. Thus the table generates $4 + 6 + 4 + 2 = 16$ helicity-flip coefficients, not just seven examples.

Table 4: All gluon linear-polarization angular kernels. Each row applies to every allowed $K \geq m$; its parity selects cosine or sine through Eq. (17.9). The scalar number/helicity kernels are already completely given by Eqs. (8.1)–(8.2).

m	n	c	Θ
0	2	1/2	$2\phi_k - 2\phi_B$
1	1	1/2	$\phi_k + \phi_{K1} - 2\phi_B$
1	3	1/4	$3\phi_k - \phi_{K1} - 2\phi_B$
2	0	1	$2\phi_{K2} - 2\phi_B$
2	4	1/4	$4\phi_k - 2\phi_{K2} - 2\phi_B$
3	1	1	$3\phi_{K3} - \phi_k - 2\phi_B$
3	5	1/4	$5\phi_k - 3\phi_{K3} - 2\phi_B$

For the eight f terms the weight is $b_U t^m$ times the cosine or sine in Eq. (4.16); for the eight g terms it is $b_G t^m$ with the opposite parity choice. These statements, together with the table, are the full gluon response catalogue. A hard analyzer may itself depend on the lepton–jet azimuth; expanding that dependence produces process-specific combinations of angles. It is not legitimate to identify ϕ_B with a jet angle without calculating B .

17.5 Physical electron helicity and ordered current tensors

The index order of a polarized current must be fixed before inserting a leptonic tensor into a squared amplitude. For the positive-energy incoming electron define $\lambda = \pm 1$ by

$$\text{diag}(\boldsymbol{\sigma} \cdot \hat{\mathbf{l}}, \boldsymbol{\sigma} \cdot \hat{\mathbf{l}}) u_\lambda(l) = \lambda u_\lambda(l), \quad j_\mu = \bar{u}_{\lambda'}(l') \gamma_\mu u_\lambda(l). \quad (17.10)$$

The label is twice the physical helicity. In the massless, positive-energy convention, $\gamma_5 u_\lambda = \lambda u_\lambda$. The plus-i tensor already used in Eq. (15.9) is *conjugate-current first*:

$$L_{\mu\nu} = \sum_{\lambda'} j_\mu^* j_\nu, \quad J_{\mu\nu}^\ell = \sum_{\lambda'} j_\mu j_\nu^* = L_{\nu\mu}. \quad (17.11)$$

With $\epsilon^{0123} = +1$, lowering all indices gives $\epsilon_{0123} = -1$. The four-dimensional gamma trace is

$$J_{\mu\nu}^\ell = 2 \left(l_\mu l'_\nu + l_\nu l'_\mu - g_{\mu\nu} l \cdot l' - i \lambda \epsilon_{\mu\nu\alpha\beta} l^\alpha l'^\beta \right). \quad (17.12)$$

For example, $l = (5, 0, 0, 5)$ and $l' = (5, 3, 0, 4)$ in GeV are on shell, with $(l - l')^2 = -10 \text{ GeV}^2$. A physical $\lambda = +1$ spinor gives $J_{xy}^\ell = -10i \text{ GeV}^2$ and $L_{xy} = +10i \text{ GeV}^2$. This example exposes a current-order interchange that an unpolarized test cannot detect.

Write the heavy-quark current as $Q_i^\mu = \bar{u}(p_1) V^{\mu i} v(p_2)$. Squaring the scalar amplitude $\mathcal{M}_i = j_\mu Q_i^\mu$ gives

$$B_{ij} = N_B \sum_{\mu\nu} J_{\mu\nu}^\ell \sum_{s_1 s_2} Q_i^\mu Q_j^{\nu*}, \quad N_B = \frac{e^4 e_Q^2 g_s^2 T_F}{Q^4}. \quad (17.13)$$

The heavy trace in Eq. (17.16) is amplitude first, so it requires $L_{\nu\mu}$, not $L_{\mu\nu}$. This Lorentz-index interchange is separate from the gluon polarization indices i, j and from the partial transpose used in an auxiliary target–parton coordinate map.

The SIDIS trace has the opposite hard-current order: $\text{Tr}(\Phi \gamma^\mu \Delta \gamma^\nu) = \text{Tr}(\Delta \gamma^\nu \Phi \gamma^\mu)$. Its contraction with the previously defined $L_{\mu\nu}$ is already consistent. Therefore no global reversal of the SIDIS beam helicity, no redefinition of a target helicity TMD, and no change of a Wilson-line relation is required. Independent current and spinor-projector comparisons test both orderings without choosing the benchmark sign in advance; their role is summarized in Sec. 21.6.

17.6 The two-diagram heavy-quark Born analyzer

Consider $\ell g \rightarrow \ell' Q \bar{Q}$, with heavy-quark mass m_Q , photon q , incoming collinear gluon k , and outgoing momenta p_1, p_2 . Strip couplings and color from the two QCD diagrams and define

$$V^{\mu\alpha} = \gamma^\mu \frac{\not{p}_1 - \not{q} + m_Q}{(p_1 - q)^2 - m_Q^2} \gamma^\alpha + \gamma^\alpha \frac{\not{p}_1 - \not{k} + m_Q}{(p_1 - k)^2 - m_Q^2} \gamma^\mu, \quad \bar{V} = \gamma^0 V^\dagger \gamma^0. \quad (17.14)$$

The amplitude is proportional to $\bar{u}(p_1) t^a V^{\mu\alpha} v(p_2)$ contracted with the lepton current and incoming gluon polarization. The two terms are both necessary. On shell,

$$\bar{u}(p_1) q_\mu V^{\mu\alpha} v(p_2) = 0, \quad \bar{u}(p_1) k_\alpha V^{\mu\alpha} v(p_2) = 0. \quad (17.15)$$

These Ward identities follow by inserting $\not{p}_1 u = m_Q u$, $\not{p}_2 v = -m_Q v$, and $p_1 + p_2 = q + k$ into the two diagrams. They are essential tests of an implementation.

After summing final spins and colors and averaging the incoming color, the hard matrix in the ordering of Eq. (17.2) is

$$B_{ij} = \frac{e^4 e_Q^2 g_s^2 T_F}{Q^4} L_{\nu\mu} \text{tr}_D [(\not{p}_1 + m_Q) V^{\mu i} (\not{p}_2 - m_Q) \bar{V}^{\nu j}], \quad T_F = \frac{1}{2}. \quad (17.16)$$

The initial gluon spin is *not* averaged a second time: $D_g = I/2$ gives the unpolarized partonic spin average. The color factor follows from $(N_c^2 - 1)^{-1} \sum_a \text{tr}(t^a t^a) = T_F$. For the exact on-shell partonic reaction the differential normalization is

$$d\hat{\sigma}[D_{\text{in}}] = \frac{d\Pi_3}{2\hat{s}_{\ell g}} \text{Tr}_2(BD_{\text{in}}), \quad \hat{s}_{\ell g} = 2l \cdot k, \quad d\Pi_3 = (2\pi)^4 \delta^{(4)}(l + k - l' - p_1 - p_2) \prod_{r=l', p_1, p_2} \frac{d^3 \mathbf{r}}{(2\pi)^3 2r^0}. \quad (17.17)$$

Equations (15.9), (17.14), (17.16), and (17.4) are a finite, explicit algorithm for all four hard coefficients. They provide an explicit implementation of the Born polarization response. The implementation in the project repository [26] evaluates this Dirac trace, checks Hermiticity and the Ward identities, and reports the four coupling-stripped coefficients for a physical test kinematic point. It is a hard-kernel calculator, not a simulated ⁷Li TMD extraction.

17.7 An explicit derivation of both Ward identities

Gauge invariance can be checked before evaluating any large Dirac trace. Define

$$r = p_1 - q = k - p_2, \quad s = p_1 - k = q - p_2, \quad D_r = r^2 - m_Q^2, \quad D_s = s^2 - m_Q^2. \quad (17.18)$$

The equality of the two forms for each internal momentum follows from $p_1 + p_2 = q + k$. The external spinors obey $\bar{u}(p_1) \not{p}_1 = m_Q \bar{u}(p_1)$ and $\not{p}_2 v(p_2) = -m_Q v(p_2)$. For a photon-momentum contraction, the first diagram gives

$$\begin{aligned} \bar{u} \not{q} \frac{\not{r} + m_Q}{D_r} \gamma^\alpha v &= \bar{u} (m_Q - \not{r}) \frac{\not{r} + m_Q}{D_r} \gamma^\alpha v \\ &= -\bar{u} \gamma^\alpha v. \end{aligned} \quad (17.19)$$

The second diagram gives

$$\begin{aligned} \bar{u} \gamma^\alpha \frac{\not{s} + m_Q}{D_s} \not{q} v &= \bar{u} \gamma^\alpha \frac{\not{s} + m_Q}{D_s} (\not{s} - m_Q) v \\ &= +\bar{u} \gamma^\alpha v. \end{aligned} \quad (17.20)$$

They cancel. For the gluon-momentum contraction, the first diagram uses $\not{k} v = (\not{r} - m_Q) v$ and gives $+\bar{u} \gamma^\mu v$. The second uses $\bar{u} \not{k} = \bar{u} (m_Q - \not{s})$ and gives $-\bar{u} \gamma^\mu v$. They also cancel.

This calculation identifies exactly what should be tested in a program. The sum of the two diagrams vanishes between on-shell external spinors after contraction with q_μ or k_α . Neither individual diagram satisfies this condition, and the unprojected matrix $q_\mu V^{\mu\alpha}$ need not vanish as a bare Dirac matrix. A spinor-free numerical check uses

$$(\not{p}_1 + m_Q) q_\mu V^{\mu\alpha} (\not{p}_2 - m_Q) = 0, \quad (17.21)$$

and its gluon analogue. The outside factors are the completeness projectors for the physical final spin states. Keeping this distinction avoids reporting a false failure of gauge invariance from an incorrectly stronger matrix condition.

The same on-shell spin sums give the trace in Eq. (17.16). Summing final colors and averaging the incoming gluon color gives $(N_c^2 - 1)^{-1} \sum_a \text{tr}(t^a t^a) = T_F = 1/2$. The incoming spin is still carried by D_{in} , so the unpolarized spin average is obtained once, by setting $D_{\text{in}} = I/2$. A second factor 1/2 inserted into B would double-count that average.

17.8 Building a physical phase-space point from scalar inputs

A transparent implementation starts with a physical center-of-mass configuration instead of arbitrary four-vectors. Let $W = \sqrt{(q+k)^2} > 2m_Q$, $Q^2 > 0$, and choose the incoming massless gluon along $+z$. Then

$$E_g = \frac{W^2 + Q^2}{2W}, \quad E_\gamma = \frac{W^2 - Q^2}{2W}, \quad (17.22)$$

$$k^\mu = (E_g, 0, 0, E_g), \quad q^\mu = (E_\gamma, 0, 0, -E_g). \quad (17.23)$$

Thus $k^2 = 0$, $q^2 = -Q^2$, and $q+k = (W, \mathbf{0})$. The virtual photon is spacelike; E_γ need not be positive in every allowed frame. Its invariant virtuality, not a real-photon mass shell, is the input.

With $p_* = (W^2/4 - m_Q^2)^{1/2}$, choose

$$p_1^\mu = \left(\frac{W}{2}, p_* \sin \theta \cos \phi, p_* \sin \theta \sin \phi, p_* \cos \theta \right), \quad (17.24)$$

$$p_2^\mu = (W, \mathbf{0}) - p_1^\mu. \quad (17.25)$$

Both final momenta are on shell. To construct a massless lepton with energy E_ℓ and transverse momentum in the $+x$ direction, impose $l'^2 = (l-q)^2 = 0$. Since $l^2 = 0$, this gives $l \cdot q = -Q^2/2$, and hence

$$l_z = \frac{-Q^2/2 - E_\ell E_\gamma}{E_g}, \quad l_x = \sqrt{E_\ell^2 - l_z^2}, \quad l = (E_\ell, l_x, 0, l_z), \quad l' = l - q. \quad (17.26)$$

A nondegenerate real l_x requires $E_\ell > W/2$ for these $Q^2 > 0$ inputs. Indeed the quantity under the square root is proportional to $Q^2(E_\ell - W/2)(E_\ell + Q^2/(2W))$. At the equality the transverse lepton direction becomes degenerate, so an azimuthal analyzer convention based on it must be treated by a limit. These kinematic checks should be made before evaluating amplitudes.

The propagator denominators provide additional simple tests. Defining $\beta_* = 2p_*/W$, one finds

$$D_r = -WE_g(1 + \beta_* \cos \theta), \quad D_s = -WE_g(1 - \beta_* \cos \theta). \quad (17.27)$$

For a massive pair above threshold, $0 < \beta_* < 1$, so these denominators are nonzero for all real polar angles. This makes the massive Born example a convenient numerical diagnostic. Massless or threshold limits require separate care and should not be used accidentally as generic points.

17.9 A reproducible benchmark and the scope of its interpretation

For the input

$$W = 5 \text{ GeV}, \quad Q^2 = 4 \text{ GeV}^2, \quad m_Q = 1.5 \text{ GeV}, \quad E_\ell = 10 \text{ GeV}, \quad \theta = 0.8, \quad \phi = 0.4, \quad \lambda = +1, \quad (17.28)$$

the direct two-diagram trace, with the common couplings, color factor, and Q^{-4} removed, gives

$$(b_U, b_G, b_C, b_S) \simeq (1684, +123.6, 110.2, 5.977) \text{ GeV}^2. \quad (17.29)$$

The two eigenvalues are approximately $(1.518, 1.850) \times 10^3 \text{ GeV}^2$. These values specify a check of the normalization and index ordering in this note; the implementation is available in the project repository [26]. The stripped trace has dimension $+2$: the spin projectors and propagator vertices balance, leaving the leptonic tensor's dimension. Restoring the factor in Eq. (17.16) gives the hard matrix appropriate to the partonic cross section and phase-space formula.

At this Born order the matrix satisfies $B(-\lambda) = B(\lambda)^*$ for the specified axes and physical momenta. Consequently b_U, b_C, b_S are beam-helicity even, whereas b_G is odd and vanishes for

an unpolarized beam. This follows because the symmetric and antisymmetric leptonic tensors select the real and imaginary parts of the corresponding hard response. The octupole gluon helicity functions therefore need the beam-helicity analyzer in this example. Absorptive phases and different processes must be analyzed with their own hard tensors rather than assumed to obey every Born-specific relation.

A positive hard matrix and small Ward residuals establish consistency of this finite perturbative calculation. They do not predict a nuclear TMD or validate an all-orders factorization formula for a chosen heavy-hadron or jet measurement. Those additional steps require specified final-state measurements, color flow, soft recoil, and nonperturbative matrix elements.

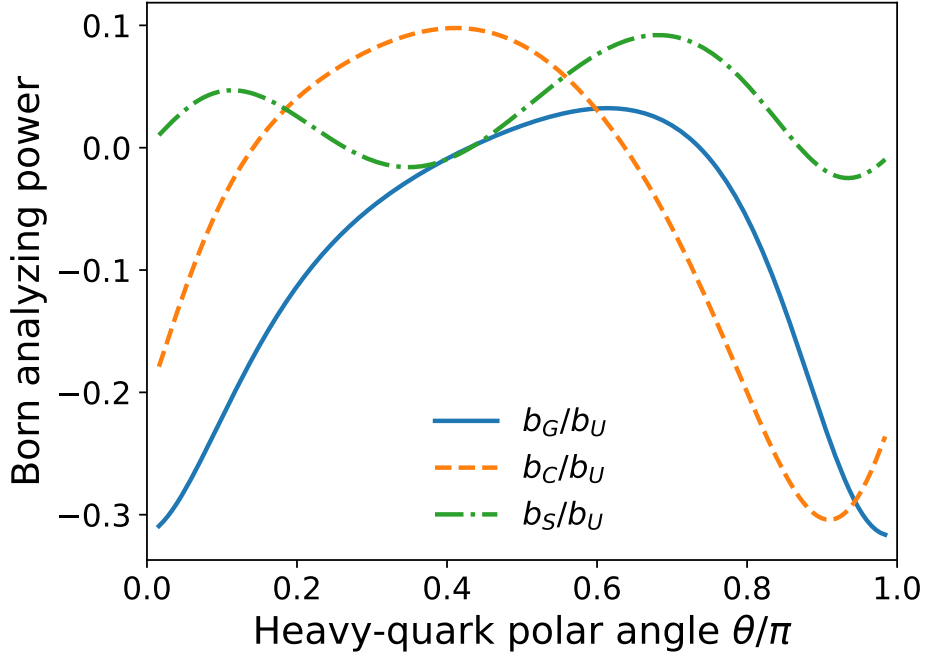


Figure 1: Corrected Born hard analyzing powers for physical incoming electron helicity +1, with the kinematics of Eq. (17.29) except for the varying heavy-quark polar angle. The circular curve uses the $L_{\nu\mu}$ contraction. The curves characterize the hard analyzer, not a lithium-7 TMD model.

17.10 From the Born analyzer to a measured small-imbalance SF

For a nearly back-to-back pair define

$$\mathbf{q}_{\text{pair},T} = \mathbf{p}_{1T} + \mathbf{p}_{2T}, \quad \mathbf{p}_{\text{rel},T} = \frac{1}{2}(\mathbf{p}_{1T} - \mathbf{p}_{2T}), \quad |\mathbf{q}_{\text{pair},T}| \ll Q, |\mathbf{p}_{\text{rel},T}|, \sqrt{m_Q^2 + p_{\text{rel},T}^2}. \quad (17.30)$$

At leading-power Born level, the active fraction is

$$x_g = x_{\text{Bj}} \left(1 + \frac{M_{Q\bar{Q}}^2}{Q^2} \right), \quad M_{Q\bar{Q}}^2 = (p_1 + p_2)^2, \quad (17.31)$$

neglecting target-mass power corrections. It is x_g , not generally x_{Bj} , that is the gluon TMD argument. Separating transverse momentum conservation before the collinear expansion of the hard part gives $\delta^{(2)}(\mathbf{q}_{\text{pair},T} - \mathbf{k}_T)$ at Born level. Therefore the response coefficients are the corresponding hard/phase-space factors multiplying the kernels of Eq. (17.5) evaluated at $(x_g, \mathbf{k}_T) = (x_g, \mathbf{q}_{\text{pair},T})$.

This is a controlled Born statement. One must not set $\mathbf{k}_T = 0$ inside an exact three-body momentum delta function and then also claim a nonzero measured pair imbalance. Beyond Born order the imbalance receives radiation and soft recoil; a schematic term becomes

$$\int d^2\mathbf{k}_T d^2\ell_T \delta^{(2)}(\mathbf{q}_{\text{pair},T} - \mathbf{k}_T - \ell_T) \times [\text{hard/measurement factors}] [\text{gluon correlator}] [\text{soft/jet factors}]. \quad (17.32)$$

The color representation, link structure, subtraction, and measured jets or heavy hadrons must be specified before this schematic formula is promoted to a factorization theorem. Colored final states do not justify an automatic universal replacement by the quark SIDIS convolution. We give no unverified all-orders hard/soft coefficients for such a process here.

17.11 Deriving the active fraction and preserving the imbalance

At leading power take the active gluon momentum in the hard amplitude to be $k = x_g P$ with the target mass neglected in that amplitude. Then

$$M_{Q\bar{Q}}^2 = (q + x_g P)^2 = -Q^2 + 2x_g P \cdot q = Q^2 \left(\frac{x_g}{x_{Bj}} - 1 \right), \quad (17.33)$$

which derives Eq. (17.31). This is why a gluon TMD is not generally sampled at x_{Bj} in pair production. A proposed kinematic bin with $x_g > 1$ has no support in this leading-power nuclear parton description.

For a small pair imbalance, the leading-power expansion is performed on the hard amplitude at fixed large scales while the measurement's transverse momentum conservation retains the small recoil. At Born level it therefore gives $\delta^{(2)}(\mathbf{q}_{\text{pair},T} - \mathbf{k}_T)$. Replacing $\mathbf{k}_T$ by zero inside the measurement delta function as well would restrict the measured pair to zero imbalance and discard the very TMD dependence being investigated. This is an example of a common distinction in factorization: some appearances of a small momentum are expanded in the hard subgraph, while its appearance in a differential measurement is retained.

Beyond Born order, soft and collinear radiation also contributes to the imbalance. The process-dependent soft and measurement factors must be defined before a cross section is inferred. At a fixed hard point the target kernels in Table 4 remain a useful polarization decomposition, but the physical measured azimuths can mix with the hard analyzer and recoil convolutions. The tensor catalogue and a factorization theorem therefore answer different questions, and the latter cannot be obtained by replacing quark charges in a SIDIS formula with a guessed gluon color factor.

18 Rank-selective target preparation and response tomography

Provenance and task. Rotation matrices and density-matrix tomography are standard quantum mechanics. The explicit octupole-only probe family and its use to separate the rank-three scattering response are constructions developed in this work. They establish what target-state positivity allows. They do not claim that a particular polarized-target apparatus already prepares these states or reconstructs all their coherences. The task is to exhibit physical probes, rotate them with controlled normalization, and determine exactly which response can be inverted from them.

18.1 Constructing positive states with only an octupole moment

Start with the normalized rank-three operator $\tau_{30} = \sqrt{5} O_{zzz}/3$. Its eigenvalues are $\sqrt{5}(1, -3, 3, -1)/10$. Its squared Hilbert–Schmidt norm is $\text{Tr}(\tau_{30}^\dagger \tau_{30}) = 1$. For a unit

direction $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, rotate it to

$$\begin{aligned} X_3(\hat{n}) &= D^{3/2}(\phi, \theta, 0) \tau_{30} D^{3/2}(\phi, \theta, 0)^\dagger \\ &= \frac{\sqrt{5}}{3} \left[(\hat{n} \cdot \mathbf{J})^3 - \frac{41}{20} \hat{n} \cdot \mathbf{J} \right], \end{aligned} \quad (18.1)$$

$$\rho_\pm(\hat{n}) = \frac{I}{4} \pm \beta_3 X_3(\hat{n}), \quad 0 < \beta_3 \leq \frac{\sqrt{5}}{6}. \quad (18.2)$$

Rotations preserve eigenvalues. Positivity of both signs is equivalent to $1/4 - \beta_3 \|\tau_{30}\|_{\text{op}} \geq 0$, and $\|\tau_{30}\|_{\text{op}} = 3\sqrt{5}/10$ gives the stated bound. All vector and quadrupole moments vanish because their defining operators are trace orthogonal to the entire rank-three subspace. Normalization is unchanged because X_3 is traceless.

Along $\hat{n} = \hat{z}$, the positive state's longitudinal octupole is

$$O = \text{Tr}(\rho_+ O_{zzz}) = \frac{3\beta_3}{\sqrt{5}}, \quad 0 < O \leq \frac{1}{2}. \quad (18.3)$$

At the endpoint $O = 1/2$, its populations are $(1/3, 0, 1/2, 1/6)$. Their purity is $7/18$, not one. More generally $\text{Tr} \rho_\pm^2 = 1/4 + \beta_3^2$. Thus “octupole-only” means that the polarized part has only rank three; the state is an ensemble with an unpolarized part and is not a pure state. This avoids a common ambiguity in the phrase pure polarization.

18.2 Deriving the rotated Cartesian components

The tensor of the positive state must be axially symmetric about $\hat{n}$, symmetric in its indices, and three-dimensionally traceless. Those conditions allow one rank-three form:

$$R_{abc} = c \left[n_a n_b n_c - \frac{1}{5} (n_a \delta_{bc} + n_b \delta_{ac} + n_c \delta_{ab}) \right]. \quad (18.4)$$

At $\hat{n} = \hat{z}$ its zzz component is $2c/5$. Setting that to O gives $c = 5O/2$, or

$$R_{abc} = \frac{5O}{2} \left[n_a n_b n_c - \frac{1}{5} (n_a \delta_{bc} + n_b \delta_{ac} + n_c \delta_{ab}) \right]. \quad (18.5)$$

This derivation is shorter than expanding all entries of an Euler-angle spin matrix and also provides a check of that matrix calculation. Writing $c_\theta = \cos \theta$ and $s_\theta = \sin \theta$, its components in the adopted light-front normalization are

$$S_{LLL} = O P_3(c_\theta), \quad P_3(c) = (5c^3 - 3c)/2, \quad (18.6)$$

$$S_{LLT,+} = \frac{O}{2} (5c_\theta^2 - 1) s_\theta e^{i\phi}, \quad (18.7)$$

$$S_{LTT,+} = 5O c_\theta s_\theta^2 e^{2i\phi}, \quad (18.8)$$

$$S_{TTT,+} = \frac{5O}{2} s_\theta^3 e^{3i\phi}. \quad (18.9)$$

For example, $R_{zzi} = (5O/2)(c_\theta^2 - 1/5)n_i$ gives the second line. Removing the transverse trace from R_{zij} and multiplying by four gives the third. Removing the transverse rank-three traces from R_{ijk} gives the fourth. The factors in these expressions therefore come from Eq. (3.18), not from independently chosen polarization fractions.

At $\theta = 0$ only S_{LLL} is present. At $\theta = \pi/2$, $S_{LLL} = S_{LTT} = 0$, but $S_{LLT,+} = -(O/2)e^{i\phi}$ and $S_{TTT,+} = (5O/2)e^{3i\phi}$. A transverse octupole preparation is consequently not a state with only the TTT component. It remains entirely rank three, because rotations mix components within a rank but never generate a lower rank from it. Angles satisfying $\cos^2 \theta = 1/5$ remove the LLT component without removing the entire rank-three tensor. These geometric facts are useful when designing target orientations.

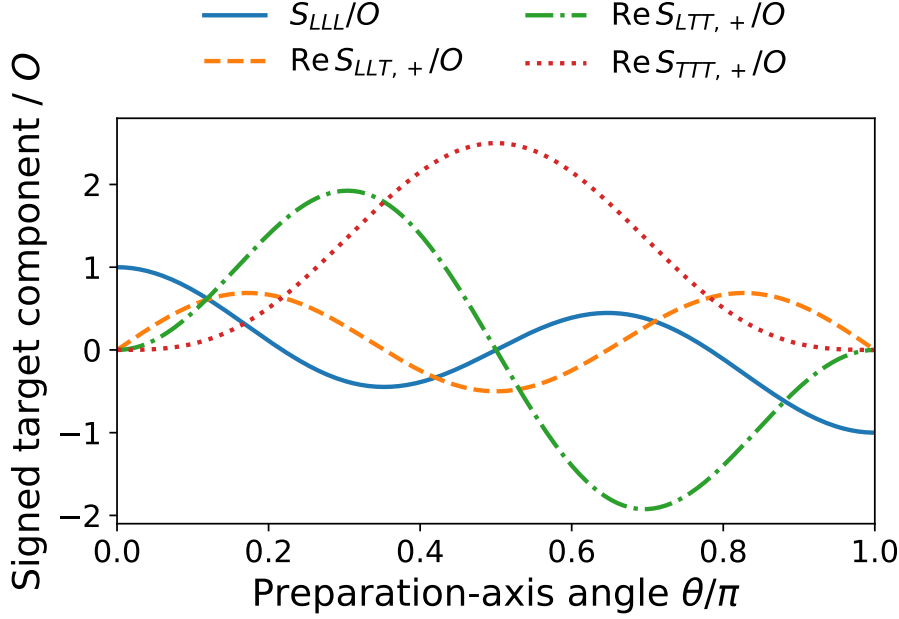


Figure 2: Signed components of a rotated octupole-only state, normalized to O , at zero preparation azimuth. Equation (18.9) follows from the spin density matrix. A transverse preparation has both LLT and TTT components; every point still has vanishing vector and quadrupole moments.

18.3 Continuous inversion of the rank-three response

At fixed hard kinematics, beam state, and specified angular projection, write the measured response as $\mathcal{W}[\rho] = \text{Tr}(\rho \widehat{W})$, where $\widehat{W}$ is a Hermitian target-spin response operator. It can include a differential cross section, a Fourier-weighted cross section, or another stated linear response. Define $w_{3Q} = \text{Tr}(\tau_{3Q}^\dagger \widehat{W})$. The paired target difference is

$$\Delta\mathcal{W}(\Omega) = \frac{\mathcal{W}[\rho_+(\Omega)] - \mathcal{W}[\rho_-(\Omega)]}{2\beta_3} = \text{Tr}[X_3(\Omega)\widehat{W}]. \quad (18.10)$$

The unpolarized contribution cancels, and trace orthogonality removes all ranks other than three. Expanding $X_3(\Omega) = \sum_Q D_{Q0}^3(\phi, \theta, 0)\tau_{3Q}$ gives

$$\Delta\mathcal{W}(\Omega) = \sum_{Q=-3}^3 D_{Q0}^{3*}(\Omega)w_{3Q}. \quad (18.11)$$

To verify the complex conjugation, use $\text{Tr}(\tau_{3Q}\widehat{W}) = (-1)^Q w_{3,-Q}$ and then $w_{3,-Q} = (-1)^Q w_{3Q}^*$ together with Hermiticity. Either equivalent real form gives the same response. The spherical rotation functions satisfy

$$\int d\Omega D_{Q0}^3(\Omega)D_{Q'0}^{3*}(\Omega) = \frac{4\pi}{7}\delta_{QQ'}. \quad (18.12)$$

Multiplying the response by D_{Q0}^3 and integrating therefore yields

$$w_{3Q} = \frac{7}{4\pi} \int d\Omega D_{Q0}^3(\Omega)\Delta\mathcal{W}(\Omega). \quad (18.13)$$

Hermiticity leaves seven real parameters: a real $Q = 0$ component and the real and imaginary parts for $Q = 1, 2, 3$.

18.4 Discrete preparations and what tomography can identify

For finitely many prepared directions, expand X_3 in any real orthonormal Hermitian rank-three basis T_j , $j = 1, \dots, 7$. Then the observed differences obey

$$\Delta\mathcal{W}_a = \sum_{j=1}^7 D_{aj} w_j, \quad D_{aj} = \text{Tr}[X_3(\hat{n}_a)T_j]. \quad (18.14)$$

At least seven independent response combinations are needed. One exact full-rank example uses the directions

$$\begin{aligned} (0, 0, 1), \quad (1, 0, 0), \quad (0, 1, 0), \quad \frac{(2, 2, -1)}{3}, \\ \frac{(1, 2, -2)}{3}, \quad \frac{(2, 1, -2)}{3}, \quad \frac{(2, 6, -9)}{11}. \end{aligned} \quad (18.15)$$

They are unit vectors, and their real rank-three response matrix has rank seven. This is an existence construction, not an optimization for a specific detector. Directions can be nearly linearly dependent even when their exact rank is seven, so singular values and propagated uncertainties should be examined in an experimental design.

The inversion reconstructs seven real components of one specified *target response operator*. It does not recover fourteen octupole TMDs from seven yields. The parton channel, quark flavor, fragmentation function, and momentum dependence remain inside that response. Beam helicity, different azimuthal projections, final states, and kinematics provide the additional independent information. State tomography, response tomography, and a TMD extraction are three distinct inverse problems.

18.5 A spin–orbital mechanism for a composite target

An illustrative angular-momentum calculation shows why a rank-three nuclear response need not require an intrinsic octupole of a constituent. Take $L = 1$ and constituent spin $s = 1/2$. The $J = 3/2$ subspace is embedded by the four Clebsch–Gordan coupled states

$$|\frac{3}{2}, \frac{3}{2}\rangle = |1, 1\rangle|+\rangle, \quad (18.16)$$

$$|\frac{3}{2}, \frac{1}{2}\rangle = \sqrt{\frac{2}{3}}|1, 0\rangle|+\rangle + \sqrt{\frac{1}{3}}|1, 1\rangle|-\rangle, \quad (18.17)$$

$$|\frac{3}{2}, -\frac{1}{2}\rangle = \sqrt{\frac{2}{3}}|1, 0\rangle|-\rangle + \sqrt{\frac{1}{3}}|1, -1\rangle|+\rangle, \quad (18.18)$$

$$|\frac{3}{2}, -\frac{3}{2}\rangle = |1, -1\rangle|-\rangle. \quad (18.19)$$

Here $|\pm\rangle$ have spin projection $\pm 1/2$. Let the columns of V be these states in the product basis. Then $V^\dagger V = I_4$, and an operator on the product space is projected into the target-spin subspace by $V^\dagger X V$.

The spin-only operator has expectation values $(1/2, 1/6, -1/6, -1/2)$, so

$$V^\dagger s_z V = \frac{1}{3} J_z. \quad (18.20)$$

For the orbital-quadrupole-weighted spin operator $(3L_z^2 - 2)s_z$, the four diagonal entries are $1/2$, $-5/6$, $5/6$, and $-1/2$. Comparing them with the diagonal of O_{zzz} and J_z gives

$$V^\dagger [(3L_z^2 - 2)s_z] V = O_{zzz} + \frac{2}{15} J_z. \quad (18.21)$$

Thus resolving a correlation between constituent spin and an orbital quadrupole introduces a target octupole response. In the language of angular momentum, the orbital rank two and spin rank one can couple to rank three. A constituent-spin analyzer that has no orbital resolution instead gives only rank one in this subspace.

This construction is an exact finite-dimensional example, not a wave function for ⁷Li. The resolving weight in a nuclear TMD is supplied by the actual momentum-dependent QCD matrix element and its spin recoupling, not literally by $3L_z^2 - 2$ in all processes. The example explains an allowed mechanism and identifies why nuclear orbital motion matters; the next section states what additional nuclear information a quantitative calculation needs.

19 Separating all fourteen octupole gluon responses

Provenance and task. The hard subprocess and Stokes decomposition are established tools. The construction developed here combines them with positive rank-three preparations to obtain a complete, explicitly invertible Born response. The distinction between seven target-response components and fourteen TMD coefficients in Sec. 18.3 is resolved by using beam helicity and joint target/recoil angles as additional analyzers. The independent response and amplitude comparisons are summarized in Sec. 21.6.

19.1 Fix the hard event while varying the target and recoil directions

Fix Q^2 , the pair mass, the heavy-quark direction, and the lepton kinematics. The Born hard matrix is computed at the collinear hard point. The small measured imbalance is retained in the target correlator, with $t = |\mathbf{q}_{\text{pair},T}|/M_A > 0$ and azimuth φ . The preparation axis has angles (ϑ, ψ) , distinct from the heavy-quark angles (θ, ϕ) and the calculated hard-analyzer phase ϕ_B . At the retained power the hard matrix does not vary with φ . All gluon coefficients in this example use the same future-pointing $[+, +]$ link class, common scales, and the measurement specification of heavy-pair leptonproduction [23].

The two states in Eq. (18.2) have moments

$$A_{30} = \eta O \alpha_0, \quad A_{3m,+} = \eta O \alpha_m e^{im\psi} \quad (m > 0), \quad (19.1)$$

$$\alpha_0 = P_3(c_\vartheta), \quad \alpha_1 = \frac{1}{2}(5c_\vartheta^2 - 1)s_\vartheta, \quad (19.2)$$

$$\alpha_2 = 5c_\vartheta s_\vartheta^2, \quad \alpha_3 = \frac{5}{2}s_\vartheta^3, \quad (19.3)$$

where $c_\vartheta = \cos \vartheta$, $s_\vartheta = \sin \vartheta$, and $\eta = \pm 1$. The factors are signed. For example, $c_\vartheta = 3/5$ gives

$$(\alpha_0, \alpha_1, \alpha_2, \alpha_3) = (-9/25, 8/25, 48/25, 32/25). \quad (19.4)$$

Replacing the negative α_0 by its absolute value changes the physical state. These factors are derived from the spin matrix, not adjusted to make an inversion nonsingular.

19.2 Four spin settings and two reduced responses

For the differential rate corrected for luminosity and acceptance write

$$\sigma_{\lambda,\eta} = \mathcal{J}_g [\mathcal{W}_0^g + \eta O(\mathcal{W}_U + \lambda \mathcal{W}_L)]. \quad (19.5)$$

The common flux/measurement factor $\mathcal{J}_g$ is independent of φ, ψ at these fixed Born variables. In the physical-helicity convention, $b_G(\lambda) = \lambda b_G(1)$ and b_U, b_C, b_S are helicity even. The unpolarized reference includes a linear-gluon term,

$$\mathcal{W}_0^g = b_U f_{00}^g + \frac{b_{\text{lin}} t^2}{2} h_{00}^{g[2]} \cos(2\varphi - 2\phi_B). \quad (19.6)$$

Summing the four rates with the target sign gives

$$\mathcal{W}_U = \frac{1}{4O\mathcal{J}_g} \sum_{\lambda,\eta} \eta \sigma_{\lambda,\eta}, \quad \mathcal{W}_L = \frac{1}{4O\mathcal{J}_g} \sum_{\lambda,\eta} \lambda \eta \sigma_{\lambda,\eta}. \quad (19.7)$$

The scalar reference cancels in both. The first response contains the three number and seven linear-polarization octupole coefficients; the second contains the four helicity coefficients. The factors four and O follow from the actual spin sums, not from the number of Fourier modes.

19.3 Derive every angular coefficient

The helicity response is

$$\mathcal{W}_L = b_G(1) \left[\alpha_0 g_{30}^g + \sum_{m=1}^3 \alpha_m t^m g_{3m}^g \cos m(\varphi - \psi) \right]. \quad (19.8)$$

For linear polarization define

$$\Psi_m^- = (2-m)\varphi + m\psi - 2\phi_B, \quad \Psi_m^+ = (m+2)\varphi - m\psi - 2\phi_B, \quad c_1 = 1/2, \quad c_2 = c_3 = 1. \quad (19.9)$$

The remaining response is

$$\begin{aligned} \mathcal{W}_U = b_U \sum_{m=1}^3 \alpha_m t^m f_{3m}^g \sin m(\varphi - \psi) + \frac{b_{\text{lin}} \alpha_0 t^2}{2} h_{30}^{g[2]} \sin(2\varphi - 2\phi_B) \\ + b_{\text{lin}} \sum_{m=1}^3 \alpha_m \left[c_m t^{|m-2|} h_{3m}^{g[|m-2|]} \sin \Psi_m^- + \frac{t^{m+2}}{4} h_{3m}^{g[m+2]} \sin \Psi_m^+ \right]. \end{aligned} \quad (19.10)$$

For example, $m = 3$ in the low branch gives $A_{33,+} \kappa^* h_{33}^{g[1]}$ before the odd- K dual. Multiplication by $-ib_+^*$ and extraction of the real part gives $\sin(3\psi - \varphi - 2\phi_B)$ with one momentum power. The high branch instead carries $A_{33,+}^* \kappa^5 h_{33}^{g[5]}/4$, hence the fifth harmonic and factor $1/4$. There is no independent intermediate orbital-rank-three term to insert between them.

Table 5: Independent octupole modes. Rows with m expand over 1, 2, 3. The full mode is $y_\nu = \lambda^{\epsilon_\nu} \mathcal{Y}_\nu$. The response amplitude is the TMD times the prefactor shown.

TMD	ϵ_ν	(p, q)	Angular function; prefactor
g_{30}^g	1	(0, 0)	1; $\alpha_0 b_G(1)$
$h_{30}^{g[2]}$	0	(2, 0)	$\sin(2\varphi - 2\phi_B)$; $\alpha_0 b_{\text{lin}} t^2/2$
f_{3m}^g	0	$(m, -m)$	$\sin m(\varphi - \psi)$; $\alpha_m b_U t^m$
g_{3m}^g	1	$(m, -m)$	$\cos m(\varphi - \psi)$; $\alpha_m b_G(1) t^m$
$h_{3m}^{g[m-2]}$	0	$(2-m, m)$	$\sin \Psi_m^-$; $\alpha_m b_{\text{lin}} c_m t^{ m-2 }$
$h_{3m}^{g[m+2]}$	0	$(m+2, -m)$	$\sin \Psi_m^+$; $\alpha_m b_{\text{lin}} t^{m+2}/4$

The pair (p, q) identifies the recoil and preparation frequencies in $p\varphi + q\psi + \delta$. Two terms may share their recoil frequency but differ in q ; varying only φ at one fixed preparation can therefore lose information that exists on the two-angle space.

19.4 An exact angular certificate and its physical qualifications

Use the normalized inner product

$$\langle F \rangle = \frac{1}{2} \sum_{\lambda=\pm 1} \frac{1}{(2\pi)^2} \int_0^{2\pi} d\varphi \int_0^{2\pi} d\psi F. \quad (19.11)$$

The sum removes products of opposite beam parity. The angular integral of $e^{i(p\varphi+q\psi)}$ vanishes unless $p = q = 0$. Writing each sine and cosine as two exponentials therefore gives, in the listed mode order,

$$G_{\nu\mu} = \langle y_\nu y_\mu \rangle = \text{diag}(1, 1/2, \dots, 1/2), \quad \det G = 2^{-13} = 1/8192. \quad (19.12)$$

The constant angular function belongs to the beam-odd channel, so its full mode is λ , not a beam-independent constant. Its squared norm is still one. This is an instructive example of why the beam channel must remain in a response certificate.

The Gram matrix certifies independent *normalized angular modes*. Recovering TMD coefficients also requires their diagonal amplitude factors in Table 5 to be nonzero: $\alpha_m \neq 0$, $t > 0$, and $b_U b_G(1) b_{\text{lin}} \neq 0$. These are identifiability conditions, not a statement about achievable statistical precision. Very small factors produce ill conditioning even before an exact rank is lost.

A discrete product quadrature with eleven recoil angles and eight preparation angles integrates these mode products exactly. Frequencies in the products have $|p| \leq 10$ and $|q| \leq 6$; no nonzero one is a multiple of 11 or 8, respectively. The discrete root-of-unity sums therefore implement the same zero-mode projection. This is exactness for a finite harmonic bandwidth, not for arbitrary acceptance functions.

19.5 Normalized moments and the distinction between a ratio and a TMD

Let C_ν denote the response amplitude including its TMD. Define

$$\mathcal{A}_\nu = \frac{Z_\nu \int d\varphi d\psi \mathcal{Y}_\nu \sum_{\lambda,\eta} \eta \lambda^{\epsilon_\nu} \sigma_{\lambda,\eta}}{O \int d\varphi d\psi \sum_{\lambda,\eta} \sigma_{\lambda,\eta}}, \quad Z_\nu = \begin{cases} 1, & \mathcal{Y}_\nu = 1, \\ 2, & \text{otherwise.} \end{cases} \quad (19.13)$$

Substituting Eq. (19.5) shows that the common $\mathcal{J}_g$ cancels. The linear-gluon modulation of the unpolarized reference averages to zero. Equation (19.12) then gives

$$\mathcal{A}_\nu = \frac{C_\nu}{b_U f_{00}^g}. \quad (19.14)$$

Thus normalized moments determine TMD *ratios*, after dividing the known target and analyzer factors. Absolute normalization requires an independently normalized reference response or f_{00}^g . A finite moment can remain well defined when an analyzer vanishes, but it then contains no information on the corresponding coefficient.

For the longitudinal preparation,

$$\mathcal{A}_L^{(0)} = \frac{b_G(1) g_{30}^g}{b_U f_{00}^g}, \quad \mathcal{A}_U^{(2)} = \frac{b_{\text{lin}} t^2 h_{30}^{g[2]}}{b_U 2 f_{00}^g}. \quad (19.15)$$

The corrected physical-helicity Born calculation changes the numerical value of $b_G(1)$ relative to the historical current ordering; it does not add another minus sign to Eq. (19.15) or redefine g_{30}^g .

19.6 A transverse preparation and the separate rank-one and rank-five terms

For a transverse cone, $(\alpha_0, \alpha_1, \alpha_2, \alpha_3) = (0, -1/2, 0, 5/2)$. The simultaneous *LLT* contribution is removed by

$$(\mathcal{P}_3 \mathcal{W})(\psi) = \frac{1}{\pi} \int_0^{2\pi} d\psi' \cos[3(\psi - \psi')] \mathcal{W}(\psi'). \quad (19.16)$$

On the ideal preparation bandwidth $|m| \leq 3$, the integral equals $\frac{1}{4} \sum_{j=0}^7 \cos[3(\psi - \psi_j)] \mathcal{W}(\psi_j)$, where $\psi_j = 2\pi j/8$. The surviving beam-even response is

$$\begin{aligned} \mathcal{P}_3 \mathcal{W}_U = & \frac{5}{2} [b_U t^3 f_{33}^g \sin(3\varphi - 3\psi) + b_{\text{lin}} t h_{33}^{g[1]} \sin(3\psi - \varphi - 2\phi_B) \\ & + \frac{1}{4} b_{\text{lin}} t^5 h_{33}^{g[5]} \sin(5\varphi - 3\psi - 2\phi_B)], \end{aligned} \quad (19.17)$$

and $\mathcal{P}_3 \mathcal{W}_L = (5/2) b_G(1) t^3 g_{33}^g \cos(3\varphi - 3\psi)$. Weighting the beam-even response by its rank-one, rank-three, or rank-five sine and integrating over φ with prefactor $2/(5\pi)$ yields, respectively,

$$b_{\text{lin}} t h_{33}^{g[1]}, \quad b_U t^3 f_{33}^g, \quad \frac{1}{4} b_{\text{lin}} t^5 h_{33}^{g[5]}. \quad (19.18)$$

This derives a separate observable for each linear-polarization branch. A transverse target by itself does not remove *LTT*; the preparation projection is a necessary part of the construction.

19.7 Acceptance, conditioning, and the boundary of the Born statement

With a known acceptance $\mathcal{E}(\varphi, \psi)$, the relevant Gram matrix is $G_{\nu\mu}^{\mathcal{E}} = \langle \mathcal{E} y_\nu y_\mu \rangle$. Full-coverage moments can mix modes when this matrix is not diagonal. A response fit with the measured acceptance restores the intended projection only when the resulting design has sufficient rank. Report the singular values of both the physically scaled columns and any column-normalized design; their condition numbers are not invariant under arbitrary rescaling of the coefficient coordinates.

The repository generates synthetic rates from density operators and Cartesian covariants, then reconstructs coefficients with independently specified Fourier modes. It separately tests missing analyzers, restricted preparations, aliasing, and a known nonuniform acceptance. These are controlled inverse-problem examples, not a predicted target performance. Beyond Born order the target-rank difference still follows by linearity in ρ , but recoil, soft, and final-state measurement factors change the angular convolution. The pointwise ratios in Eq. (19.14) are not thereby promoted to all-order ratios.

20 Nuclear matrix elements, target calibration, and experimental inference

Provenance and task. The spin algebra and target-level correlator definitions are independent of a nuclear approximation. Impulse convolutions, spin-dependent spectral functions, and rescattering are established ingredients of nuclear structure studies; nuclear TMD calculations illustrate their importance [37]. The present construction organizes these ingredients for a spin-3/2 target without specifying a numerical ⁷Li wave function. Here we make that setup and the subsequent polarization calibration explicit. The output is a controlled list of required inputs and response projections, not a numerical TMD prediction.

20.1 Nuclear spin structure and an impulse-approximation organization

The operator definitions apply to the entire ⁷Li nucleus. They do not assert that a nuclear TMD is a proton TMD multiplied by a spin population. Nuclear binding, orbital motion, spin recoupling, and the gauge link all remain inside the matrix element. The distinction is especially important for octupole moments: a spin-1/2 nucleon has no intrinsic rank-three density-matrix multipole, but a nucleon embedded in a nucleus can participate in a rank-three *nuclear* response through spin-orbital correlations. For example, coupling an $L = 1$ orbital sector to a spin-1/2 sector permits a total $J = 3/2$ state. An octupole signal is therefore not, by itself, a proof of an exotic partonic constituent or a uniquely identifiable many-nucleon mechanism.

A useful starting model, not an exact QCD identity, is a light-front impulse approximation. Let $y = p_N^+/P_A^+$ and let $\mathbf{p}_{NT}$ denote the transverse momentum of the active nucleon. Then

$$x_N = \frac{x}{y}, \quad \mathbf{k}_{T,\text{rel}} = \mathbf{k}_T - \frac{x}{y} \mathbf{p}_{NT}. \quad (20.1)$$

The second equation follows by resolving the quark momentum into its fraction x_N of the moving nucleon momentum plus its momentum relative to that nucleon. With a convention in which the spin spectral density $\mathcal{S}_{ss'}^{N,\rho}$ absorbs removal-energy integration and specified light-front spin rotations, a schematic correlator is

$$\begin{aligned} \Phi_A[\rho](x, \mathbf{k}_T) = & \sum_{N=p,n} \sum_{ss'} \int_x^1 \frac{dy}{y} \int d^2\mathbf{p}_{NT} \mathcal{S}_{ss'}^{N,\rho}(y, \mathbf{p}_{NT}) \Phi_N^{s's} \left(\frac{x}{y}, \mathbf{k}_T - \frac{x}{y} \mathbf{p}_{NT} \right) \\ & + \delta\Phi_{\text{rescatter}} + \delta\Phi_{\text{off-shell}} + \delta\Phi_{\text{nonnucleonic}}. \end{aligned} \quad (20.2)$$

The spin indices in this formula belong to the nucleon, while the nuclear spin state enters through $\mathcal{S}^{N,\rho}$. Its number normalization may be chosen as $\int dy d^2\mathbf{p}_{NT} \text{Tr}_s \mathcal{S}^{p,\rho} = 3$ and $\int dy d^2\mathbf{p}_{NT} \text{Tr}_s \mathcal{S}^{n,\rho} = 4$. A realistic implementation must state the wave function, spectral measure, spin basis, and treatment of binding; changing these conventions changes the factors assigned to $\mathcal{S}$.

In particular, multiplying two separately gauge-invariant nucleon objects does not automatically reproduce the nuclear staple and rescattering. Those effects require a controlled approximation or a separate calculation. Nuclear TMD studies explicitly demonstrate the importance of spin-orbit and rescattering effects, although a large-nucleus approximation is not automatically quantitative for $A = 7$ [37]. Equation (20.2) is a well-defined place to introduce and test a nuclear model; it is not used as a proof of the operator basis.

The momentum fraction throughout this note is per *whole nucleus*. A variable quoted per average nucleon is approximately $x_N^{\text{exp}} = Ax$, not the same variable as x . In pure QCD the nuclear momentum sum rule has the conventional form

$$\int_0^1 dx x \left[\sum_q (f_1^{q/A} + f_1^{\bar{q}/A}) + f_1^{g/A} \right] = 1. \quad (20.3)$$

The total momentum generator is spin independent; therefore the same weighted sum with $f_1 \rightarrow f_{1LL}$ vanishes. Individual quark and gluon tensor distributions need not vanish. Electromagnetic partons, if included in the factorization scheme, must be added consistently to the total momentum sum.

20.2 Setting up a nuclear calculation before projecting TMDs

A nuclear impulse calculation needs a nuclear light-front state or a specified approximation to it, the spin-dependent probability amplitude for removing a proton or neutron, and a nucleon correlator in a compatible spin and momentum convention. The nuclear spin indices have four values; the active nucleon's spin indices have two. They are connected by the nuclear wave function and spin recoupling, not by equating their polarization labels.

The factor $1/y$ in Eq. (20.2) follows from longitudinal momentum conservation. Before eliminating the nucleon parton fraction, insert $\delta(x - yx_N)$ and integrate over x_N . The Jacobian is $1/y$ and the argument becomes $x_N = x/y$. The support $0 \leq x_N \leq 1$ then sets $x \leq y \leq 1$ for the positive- x contribution. Resolving the parton transverse momentum relative to the moving nucleon gives $\mathbf{k}_T = x_N \mathbf{p}_{NT} + \mathbf{k}_{T,\text{rel}}$, proving the shift in Eq. (20.1). These are kinematic relations; they do not determine the spectral density.

To extract a nuclear coefficient, first compute the full spin-dependent nuclear correlator for a complete set of target responses. Then apply the Dirac or gluon projection, the nuclear K, Q projection, and the transverse harmonic inverse. Projecting an individual nucleon TMD and multiplying by a nuclear scalar polarization factor generally misses spin-orbital correlations and kinematic mixing caused by the shifted transverse momentum. Even a nucleon correlator with only lower-spin components can contribute to a nuclear octupole once those correlations are retained.

A reproducible approximation should state the spectral integration measure, removal-energy treatment, normalization to three protons and four neutrons, spin basis and any Wigner/Melosh rotations, and the handling of off-shell and nonnucleonic pieces. It should also state how the nuclear Wilson line and rescattering are treated. Multiplying separately gauge-invariant nucleon factors is not a derivation of the nuclear gauge link. Finally, test momentum and local-moment constraints and the appropriate positivity properties before interpreting a calculated coefficient as a physical nuclear prediction.

20.3 Vector polarization and independent higher-rank calibration

With the normalization of Section 3, the commonly quoted vector polarization fraction is

$$P_z = \frac{\langle J_z \rangle}{J} = \frac{2}{3} S_L. \quad (20.4)$$

It does not independently determine S_{LL} and S_{LLL} . For example, the spin-temperature ansatz

$$p_m = \frac{\exp(\beta m)}{Z(\beta)}, \quad Z = 2 \cosh(3\beta/2) + 2 \cosh(\beta/2) \quad (20.5)$$

ties all three longitudinal moments to one parameter. That is a preparation model, not a density-matrix identity. Selective RF saturation or other nonequilibrium manipulations need not satisfy it. A reported vector polarization alone cannot establish a calibrated quadrupole or octupole target state. No quantitative performance claim for a particular ⁷Li material is assumed in this note.

20.4 An ideal NMR population inversion, with its assumptions

For resolved, weakly probed transitions in the high-field secular limit, the transverse magnetic-dipole operator connects adjacent levels. The squared J_+ matrix elements are 3, 4, 3. Define signed population differences in descending m order by

$$d_1 = p_{3/2} - p_{1/2}, \quad d_2 = p_{1/2} - p_{-1/2}, \quad d_3 = p_{-1/2} - p_{-3/2}. \quad (20.6)$$

If the three line integrals have been corrected for frequency-dependent response, and the common calibration constant is C , their ideal signed areas are

$$I_1 = 3Cd_1, \quad I_2 = 4Cd_2, \quad I_3 = 3Cd_3. \quad (20.7)$$

Here the sign is a specified absorption/population convention; it is not inferred from an uncalibrated circuit voltage. Starting with $p_{-3/2} = t$, the other populations are $t + d_3$, $t + d_3 + d_2$, and $t + d_3 + d_2 + d_1$, where normalization gives $t = (1 - d_1 - 2d_2 - 3d_3)/4$. Substitution gives

$$S_L = \frac{3d_1 + 4d_2 + 3d_3}{2}, \quad S_{LL} = d_1 - d_3, \quad S_{LLL} = \frac{3}{10}(d_1 - 2d_2 + d_3). \quad (20.8)$$

Thus the octupole estimator uses transition-strength-corrected populations; it is not the same weighted sum as the total absorption. The formulas also provide a direct positivity diagnostic: the four reconstructed populations must lie between zero and one.

A real solid can have overlapping quadrupolar lines, a powder orientation distribution, background phase and gain, unequal RF excitation, or significant state mixing. Then the appropriate forward NMR response must be fitted or inverted before applying the population formulas. These complications affect the measurement of ρ , not the spin algebra. A diagonal-population measurement determines only S_L, S_{LL}, S_{LLL} in that axis. The $m > 0$ target components require controlled rotations or independent coherence tomography. Under a rotation the irreducible rank K is preserved; rotations mix its $2K + 1$ components but cannot manufacture rank-three polarization from a state containing only rank one.

20.5 Spin temperature is an additional relation among populations

The spin-temperature ansatz in Eq. (20.5) gives one parameter for all four populations. It is useful when the preparation is well described by that equilibrium form, but it is not a consequence of the spin-3/2 density matrix. To see its content, expand $\exp(\beta m_J)$ around $\beta = 0$ and normalize by Z . Using the eigenvalues in Eq. (3.24) gives

$$S_L = \frac{5}{4}\beta - \frac{17}{48}\beta^3 + O(\beta^5), \quad (20.9)$$

$$S_{LL} = \frac{1}{2}\beta^2 + O(\beta^4), \quad (20.10)$$

$$S_{LLL} = \frac{3}{40}\beta^3 + O(\beta^5). \quad (20.11)$$

The first nonzero orders differ. This explains why a small vector polarization can coexist with still smaller higher moments in this *particular preparation model*. The density-matrix space itself allows an octupole-only state with zero vector polarization. In the opposite limit $\beta \rightarrow +\infty$, the state approaches $m_J = 3/2$, with $(S_L, S_{LL}, S_{LLL}) = (3/2, 1, 3/10)$. Even full vector polarization does not maximize the unconstrained magnitude $|S_{LLL}|$.

Selective saturation, coherent rotations, and nonequilibrium populations need not obey these spin-temperature relations. A measurement that quotes only P_z determines only one moment unless the preparation model is independently justified. The purpose of the three-transition NMR inversion is to obtain the other longitudinal moments directly from calibrated population information.

20.6 Deriving transition strengths and propagating calibration errors

For a weak transverse probe, the transition probability is proportional to a squared matrix element of J_x or its rotating component J_+ . The ladder identity gives strengths 3, 4, 3 for the adjacent transitions. The common rotating-field normalization is absorbed into the calibrated constant C . Net absorption is proportional to the appropriate population difference, so the unequal strengths must be divided out before combining line areas into moments. The signed convention and frequency-dependent circuit response are part of that calibration.

The inversion in Eq. (20.8) can be written directly in terms of corrected line areas:

$$S_L = \frac{I_1 + I_2 + I_3}{2C}, \quad S_{LL} = \frac{I_1 - I_3}{3C}, \quad S_{LLL} = \frac{2I_1 - 3I_2 + 2I_3}{20C}. \quad (20.12)$$

Only the vector moment is proportional to the total signed absorption in this ideal limit. The octupole uses a different contrast, especially sensitive to the relative central-line area. A baseline or overlap error that leaves the total area almost unchanged can therefore bias the inferred octupole substantially.

For an explicit uncertainty calculation, put $\mathbf{d} = (d_1, d_2, d_3)^T$ and $\mathbf{s} = (S_L, S_{LL}, S_{LLL})^T$. Then

$$\mathbf{s} = \mathbf{M}\mathbf{d}, \quad \mathbf{M} = \begin{pmatrix} 3/2 & 2 & 3/2 \\ 1 & 0 & -1 \\ 3/10 & -3/5 & 3/10 \end{pmatrix}, \quad \Sigma_s = \mathbf{M}\Sigma_d\mathbf{M}^T. \quad (20.13)$$

The covariance of $\mathbf{d}$ follows from that of the line integrals and C , including common calibration correlations. In particular, an uncertainty in the common scale affects all three moments together. For resolved lines and known C , the four populations follow from normalization and the three differences. Their nonnegativity is an immediate check of the reconstructed state.

In a solid, overlapping quadrupolar lines, orientation distributions, RF inhomogeneity, circuit phase, saturation, and state mixing can invalidate a simple assignment of three measured areas to three population differences. Then first construct and fit the appropriate NMR forward model, and propagate its parameter covariance into ρ . The spin-moment identities remain valid after that step. Population spectroscopy in one axis does not determine arbitrary complex coherences; controlled rotations or additional tomography measurements are needed for the $m > 0$ components.

20.7 An explicit octupole-only response combination

There is a simple physically allowed preparation pair that illustrates why the third rank is independently measurable. Use longitudinal states with

$$S_L = S_{LL} = 0, \quad S_{LLL} = \eta O_0, \quad \eta = \pm 1, \quad 0 < O_0 \leq \frac{1}{2}, \quad (20.14)$$

and all transverse components zero. At $O_0 = 1/2$, the positive state has the population vector

$$(p_{3/2}, p_{1/2}, p_{-1/2}, p_{-3/2}) = (1/3, 0, 1/2, 1/6). \quad (20.15)$$

Reversing the level order gives the negative state. These are density-matrix examples, not a claim that a specified apparatus already prepares them.

Let $\sigma_{\lambda,\eta}$ denote the differential SIDIS cross section in the common phase-space convention, with fully polarized lepton helicity $\lambda = \pm 1$. Table 3 gives

$$\sigma_{\lambda,\eta} = \mathcal{N} \left[\mathcal{U} + \lambda \eta O_0 D_\ell F_{30}^L + \eta O_0 \varepsilon \sin(2\phi_h) F_{30}^{T[1]} \right], \quad (20.16)$$

where $\mathcal{U}$ includes the unpolarized-target responses. Therefore

$$\frac{1}{4} \sum_{\lambda=\pm 1} \sum_{\eta=\pm 1} \lambda \eta \sigma_{\lambda,\eta} = \mathcal{N} O_0 D_\ell \mathcal{C}[g_{30}^q D_1]. \quad (20.17)$$

This double difference isolates an octupole helicity *structure function*, not the quark TMD by itself. The beam-averaged target difference instead selects the $\sin 2\phi_h$ octupole Collins response. For partial beam polarization the measured helicity factor must be included. Unequal luminosities, efficiencies, or residual vector polarization require calibration; in particular, a nonzero residual S_L leaks the ordinary helicity response into Eq. (20.17).

20.8 A staged determination of TMDs from data

The operator decomposition determines which functions can exist. Their numerical extraction is a separate inverse problem. First fit the *structure functions* by combining angular distributions, beam states, and independently measured target density matrices. For bin b and preparation a , a schematic response model is

$$N_{ab} = \mathcal{L}_a \int_b d\mathcal{P} \mathcal{A}_a(\mathcal{P}) \mathcal{N}(\mathcal{P}) \sum_\nu M_{a\nu}(\mathcal{P}; \rho_a, \lambda_a) F_\nu(\mathcal{P}) + N_{ab}^{\text{background}}, \quad (20.18)$$

where $\mathcal{L}$ is integrated luminosity, $\mathcal{A}$ is acceptance, and M contains the known spin amplitudes, angular functions, and lepton factors. Binning, resolution migration, and uncertainties in ρ_a belong in this model, rather than in redefined TMDs.

The response matrix must have adequate rank over the actual target preparations and acceptance. Distinct correlator coefficients may share an azimuthal harmonic in one fixed state, so merely fitting more Fourier modes cannot replace target-state variation. The singular values or Fisher information of the response design provide a quantitative identifiability check.

Only after that step should one fit flavor-dependent TMDs through the convolution, fragmentation functions, scale evolution, and specified factorization scheme. Different detected hadrons, kinematics, and processes constrain different combinations. A single measured structure function generally does not uniquely determine both a TMD and a fragmentation function. Finite-impact-parameter or Bessel-weighted methods can simplify a convolution, but do not remove these flavor and nonperturbative ambiguities. Unregulated polynomial k_T moments must not be treated as automatically UV-finite data products. Target-density positivity is exact; correlator positivity must retain the scheme qualifications in Section 13.

20.9 Designing an identifiable structure-function extraction

At fixed kinematic bin, and after correcting luminosity and acceptance with a specified forward model, the response can be organized as a linear system $\mathbf{y} = \mathbf{X}\mathbf{F} + \boldsymbol{\eta}$, where $\mathbf{F}$ contains the structure functions and $\boldsymbol{\eta}$ contains noise and modeled backgrounds. The rows of $\mathbf{X}$ contain known target moments, beam factors, and angular weights, integrated over the same bin and detector response as the data. For Gaussian errors with covariance V , an identifiable linear problem has estimator

$$\hat{\mathbf{F}} = (\mathbf{X}^T V^{-1} \mathbf{X})^{-1} \mathbf{X}^T V^{-1} \mathbf{y}, \quad \text{Cov}(\hat{\mathbf{F}}) = (\mathbf{X}^T V^{-1} \mathbf{X})^{-1}. \quad (20.19)$$

For low counts one instead uses the corresponding Poisson likelihood. These expressions assume that the response matrix and background model are specified; uncertainties in the target moments or acceptance must be fitted or propagated rather than treated as exact.

A simple example makes the rank issue concrete. Suppose only the azimuth-independent helicity response is measured in several diagonal target states. The two columns for ordinary and octupole helicity are proportional to $S_L^{(a)}$ and $S_{LLL}^{(a)}$. If every state has $S_{LLL}^{(a)} = c S_L^{(a)}$ with the same c , the two columns are linearly dependent. No amount of statistics distinguishes the two functions in that design. Independent population control, or a calibrated octupole-only state, supplies a new direction in the response space. More angular bins alone cannot remove this degeneracy.

A useful sequence for an analysis is therefore to define the measured phase space and axes; reconstruct or constrain the density matrix for each preparation; calculate the response weights; examine the design matrix singular values; fit the structure functions; and only then fit flavor-dependent TMDs through the fragmentation or gluon-sensitive factorization formula. The final TMD problem is not the same linear inversion as Eq. (20.19): it can contain unknown fragmentation functions, evolution, flavor sums, and nonlinear shape parameters. Additional final states and kinematics supply the missing independent information.

The difference between formal coefficient independence and experimental identifiability is essential. The correlator basis is complete at the operator level. A particular experiment may determine only a subset or certain linear combinations. Reporting those combinations with their conventions and covariance is a physical result, not a failure of the tensor construction.

21 Computational implementation and independent validation

The computational companion [26, 38] implements the finite constructions used in this note. Its role is to make the coefficient maps, response kernels, independent comparisons, and assumptions inspectable. It does not supply the unknown radial TMDs. SymPy provides exact matrix and polynomial arithmetic [24]; NumPy provides numerical arrays and linear algebra [25]; mpmath

provides independently recomputed higher-precision values [39]. The implementation description below summarizes scientific methods and evidence rather than installation instructions or programming lessons. The repository maintains the runnable interfaces, reproduction commands, and versioned reports separately.

21.1 Versioned scope and the status of the evidence

The public baseline documented here is revision `19fa1a9719ca71fe5325243cf695d2ded2d890b4`, with reported scientific-source digest `2e5bfeb14c88854c1da992a8c429cd99fcd07a1f7248ff20e96dfed30abff13d`. The corrected companion paper is identified by manuscript digest `7436c7b7dab536999f7caf42af54c5820fa474ca81ab4ff8e4cf15259c9a24ee`. This note is a separate document and has its own source fingerprint. A commit or digest identifies an artifact; it does not establish a mathematical result merely by being recorded.

The release record supplied for that public revision reports 591 required computational records, 64 software tests, independent certificates, strict documentation checks, and a deployment of the same merged revision. These are release-specific observations, not a newly executed 591-record run accompanying this note. Source-level inspection available for this revision used the supplied corrected scientific archive at `1ede3762e6e3f2cf3a0809fd170b4b19b8f5da62`. The intervening release records report unchanged scientific results and improvements to evidence validation, metadata, documentation, and deployment. A fresh retrieval of GitHub’s moving `main` branch was unavailable during preparation. Accordingly, the scientific construction is identified by its reviewed source definitions, while the final release inventory and maintenance status are attributed to that supplied publication record. Neither is silently presented as a fresh audit of a later revision.

The cumulative profiles organize the implemented coverage as follows. Their totals are not additive, and one record may contain many component identities or kinematic evaluations.

Table 6: Versioned computational-profile inventory. Each row includes all preceding layers; records and software tests are different units.

Profile	Records	Additional scientific scope
<code>baseline</code>	58	Historical symbolic checks, canonical Born point and grid, and software regressions.
<code>foundations</code>	194	Independent spin/STF constructions, dictionary, covariant certificates, projectors, and physical target states.
<code>quark-processes</code>	371	All SIDIS/DY rows, independent convolutions, current normalization, and conditional reversal algebra.
<code>gluon-processes</code>	437	Gluon responses and separation, synthetic recovery, independent Born amplitudes, precision checks, and scans.
<code>limits-positivity</code>	579	Collinear/Fourier checks, local moments, nuclear variables, and conditional positivity.
<code>full</code>	591	Physical-helicity/current-order closure and source/spectral versus auxiliary matrix conventions.

The older 49-check validator remains a reproducible baseline, not the scope of the present companion. Likewise, successful completion of the full computational profile does not mean that

all analytical QCD statements have been proved by finite computation. The distinction between executable claims and analytical assumptions is part of the validation definition.

21.2 Independent construction versus repetition

A useful comparison changes the route by which a mathematical object is constructed. The spin sector starts from the ladder matrix elements and solves the quadratic and cubic trace-removal constraints, giving $5/4$ and $41/60$. A separate Clebsch–Gordan construction gives spherical operators and their trace metric. Their Cartesian conversion checks normalization, conjugation, phase conventions, and all sixteen Hermitian density coordinates, including imaginary off-diagonal entries. Traces of the physical preparations verify their vanishing vector/quadrupole moments and their positivity interval; a real rank-three response basis and seven directions give the exact tomography determinant $-1400/35937$ in the stated ordering and normalization.

For the transverse sector, Cartesian harmonic projection of a raw monomial is compared with its two complex-helicity components through rank five. All traces, permutation symmetries, rotation phases, duals, and powers of the positive symbolic mass M_A are retained. In particular, a calculation at $M_A = 1$ alone cannot expose a missing M_A^{-n} factor. The reference-STF and dyadic contractions in Eqs. (6.9) and (6.10) are constructed separately. The dyadic has the nonzero residual $-\mathbf{k}_T^2 S_{TT}^{ij}$; it cannot pass as the null STF structure.

The lower-spin gluon dictionary is checked by independently entering the original reference tensors, including Lorentz index lowering, and the minimal Cartesian correlator. Trace, antisymmetric, and STF projections then recover Eqs. (6.18) and (6.23). The coefficient conversion has rank eighteen and the common TT -shift kernel described in Sec. 6.8. Applying a dictionary and its inverse from the same function would not provide this independent comparison. Equal named polarization components, equal Cartesian tensors, operator normalization, and mass normalization remain separately tested choices.

21.3 Completeness, exact certificates, and coefficient recovery

The explicit Cartesian covariants are compared with an independent transverse-helicity and reflection-parity enumeration. This tests the actual tensor columns, not only the expected totals $N_q = N_g = 32$ and $(N_0, N_1, N_2, N_3) = (2, 6, 10, 14)$. The four real parton components are paired with sixteen Hermitian target coordinates, giving a 64×32 real linear map for each species. A normalized physical state still has only fifteen independent polarization coordinates; the extra scalar coordinate in this linear-map construction is the trace sector, not an additional polarization.

At the exact nonzero point $(k_x, k_y, M_A) = (2, 1, 3)$, the stored row and column conventions give maximal-minor determinants

$$\det C_q^{\text{minor}} = \frac{31719424}{12157665459056928801}, \quad (21.1)$$

$$\det C_g^{\text{minor}} = \frac{1936000}{79766443076872509863361}. \quad (21.2)$$

Their nonzero values establish a rank lower bound. The determinant magnitudes depend on coordinate normalization and are not observables. The separately derived reflection-invariant Hermitian matrix space has real dimension 32, supplying the upper bound and spanning statement. A specialized nonzero minor alone would not supply that argument or prove invertibility at exceptional zero-momentum points.

Certificate verification reconstructs the relevant entries from the alternative helicity route and checks the row/column order, evaluation point, exact arithmetic, and provenance. It does not

merely read a stored `rank:32`. Analytical coefficient projectors are also applied to correlators built from each independent coefficient and to mixed exact combinations. Their results agree with recovery through the separate Cartesian linear map. These comparisons check signs and normalization that rank alone cannot determine: a sign error in one nonzero column can preserve full rank. The source–auxiliary transformation in Eq. (13.9) is permitted for coordinate-map checks, not as a positivity-preserving transformation of physical inputs.

21.4 Quark response kernels and independent convolution checks

The Dirac algebra is checked in independently built Dirac and chiral representations with an explicit conversion. All twelve distribution–fragmentation projection pairs in SIDIS and all sixteen distribution–distribution pairs in DY are evaluated for every transverse photon-index choice. The comparisons include vanishing and antisymmetric terms. The current and lepton contractions are then applied to the constructed target covariants and compared with independently curated response fixtures: 32 SIDIS rows and fourteen octupole DY rows, rather than a table regenerated by the production routine being tested.

Every row retains its target amplitude, beam factor, angular phase, orbital weight, flavor labels, and partner distribution or fragmentation function. The high-rank TTT Collins factor $1/2$, the DY Boer–Mulders partner sign, and the physical antiquark axial mapping are consequently checked at the complete-response level. Direct on-shell massless QED currents corroborate the DY $1 + \cos^2 \theta_\ell$ and $\sin^2 \theta_\ell$ terms, the physical longitudinal analyzing sign, and the flux/color normalization. These current checks are separate from Wilson-line reversal.

Three routes compare the integrated responses: the full Cartesian correlator contraction after direct delta-function elimination, the independently curated harmonic weights, and Gaussian generating-function moments. These convergent radial inputs are integration fixtures, not nuclear predictions. Their unequal widths, nonunit masses and z , and independent quark/antiquark flavor weights prevent accidental equality from concealing missing factors.

The momentum-conservation sign has a particularly discriminating test. For positive widths w_A, w_B and Gaussian product $\exp(-k^2/w_A - p^2/w_B)$ subject to $\mathbf{k} + \sigma_p \mathbf{p} = \mathbf{r}$, completing the square gives

$$\langle \mathbf{k} \rangle = \frac{w_A}{w_A + w_B} \mathbf{r}, \quad \langle \mathbf{p} \rangle = \sigma_p \frac{w_B}{w_A + w_B} \mathbf{r}. \quad (21.3)$$

The manuscript’s SIDIS variable has $\sigma_p = -1$, whereas incoming DY momenta have $\sigma_p = +1$. The unweighted radial integral is identical for the two signs. An odd moment in the second momentum, or a nonzero Collins/Boer–Mulders weighted response, distinguishes them. Thus a successful unpolarized Gaussian check is not sufficient evidence for the fragmentation momentum convention.

For Gaussian-weighted polynomial fixtures, sufficiently high-order Gauss–Hermite quadrature is exact before floating-point rounding; an N -node one-dimensional rule covers polynomial degree through $2N - 1$. The reported rapid convergence of those fixtures is therefore not a convergence theorem for arbitrary unknown TMD shapes. Oscillatory Fourier integrals require their own numerical convergence checks.

21.5 Reversal relations and matched operator definitions

The antiunitary check transforms the complete complex target matrix and the partonic projection, including explicit imaginary factors. Together with the stated analytical field/link transformation rule it recovers all 64 quark/gluon signs in Eq. (7.5). The octupole subsets are three odd and eleven even quark coefficients, and ten odd and four even gluon coefficients. Flavor, antiquark

status, parent hadron, orbital rank, and link/color class are distinct labels. Reversing a simple gluon pair twice restores the same pair; it never identifies $[+, +]$ with $[+, -]$.

Scale/subtraction matching is part of the comparison. The conditional evolution check uses the intertwining relation in Eq. (7.11), not an arbitrary common mixing matrix. A synthetic two-process comparison applies the target sign relation inside the independently defined SIDIS and DY convolutions, retaining different partner functions. This checks that the operator null relation can hold while the complete asymmetries are neither equal nor opposite. It is not an empirical universality test or a calculation of a QCD Wilson-line matrix element.

21.6 Gluon responses, physical currents, and numerical diagnostics

The gluon response is checked both as the source-ordered matrix trace $\text{Tr}(BD_g)$ and through complex helicity tensors. Real, circular, and elliptic incoming polarizations distinguish a one-sided transpose that real-only tests can miss. All 32 target coefficients and their separate scalar, circular, and low/high linear-polarization terms are compared with independently specified expectations. The fourteen octupole cone responses are evaluated using physical density operators, whereas their reconstruction uses independently specified Fourier modes.

The angular certificate reconstructs the Gram matrix of Eq. (19.12) with beam-helicity averaging and verifies its determinant $1/8192$. Coefficient recovery additionally checks the nonzero amplitude factors. Continuous and eight-angle transverse-target projectors remove the accompanying LLT term. Missing circular or linear analyzers, zero cone amplitudes, beam averaging, insufficient preparation variation, and aliased sampling produce explicit losses of information. A known synthetic acceptance tests leakage under naive moments and recovery through the appropriate response matrix; it does not determine unknown experimental acceptance or target performance.

The Born comparison uses independently coded massive-quark spinors, massless lepton currents, and the sum of the two diagram amplitudes, against the production Dirac trace. Spinor completeness, color traces, the single initial-gluon spin average, current ordering, and overall scaling are checked before ratios are formed. A common multiplicative error can preserve both Ward cancellations and normalized analyzing powers, so agreement of those diagnostics alone does not establish the absolute hard-matrix normalization.

Physical electron helicity is anchored to the eigenvalue along the incoming momentum. The corrected production tensor satisfies $J_{\mu\nu}^\ell(\lambda) = L_{\nu\mu}(\lambda)$ and agrees with direct amplitudes at the *same* helicity. The old same-label result is retained only as a version-specific negative regression. At the reference point it reverses b_G and produces an off-diagonal matrix difference of approximately 247.3 GeV^2 ; its agreement after replacing λ by $-\lambda$ is not the canonical physical-helicity test. No target g function is redefined to absorb that hard-current error.

The reference grid contains 36 specified polar-angle, azimuth, and beam polarization tuples. The separate dense scan evaluates 161 uniformly spaced angles on $[0.05, \pi - 0.05]$ at fixed $\phi = 0.4$ and each of $\lambda = -1, 0, +1$, for 483 actual amplitude evaluations. The corrected release records compare the full complex matrix with direct spinor amplitudes at every grid and scan tuple. Four further cases recompute kinematics and amplitudes at 50 and 80 decimal digits. This is a distinct arithmetic comparison of the same two-diagram physics, not a third Feynman derivation. The reported scan differences are at the 10^{-12} GeV^2 level, high-precision/double-precision comparisons at the 10^{-13} – 10^{-12} GeV^2 level, and synthetic coefficient-recovery errors at the 10^{-14} level. Exact case values and tolerance definitions belong to their versioned payloads, not to a universal floating-point precision claim.

Kinematic conservation and invariant-shell residuals carry different units and are recorded separately in the independent diagnostics. Ward contractions are tested between on-shell states or completeness projectors, not as vanishing bare vertices. Hermiticity is tested on the raw complex

matrix before symmetrization or a Hermitian eigensolver; eigenvalues and analyzing-vector norms cannot determine the sign of b_G . Near a zero amplitude, a nonvanishing diagram/norm scale and an absolute residual accompany any relative test. No negative eigenvalue is clipped away before the original input is classified. Invalid or axis-degenerate kinematics are separate domain outcomes.

21.7 Limits, moments, and conditional positivity

All 64 Cartesian covariants are angularly projected and compared with helicity harmonics before straight-link selection is applied. Seven quark and six gluon rank-zero candidates reduce to the six and five collinear structures in Sec. 12. The excluded quark $h_{21}^{q[0]}$ and gluon $h_{32}^{g[0]}$ remain legitimate finite- k_T coefficients. A synthetic radial tail $1/[\pi(k^2 + \Lambda^2)]$ has disk integral $\ln(1 + R^2/\Lambda^2)$; this explicit cutoff dependence prevents angular selection from being misreported as a convergent renormalized PDF integral.

Fourier checks compare Cartesian differentiation of the scalar transform, angular/Bessel reduction, and independently coded Cartesian integration through rank five. Both helicity branches, the STF normalization, M_A^{-n} , the positive-exponential phase, and the inverse $(2\pi)^{-2}$ measure are retained. Gaussian fixtures permit exact identities, while smooth quartic radial fixtures and nonzero transverse separations test numerical integration beyond that family. Odd-rank transforms can be imaginary; reality is expressed by $\tilde{F}(\mathbf{b})^* = \tilde{F}(-\mathbf{b})$, not by forcing every transformed component to be real.

The local-moment checks construct the rotational content of the bilinears and derivative indices by independent character and coupled-spin routes. The antisymmetric tensor bilinear contributes two rotational vectors, not a spin-two operator. Combining this content with the spin-3/2 operator space gives the absence of rank-three reduced quark moments for $N < 3$ and their allowed, but not necessarily nonzero, value at $N = 3$. Charge conjugation and the negative-to-positive- x variable change give the alternating quark–antiquark combinations. These are finite checks conditional on the identification with renormalized local operators and on any ordinary-integral convergence assumption. Nuclear support, relative transverse momentum, and the Mellin Jacobian y^{N-1} are checked without assigning nuclear wave functions or individual tensor-momentum sum-rule values.

Positivity checks distinguish physical target states, the perturbative hard Gram matrix, and a partonic correlator in a specified positive spectral prescription. The source-index joint matrix and its mixed-target contraction are checked with imaginary coherences and independently constructed spectral amplitudes. Actual collinear matrices are permuted into their coupled blocks, supplying exact certificates for the factors in Eqs. (13.13) and (13.14). Interior, saturated, zero-diagonal, and indefinite examples test both necessity and sufficiency. Pairwise-minor and leading-minor counterexamples, the partial-transpose example, and negative-trace fixed-target examples ensure that weaker criteria cannot accidentally certify positive semidefiniteness.

21.8 Implementation map and evidence integrity

The principal scientific modules are grouped in Table 7. This is a correspondence between mathematics and implementations, not a replacement for the defining operator equations. The public documentation describes the current entry points and stores the reviewed fixtures independently of the production response generators.

Table 7: Scientific implementation map; filenames are relative to `src/` in the computational companion.

Mathematical role	Implementations
Spin/state construction	<code>spin_foundations.py</code> , <code>spin_state_foundations.py</code> ; the historical entry point delegates to <code>spin32_symbolic.py</code> .
STF and reference conversion	<code>transverse_foundations.py</code> , <code>gluon_dictionary_foundations.py</code> .
Covariants and inverses	<code>correlator_foundations.py</code> , <code>projector_foundations.py</code> , <code>foundation_certificates.py</code> .
Quark hard responses	<code>process_dirac.py</code> , <code>process_responses.py</code> , <code>response_fixtures.py</code> , <code>process_normalization.py</code> , <code>process_observables.py</code> .
Convolutions and reversal	<code>process_convolutions.py</code> , <code>process_reversal.py</code> .
Gluon responses and separation	<code>gluon_stokes.py</code> , <code>gluon_responses.py</code> , <code>gluon_response_fixture.py</code> , <code>gluon_octupole.py</code> , <code>gluon_angular.py</code> , <code>gluon_reconstruction.py</code> .
Born/current comparisons	<code>gluon_born_response.py</code> , <code>born_direct.py</code> , <code>born_precision.py</code> , <code>born_validation.py</code> , <code>current_index_conventions.py</code> .
Limits and local moments	<code>collinear_limits.py</code> , <code>fourier_limits.py</code> , <code>local_moments.py</code> .
Conditional positivity	<code>conditional_positivity.py</code> , <code>source_joint_positivity.py</code> , <code>collinear_block_certificate.py</code> .
Run evidence	<code>validation_manifest.py</code> , <code>validation_evidence.py</code> ; runner, certificate, summary, plotting, scope, and publication utilities are under <code>scripts/</code> .

Each run identifies its required and executed claim sets, source/input and convention digests, arithmetic or numerical domain, dependencies, result type, and completion state. Missing, duplicated, unexpected, failed, nonfinite, or mixed-run evidence cannot be replaced by an older successful report. Certificate checkers reconstruct mathematical entries and compare their provenance. A scientific mutation test succeeds only when the intended wrong identity or input is detected; an unrelated parser or import error does not establish that detection.

The final reported maintenance updates require the exact diagnostic-key set in every Born scan case, with explicitly Boolean success values. An empty collection cannot pass through the vacuous truth of a generic all-elements test. The pre-build gate also checks the *live* source tree rather than only the clean state recorded in an earlier manifest; legitimate derived reports and figures are generated after that input check. High-precision convention identifiers and explanatory metadata have separate fields, and the mass-rescaling description agrees with Eq. (14.5). The documentation distinguishes the corrected canonical calculation from explicit legacy reproduction, identifies current versus historical manuscript provenance, and includes regressions for damaged mathematical markup.

Scientific validation and permission to publish remain separate. The reported Pages procedure uses a default-disabled, manually authorized exact commit, validates that checkout, builds its reports and site, and deploys only that successful artifact. It does not grant blanket approval to a later commit. Public files comprise code, scientific fixtures, certificates, result data, diagnostic

figures, and explanatory Markdown; manuscripts and private review materials remain outside the repository. These controls improve traceability and prevent stale success from being misstated as new evidence. They are software/reproduction properties, not additional QCD identities.

21.9 What the computational agreement establishes

The independent comparisons support the finite spin algebra, physical transverse basis, coefficient inverses, Born responses, and qualified limit/positivity constructions stated in this note. Their independence is specific: Cartesian and helicity tensors differ as constructions; direct spinor and trace methods differ as implementations of the same Born diagrams; higher precision changes arithmetic rather than physics. Shared conventions, SymPy arithmetic, and previously checked spin primitives are disclosed rather than counted as new independent physical inputs.

The remaining analytical inputs are parity and the field/link transformation rules, admissible UV and rapidity subtraction, process-specific factorization and soft/measurement structure, local operator identification and endpoint convergence, and the chosen positive spectral prescription where positivity is used. Loop matching coefficients, QCD evolution kernels, nuclear matrix elements, and experimental sensitivity are not outputs of these finite tests. The repository therefore supports a complete, reproducible realization of the declared computational scope, while the observable formulas and operator assumptions retain the precise qualifications stated in the analytical sections.

22 Synthesis and outlook

The formalism is a sequence of distinct constructions. A positive spin-3/2 density matrix determines fifteen real polarization components. The irreducible ranks $K = 1, 2, 3$ separate vector, quadrupole, and octupole information. A gauge-invariant quark or gluon operator then supplies four real leading-twist parton polarization projections. Coupling target helicities to two-dimensional orbital STF harmonics, and applying parity, gives thirty-two independent coefficients for each parton species. Fourteen belong to the target octupole sector. Explicit coefficient inverses and the full population/coherence map make these definitions operational.

The equality of quark and gluon totals does not make their helicity- flip channels equivalent. Quark transverse spin changes parton helicity by one unit; gluon linear polarization changes it by two. Their allowed orbital ranks, link-reversal properties, and collinear survivors differ. The minimal gluon parametrization also separates a pure-trace unprojected term from an independent symmetric-traceless response. This is a basis reduction with an explicit coefficient translation, not a loss of a physical observable.

The six quark and five gluon collinear distributions are obtained by an angular and straight-link symmetry test with UV renormalization kept explicit. Population and coherence contrasts explain their physical meaning, and local quark operators restrict the moments that an octupole can carry. The standard quark SIDIS hard trace and fragmentation correlator then convert the target basis into the complete thirty-two-response Born catalogue. A separately specified gluon hard matrix converts the same target multipoles into gluon- sensitive responses; the heavy-quark Born example shows how to construct and verify that matrix directly.

The SIDIS–Drell–Yan comparison additionally fixes which target functions are shared or sign reversed when the Wilson staples are reversed at matched scales and subtraction conventions. The three number, four helicity, and seven transverse-spin octupole DY channels require distinct partner analyzers. For gluons, positive preparations and the joint recoil–preparation Fourier basis separate fourteen responses under the stated nondegeneracy conditions. The physical beam-helicity dependence uses the corrected $L_{\nu\mu}$ Born contraction, without changing operator

universality or adding a compensating sign to a target TMD.

The established ingredients are identified throughout: spin representation theory, lower-spin distributions, the fragmentation classification, forward GPD/PDF constraints, TMD operator definitions, standard hard amplitudes, and evolution. The project-specific construction combines them into the target-side spin-3/2 tensor basis, its minimal gluon organization, and explicit octupole response and preparation maps. The additional worked Gaussian and matrix examples are teaching devices. None is a fitted ⁷Li TMD.

A graduate calculation can now proceed in stages with definite outputs. Reproduce the spin and STF identities; recover a coefficient from a trial correlator; derive a selected SIDIS or gluon response; check its collinear, parity, and dimensional limits; and design a set of target preparations whose response matrix has adequate rank. A quantitative nuclear application then adds a specified wave function or spectral density, gauge-link/rescattering treatment, fragmentation or final-state measurement model, and scale evolution. Those dynamical choices determine sizes and shapes. The present construction determines which quantities they must calculate and how the resulting matrix elements enter a measured cross section.

Computational access. The reference implementation is collected in the Li7-TMDs repository [26]. Section 21 describes independent constructions, coefficient and angular certificates, corrected physical-current comparisons, numerical diagnostics, and the repository’s versioned validation evidence. Exact construction, basis verification, numerical diagnostics, and non-perturbative modeling remain distinct tasks. The code supports reconstruction of the stated mathematics rather than replacing its physical assumptions.

A Rank-four and rank-five transverse momentum tensors

For readers implementing the basis without tensor software, the rank-four and rank-five dimensionless STF tensors are

$$\begin{aligned} k_4^{i_1 \dots i_4} &= \frac{1}{M_A^4} \left[k^{i_1} \dots k^{i_4} - \frac{\mathbf{k}_T^2}{6} \sum_6 \delta^{i_1 i_2} k^{i_3} k^{i_4} + \frac{(\mathbf{k}_T^2)^2}{24} \sum_3 \delta^{i_1 i_2} \delta^{i_3 i_4} \right], \\ k_5^{i_1 \dots i_5} &= \frac{1}{M_A^5} \left[k^{i_1} \dots k^{i_5} - \frac{\mathbf{k}_T^2}{8} \sum_{10} \delta^{i_1 i_2} k^{i_3} k^{i_4} k^{i_5} + \frac{(\mathbf{k}_T^2)^2}{48} \sum_{15} \delta^{i_1 i_2} \delta^{i_3 i_4} k^{i_5} \right]. \end{aligned} \quad (\text{A.1})$$

Each indicated sum is over distinct index placements, without dividing by the number of terms. Contracting any two indices with δ_{ij} gives zero. These formulas agree with the complex-component rule in Section 4; in particular, at $\mathbf{k}_T = (k, 0)$ the all- x components are $k^4/(8M_A^4)$ and $k^5/(16M_A^5)$. The rank-four and rank-five expressions can be reproduced either by subtracting traces with undetermined coefficients or by the complex harmonic rule in Eq. (4.12). These are exact physical two-dimensional identities. The computational implementations and reproduction instructions are cited through the project repository [26].

B Notation and implementation checklist

Table 8: Index spaces and frequently used symbols.

Symbol	Meaning and convention
J, m_J, Λ	Total target spin and its spin/helicity projections. Here $J = 3/2$, and the four-state basis is ordered from $3/2$ to $-3/2$.
K, Q	Irreducible target-spin operator rank and spherical component. $K = 0, 1, 2, 3$, $Q = -K, \dots, K$. The spherical index Q is not the hard scale.
m	Magnitude of the target's transverse helicity component, $m = 0, \dots, K$.
r, n	Transverse parton-channel rank and orbital STF momentum rank. $r = 0, 1, 2$ as specified by the projection.
a, b, c on J, Q, O	Three-dimensional Cartesian spin indices, x, y, z . Other uses as amplitudes or Gaussian widths are local to their stated examples.
i, j, k	Euclidean transverse indices, x, y , contracted with δ_{ij} .
μ, ν, α	Lorentz indices, contracted with metric $(+ - - -)$.
ρ, S, T, R	Normalized target density matrix and its vector, quadrupole, and octupole expectation tensors.
τ_{KQ}, A_{Km}	Orthonormal spherical spin operators and the specified longitudinal/transverse target moments.
A_+, κ	Complex transverse helicity encoding and $(k_x + ik_y)/M_A$. These are not light-front plus components.
$v^\pm$	Light-front components $(v^0 \pm v^3)/\sqrt{2}$.
$E, \tilde{A}, \text{STF}_2$	$E_{xy} = +1$, one-index transverse dual, and two-dimensional symmetric trace subtraction.
$\mathbf{k}_n, \mathcal{C}_m, \mathcal{S}_m$	Dimensionless momentum STF tensor and its cosine- and sine-type contractions with a target component.
x, x_{Bj}, x_g	Target parton fraction, measured Bjorken variable, and Born heavy-pair active gluon fraction; x_A, x_B in DY label the two incoming hadrons.
y, z	SIDIS lepton energy-loss and hadron fractions. In the nuclear convolution only, y denotes the explicitly stated nucleon plus fraction.
$\mathbf{k}_T, \mathbf{p}_T, \mathbf{j}_T$	Incoming parton transverse momentum, outgoing quark momentum relative to the hadron, and hadron momentum relative to its quark; $\mathbf{j}_T = -z\mathbf{p}_T$.
Φ, Γ, Δ	Quark distribution, gluon distribution, and quark fragmentation correlators.
$\mathcal{F}, \mathcal{G}, \mathcal{T}, \mathcal{H}$	Number, helicity, quark transverse-spin, and gluon linear-polarization projections.
$f_{Km}^a, g_{Km}^a, h_{Km}^{a[n]}$	Radial TMD coefficients for species $a = q, g$. The bracket $[n]$ is an orbital rank, not a transverse moment.

Symbol	Meaning and convention
$F^U, F^L, F^{T[n]}$	SIDIS structure functions for the active quark number, helicity, and transverse-spin projections. These labels are not parton species.
$D_1, H_1^\perp$	Unpolarized and Collins fragmentation functions in the explicitly fixed $\mathbf{p}_T$ convention.
$\mu, \zeta, \mathbf{b}_T$	UV scale, rapidity scale, and transverse separation. μ as a scale is distinct from its use as a Lorentz index.
B, b_U, b_G, b_C, b_S	Hard gluon polarization matrix and its four real Stokes coefficients.
λ, h, P_ℓ	Doubled physical incoming electron helicity and its ensemble average. Historical source-label reproduction is a separate convention.
$L_{\mu\nu}, J_{\mu\nu}^\ell$	Conjugate-current-first and amplitude-first lepton tensors, related by reversing their Lorentz indices.
β_3, O, α_m	Octupole preparation strength, longitudinal moment, and signed cone factors; $O = 3\beta_3/\sqrt{5}$.
$\mathbf{p}_{\text{rel},T}, \mathcal{K}_a$	Heavy-pair relative transverse momentum and the Collins–Soper evolution kernel; neither is target rank K .
$\mathcal{W}[\rho], \mathcal{R}$	Target response and a spatial rotation, distinct from the Cartesian octupole moment R_{abc} .
$\mathbb{M}_{\text{source}}, \mathbb{M}_{\text{aux}}$	Spectral joint matrix and auxiliary coefficient coordinates; their partial-transpose relation preserves map rank, not individual-matrix positivity.

A reliable implementation can be checked at five interfaces. At the *spin interface*, reconstruct ρ from all moments and check its trace, Hermiticity, and eigenvalues. At the *tensor interface*, check traces, dimensions, reflection signs, and the sum/difference orbital ranks. At the *operator interface*, project a reconstructed correlator and recover each defining coefficient. At the *hard interface*, check the Dirac traces, momentum conservation, Ward identities, and hard-matrix ordering. At the *measurement interface*, check the fragmentation Jacobian, target and beam factors, angular moment normalization, and rank of the response design. A discrepancy at one interface should be fixed there rather than absorbed into an unnamed sign or normalization of a fitted TMD.

References

- [1] International Atomic Energy Agency, Nuclear Data Section, [Nuclear electromagnetic moments database](#) (n.d.), evaluated nuclear spin and moment data; accessed 2 October 2026.
- [2] U. Fano, Description of states in quantum mechanics by density matrix and operator techniques, *Rev. Mod. Phys.* **29**, 74 (1957).
- [3] R. L. Jaffe and A. Manohar, Deep inelastic scattering from arbitrary spin targets, *Nucl. Phys. B* **321**, 343 (1989).
- [4] A. Bacchetta and P. J. Mulders, Deep inelastic leptonproduction of spin-one hadrons, *Phys. Rev. D* **62**, 114004 (2000), [arXiv:hep-ph/0007120](#).

- [5] A. Bacchetta, M. Diehl, K. Goeke, A. Metz, P. J. Mulders, and M. Schlegel, Semi-inclusive deep inelastic scattering at small transverse momentum, *J. High Energy Phys.* **02** (2007), 093, arXiv:hep-ph/0611265 .
- [6] S. Kumano and Q.-T. Song, Transverse-momentum-dependent parton distribution functions up to twist 4 for spin-1 hadrons, *Phys. Rev. D* **103**, 014025 (2021), arXiv:2011.08583 [hep-ph] .
- [7] J. Zhao, Z. Zhang, Z.-t. Liang, T. Liu, and Y.-j. Zhou, Inclusive and semi-inclusive production of spin-3/2 hadrons in e^+e^- annihilation, *Phys. Rev. D* **106**, 094006 (2022), arXiv:2206.11742 [hep-ph] .
- [8] J. Zhao, Z. Zhang, Z.-t. Liang, T. Liu, and Y.-j. Zhou, Semi-inclusive production of spin-3/2 hadrons in deep inelastic scattering, *Phys. Rev. D* **109**, 074017 (2024), arXiv:2401.10031 [hep-ph] .
- [9] C. Lorcé and B. Pasquini, Structure analysis of the generalized correlator of quark and gluon for a spin-1/2 target, *J. High Energy Phys.* **09** (2013), 138, arXiv:1307.4497 [hep-ph] .
- [10] V. E. Lyubovitskij and I. Schmidt, New findings in gluon TMD physics, *Phys. Rev. D* **104**, 014001 (2021), arXiv:2105.07842 [hep-ph] .
- [11] I. C. Cloët, W. Bentz, and A. W. Thomas, EMC and polarized EMC effects in nuclei, *Phys. Lett. B* **642**, 210 (2006), arXiv:nucl-th/0605061 .
- [12] D. Fu, B.-D. Sun, and Y. Dong, Generalized parton distributions in spin-3/2 particles, *Phys. Rev. D* **106**, 116012 (2022), arXiv:2209.12161 [hep-ph] .
- [13] D. Fu, Y. Dong, and S. Kumano, Transversity generalized parton distributions in spin-3/2 particles, *Phys. Rev. D* **109**, 096006 (2024), arXiv:2402.11561 [hep-ph] .
- [14] D. Fu, Y. Dong, S. Kumano, and J.-J. Xie, Generalizing the Soffer bound: Positivity constraints on parton distributions of spin-3/2 particles, *Phys. Rev. D* **113**, L111901 (2026), arXiv:2602.11587 [hep-ph] .
- [15] P. J. Mulders and J. Rodrigues, Transverse momentum dependence in gluon distribution and fragmentation functions, *Phys. Rev. D* **63**, 094021 (2001), arXiv:hep-ph/0009343 .
- [16] D. Boer, S. Cotogno, T. van Daal, P. J. Mulders, A. Signori, and Y.-J. Zhou, Gluon and Wilson loop TMDs for hadrons of spin ≤ 1 , *J. High Energy Phys.* **10** (2016), 013, arXiv:1607.01654 [hep-ph] .
- [17] S. Cotogno, T. van Daal, and P. J. Mulders, Positivity bounds on gluon TMDs for hadrons of spin ≤ 1 , *J. High Energy Phys.* **11** (2017), 185, arXiv:1709.07827 [hep-ph] .
- [18] J. Zhao, A. Bacchetta, S. Kumano, T. Liu, and Y.-j. Zhou, Semi-inclusive deep inelastic scattering off a tensor-polarized spin-1 target, *J. High Energy Phys.* **12** (2025), 067, arXiv:2508.06134 [hep-ph] .
- [19] W. Cosyn and C. Weiss, Semi-inclusive deep-inelastic scattering on a polarized spin-1 target. I. Cross section and spin observables, *Phys. Rev. C* **114**, 025201 (2026), arXiv:2603.23699 [hep-ph] .
- [20] J. C. Collins, Fragmentation of transversely polarized quarks probed in transverse momentum distributions, *Nucl. Phys. B* **396**, 161 (1993), arXiv:hep-ph/9208213 .
- [21] S. Arnold, A. Metz, and M. Schlegel, Dilepton production from polarized hadron hadron collisions, *Phys. Rev. D* **79**, 034005 (2009), arXiv:0809.2262 [hep-ph] .

- [22] D. Boer, S. J. Brodsky, P. J. Mulders, and C. Pisano, Direct probes of linearly polarized gluons inside unpolarized hadrons, *Phys. Rev. Lett.* **106**, 132001 (2011), arXiv:1011.4225 [hep-ph] .
- [23] C. Pisano, D. Boer, S. J. Brodsky, M. G. A. Buffing, and P. J. Mulders, Linear polarization of gluons and photons in unpolarized collider experiments, *J. High Energy Phys.* **10** (2013), 024, arXiv:1307.3417 [hep-ph] .
- [24] A. Meurer, C. P. Smith, M. Paprocki, O. Čertík, S. B. Kirpichev, M. Rocklin, A. Kumar, S. Ivanov, J. K. Moore, S. Singh, T. Rathnayake, S. Vig, B. E. Granger, R. P. Muller, F. Bonazzi, H. Gupta, S. Vats, F. Johansson, F. Pedregosa, M. J. Curry, A. R. Terrel, Š. Roučka, A. Saboo, I. Fernando, S. Kulal, R. Cimrman, and A. Scopatz, SymPy: symbolic computing in Python, *PeerJ Comput. Sci.* **3**, e103 (2017).
- [25] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. Fernández del Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, and T. E. Oliphant, Array programming with NumPy, *Nature* **585**, 357 (2020).
- [26] D. Keller, Li7-TMDs: Spin-3/2 TMD symbolic validation and gluon hard-response toolkit, GitHub repository (2026).
- [27] T. C. Rogers and P. J. Mulders, No generalized TMD-factorization in the hadro-production of high transverse momentum hadrons, *Phys. Rev. D* **81**, 094006 (2010), arXiv:1001.2977 [hep-ph] .
- [28] R. Boussarie, M. Burkardt, M. Constantinou, W. Detmold, M. Ebert, M. Engelhardt, S. Fleming, L. Gamberg, X. Ji, Z.-B. Kang, C. Lee, K.-F. Liu, S. Liuti, T. Mehen, A. Metz, J. Negele, D. Pitonyak, A. Prokudin, J.-W. Qiu, A. Rajan, M. Schlegel, P. Shanahan, P. Schweitzer, I. W. Stewart, A. Tarasov, R. Venugopalan, I. Vitev, F. Yuan, and Y. Zhao, TMD Handbook (2023), arXiv:2304.03302 [hep-ph] .
- [29] J. C. Collins, Leading-twist single-transverse-spin asymmetries: Drell–Yan and deep-inelastic scattering, *Phys. Lett. B* **536**, 43 (2002), arXiv:hep-ph/0204004 .
- [30] J. C. Collins and A. Metz, Universality of soft and collinear factors in hard-scattering factorization, *Phys. Rev. Lett.* **93**, 252001 (2004), arXiv:hep-ph/0408249 .
- [31] X. Xie, D.-Y. Chen, and Z. Lu, Gluon TMDs for tensor polarized deuterons in a spectator model, *Phys. Rev. D* **114**, 034003 (2026), arXiv:2603.15224 [hep-ph] .
- [32] M. G. A. Buffing, A. Mukherjee, and P. J. Mulders, Generalized universality of definite rank gluon transverse momentum dependent correlators, *Phys. Rev. D* **88**, 054027 (2013), arXiv:1306.5897 [hep-ph] .
- [33] A. Peres, Separability criterion for density matrices, *Phys. Rev. Lett.* **77**, 1413 (1996), arXiv:quant-ph/9604005 .
- [34] D. Boer, L. Gamberg, B. Musch, and A. Prokudin, Bessel-weighted asymmetries in semi inclusive deep inelastic scattering, *J. High Energy Phys.* **10** (2011), 021, arXiv:1107.5294 [hep-ph] .
- [35] A. Bacchetta, U. D’Alesio, M. Diehl, and C. A. Miller, Single-spin asymmetries: the Trento conventions, *Phys. Rev. D* **70**, 117504 (2004), arXiv:hep-ph/0410050 .

- [36] S. Hino and S. Kumano, Structure functions in the polarized Drell–Yan processes with spin-1/2 and spin-1 hadrons: I. General formalism, *Phys. Rev. D* **59**, 094026 (1999), [arXiv:hep-ph/9810425](#) .
- [37] Y. V. Kovchegov and M. D. Sievert, Calculating TMDs of a large nucleus: Quasi-classical approximation and quantum evolution, *Phys. Rev. D* **93**, 014027 (2016), [arXiv:1505.01176 \[hep-ph\]](#) .
- [38] D. Keller, [Li7-TMDs: versioned computational companion](#), GitHub repository and documentation (2026), revision 19fa1a9; release provenance and validation scope recorded in the note. The release inventory is supplied publication evidence, not a new live retrieval for this revision.
- [39] F. Johansson *et al.*, [mpmath: a Python library for arbitrary-precision floating-point arithmetic](#), Software and documentation; version 1.3.0 used in the worked calculations (2023).